



DFleet 4.0

User Manual



DFleet User Manual

Version 1.0

2026-05-18

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This manual explains the handling and operating procedures, as well as warning information for the DFleet system. To take full advantage of the features of this system, read this manual thoroughly before using it. Store this manual where it can easily be referenced by anyone using this unit.

Designed and Made in Malaysia.

Table of Contents

Copyright Notice	2
Table of Contents	3
Preface	7
Section 1: Introduction	7
Safety	9
Section 1: Overview	9
Section 2: General Safety Symbol and Notes	9
Task Panel	10
Section 1. Accessibility	10
Supervisor View	11
User View	11
Call Button View	12
Guest View	12
Section 2. Task Panel Description	13
Create Tasks	13
Live Map	14
Description of colour of AMR icon in DFleet	14
Live Map 2D Features	15
Live Map 3D Features	17
Components	18
Management	21
Transaction	22
Dashboard	23
Configuration	25
Section 1. AGVs	25
Applying AMR Skillset	26
AGV Assignment policy	28
Pairing with AMR	32
Edit AGVs 3D Model	35
AGV Configuration Page	35
General Settings	35
AGV Home	36
Priority	36
Active	36
Dimension Profiles	37
Executor Settings	38
Allowed Motions	38
Task Triggers	38
Initialization Task Template	41
Initialization at a Station	41
Pre-Initialization Task Template	41
Default Initialization Task Template	42

Default Initialization Timeout	42
Default to Starting Task Runner in Paused State	42
Default App	42
Default App Timeout	42
Minimum Battery Level	42
Section 2. Component	43
Audio	44
Device Monitoring	44
Lift	44
Register	45
Remote I/O	46
Scheduler	47
Disable Call Button Scheduler	48
Add Task Scheduler	49
Add Management Task Scheduler	50
Calendar	51
Script	51
SecsGem	52
Sim AGV	52
Task Runner	53
Action Test	55
Homing	56
UPS	56
Section 3. Maps	57
Track AMR Map	58
Map Config Panel	59
Trackless AMR Map	63
Raw Map Editor	64
Map Layout Editor	67
Live App	72
Raw Map Editor Live App	73
Map Layout Editor Live App	74
Map Annotation	75
Map 3D Annotation	78
Properties	79
Import Map from AGV	79
Active Map	80
Map Teleportation	81
Map Transition Trigger	82
Example Use Case 1	85
Example Use Case 2	85
Map Parameters	87
Section 4. Task Templates	88
Task Template Editor/Creator	90
Task Template Groups and Users	96
Task Template Global Parameters	98



Task Template Variables	99
Skills of DFleet	101
Section 5. Management Task Templates	118
Creating Management Task Templates	119
Skills Available in Management Task Templates	119
Creating Management Task Trigger	138
Section 6. Transaction Manager	142
Section 7. Traffic Controller	147
Section 8. Backup & Restore	149
Backup	150
Advanced Backup	151
Restore	152
Advanced Restore	153
Section 9. DFleet Cluster	154
Section 10. DFleet Federation	157
Federation Peer	157
Federation Coordinator	158
Section 11. Network Connections	159
Network Configurations Editor	159
Hostname Editor	160
HTTPS Settings Editor	160
Section 12. Date & Time	161
Section 13. DFleet Controller	162
Section 14. Users, Groups & Permissions	163
Users	163
Groups	164
Permissions	164
Create New User	165
Delete User	165
Change User Password	165
Default Permission Group	166
Call Button	166
User	166
Supervisor	166
Identity Provider	167
Section 15. Reports	170
AGV Activities	170
Completed Tasks	174
Completed Transactions	175
System Monitor	176
Message Inspector	177
Audit Log	178
Custom Log	178
System Log	179
Download Log	179
Section 16. Search	180



Section 17. Help	181
Advance Features	182
Section 1. Custom Panel	182
Available Tag	184
Disconnected Indicator	184
Live Map	184
AMR Status	184
AMR Card	184
AMR Summary	185
AMR Battery	185
Running Tasks	185
Completed Tasks	185
Task Card	185
Task Breakdown Card	185
Recent Custom Logs	186
System Info Bar	186
Controller Info Row	186
User Group Conditional Logic	186
Live Register	187
Live Remote IO	187
Live Remote IO Error	187
Running Transactions	187
Completed Transactions	188
Transaction Card	188
Transaction Breakdown Card	188
Active Carriers	188
Blocked Ports	188
SecsGem Terminal	188
SecsGem Row	188
Live Device Monitoring	189
Device Monitoring Row	189
Javascripts API	189
DFleet.apiUrl (property)	190
DFleet.agv (object)	191
DFleet.agv.getAgvs() (function)	191
DFleet.agv.channel.subscribe(cb) (function)	191
DFleet.agv.channel.unsubscribe(cb) (function)	191
DFleet.agv.Agv (class)	191
DFleet.task (object)	192
DFleet.task.channel.subscribe(cb) (function)	192
DFleet.task.channel.unsubscribe(cb) (function)	192
DFleet.task.Task (class)	192
DFleet.CPNT.register (object)	193
DFleet.CPNT.register.getRegisters() (function)	193
DFleet.CPNT.register.channel.subscribe(cb) (function)	193
DFleet.CPNT.register.channel.unsubscribe(cb) (function)	193



DFleet.CPNT.remotelo (object)	194
DFleet.CPNT.remotelo.getDevices() (function)	194
DFleet.CPNT.remotelo.channel.subscribe(cb) (function)	194
DFleet.CPNT.remotelo.channel.unsubscribe(cb) (function)	194
Section 2. SMTP Settings	195
Section 3. Webhooks	196
Adding a new webhook	196
Content of the webhook	197
agv_update event	197
task_create & task_update	198
Additional Content	200
Section 1: AMR Transfer Between DFleet	200
Section 2: AMR in Transaction Mode	205
Disclaimer	206

Preface

Section 1: Introduction

DFleet is a fleet management system that enables multiple AMRs to execute tasks within the same working area. DFleet controls the traffic flow of AMR and manages task assignment to ensure all AMRs efficiently complete the tasks. Moreover, DFleet extends its capability by communicating to other devices to allow features such as automatic task triggering.



AGVs

DFleet Version: 4.0.0-rust (trackless)

DFleet Endpoint:

AGVs (1-1 of 1)

☐	#	Name	Home	Priority	Skillset	Active	Last Seen	Operation
☐	1	SIM 01 (00000000-0000-0000-0001-000000000000)	Home sim 1	0			a minute ago	  

 Edit AGV assignment policy

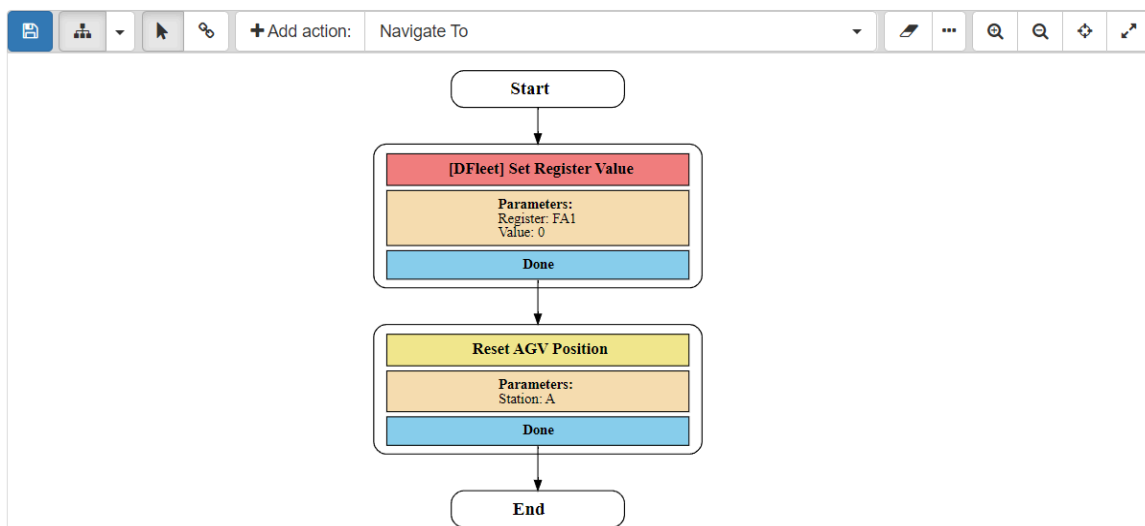
 Edit AGVs 3D Model

AGVs' Active Skillsets:



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Structure



Safety

Section 1: Overview

This chapter contains important safety information, which must be read and understood by the integrator of DF AMR. It is essential that all instructions and guidance provided in other chapters and parts of this manual are observed and followed.

Section 2: General Safety Symbol and Notes



Danger:

Indicates an imminently hazardous situation that will result in death or severe personal injury, or major damage to the equipment, if proper precautions are not taken.



Warning:

Indicates a potentially hazardous situation that **could** result in death or severe personal injury, or major damage to the equipment, if proper precautions are not taken.



Caution:

Indicates a situation that **could** result in minor personal injury or damage to the equipment if proper precautions are not taken.



Mandatory:

Indicates an action that **will** result in property damage if not conducted properly.



Notice:

Indicates a situation that **could** result in property damage if proper precautions are not taken.

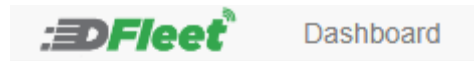


Info:

Indicates a situation that **could** result in easier access of certain functions or task completion

Task Panel

Task Panel is the landing page after login. On this page, users can monitor the list of running tasks and completed tasks. To access the DFleet task panel, users can navigate to the DFleet server IP address using a web browser such as Firefox or Chrome.



Task Panel

Section 1. Accessibility

Task Panel can be accessed by all level of users and the default users are:

- Supervisor
- User
- Call button
- Guest

Although the task panel is accessible by all types of users, the information displayed within the task panel varies depending on the user's type.



Supervisor View

Menu

Task

Live Map

Components

Management

Transaction

Device Connectivity

AGVs: 0 / 2 RTOs: 0 / 18 Others: 0 / 0

AGVs

Name:

Status:

Battery:

Location:

Task Counter:

Mileage:

Last Seen:

AMR 1

Disconnected

-%

-

-

-

-

Name:

Status:

Battery:

Location:

Task Counter:

Mileage:

Last Seen:

AMR 2

Disconnected

-%

-

-

-

-

Today's Schedule

No Event yet.

A1 Material Delivery

A1 Recovery

A1 Recovery Flow

A2 Call AGV To Supply Conveyor

A2 Call AGV To Supply Conveyor 1....

A2 Homing

A2 Material Delivery

A2 Recover Empty Pallet

A2 Recover Material Delivery

Lifter DOWN

Lifter UP

Recovery A1 Homing

Recovery A2 Homing

Remote Io Read Test

Create Task

Running Tasks

Name	Template	AGV	Status	Creation Time	Start Time	ETA Station	ETA	Owner
No running task yet.								

Completed Tasks

Name	Template	Agv	Status	Creation Time	Start Time	End Time	Duration	Owner
No completed task yet.								

View all completed tasks.

User View

Menu

Task

Live Map

Components

Management

Transaction

AGVs

Name:

Status:

Battery:

Location:

Task Counter:

Mileage:

Last Seen:

SIM 01

Not Ready

88.0%

-

-

1,716.7m

now

Today's Schedule

No Event yet.

No task templates assigned to current user.

Create Task

Running Tasks

Name	Template	AGV	Status	Creation Time	Start Time	ETA Station	ETA	Owner
No running task yet.								

Completed Tasks

Name	Template	Agv	Status	Creation Time	Start Time	End Time	Duration	Owner
Traffic Navigate To	Traffic Navigate To	SIM 03	Aborted	2:43 p.m. 07/05/2026	2:43 p.m. 07/05/2026	3:07 p.m. 07/05/2026	24 minutes	-
Traffic Navigate To	Traffic Navigate To	SIM 05	Aborted	2:43 p.m. 07/05/2026	2:43 p.m. 07/05/2026	3:07 p.m. 07/05/2026	23 minutes	-
Traffic Navigate To	Traffic Navigate To	SIM 02	Aborted	2:43 p.m. 07/05/2026	2:43 p.m. 07/05/2026	3:07 p.m. 07/05/2026	23 minutes	-
Traffic Navigate To	Traffic Navigate To	SIM 04	Aborted	2:43 p.m. 07/05/2026	2:43 p.m. 07/05/2026	3:07 p.m. 07/05/2026	23 minutes	-
Traffic Navigate To	Traffic Navigate To	SIM 01	Aborted	2:43 p.m. 07/05/2026	2:43 p.m. 07/05/2026	3:07 p.m. 07/05/2026	23 minutes	-

View all completed tasks.

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Call Button View

DFleet

Dashboard

May 18, 2026, 10:40:19 AM

CallButton

AGV Calling System

Running Tasks

Name	Template	AGV	Status	Creation Time	Start Time	ETA Station	ETA	Owner
No running task yet.								

No task templates assigned to current user.

AGV Status

SIM 01: Not Ready

Today's Schedule

Task name	Operation Mode	Start Time	Arrival Time	End Time
No scheduled event today.				

Completed Tasks

Name	Template	Agv	Status	Creation Time	Start Time	End Time	Duration	Owner
Traffic Navigate To	Traffic Navigate To	SIM 03	Aborted	2:43 p.m. 07/05/2026	2:43 p.m. 07/05/2026	3:07 p.m. 07/05/2026	24 minutes	-
Traffic Navigate To	Traffic Navigate To	SIM 05	Aborted	2:43 p.m. 07/05/2026	2:43 p.m. 07/05/2026	3:07 p.m. 07/05/2026	23 minutes	-
Traffic Navigate To	Traffic Navigate To	SIM 02	Aborted	2:43 p.m. 07/05/2026	2:43 p.m. 07/05/2026	3:07 p.m. 07/05/2026	23 minutes	-
Traffic Navigate To	Traffic Navigate To	SIM 04	Aborted	2:43 p.m. 07/05/2026	2:43 p.m. 07/05/2026	3:07 p.m. 07/05/2026	23 minutes	-
Traffic Navigate To	Traffic Navigate To	SIM 01	Aborted	2:43 p.m. 07/05/2026	2:43 p.m. 07/05/2026	3:07 p.m. 07/05/2026	23 minutes	-

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Guest View

DFleet

Dashboard

May 18, 2026, 10:39:32 AM

Login

AGV Calling System

Running Tasks

Name	Template	AGV	Status	Creation Time	Start Time	ETA Station	ETA	Owner
No running task yet.								

Today's Schedule

Task name	Operation Mode	Start Time	Arrival Time	End Time
No scheduled event today.				

Completed Tasks

Name	Template	Agv	Status	Creation Time	Start Time	End Time	Duration	Owner
Traffic Navigate To	Traffic Navigate To	SIM 03	Aborted	2:43 p.m. 07/05/2026	2:43 p.m. 07/05/2026	3:07 p.m. 07/05/2026	24 minutes	-
Traffic Navigate To	Traffic Navigate To	SIM 05	Aborted	2:43 p.m. 07/05/2026	2:43 p.m. 07/05/2026	3:07 p.m. 07/05/2026	23 minutes	-
Traffic Navigate To	Traffic Navigate To	SIM 02	Aborted	2:43 p.m. 07/05/2026	2:43 p.m. 07/05/2026	3:07 p.m. 07/05/2026	23 minutes	-
Traffic Navigate To	Traffic Navigate To	SIM 04	Aborted	2:43 p.m. 07/05/2026	2:43 p.m. 07/05/2026	3:07 p.m. 07/05/2026	23 minutes	-
Traffic Navigate To	Traffic Navigate To	SIM 01	Aborted	2:43 p.m. 07/05/2026	2:43 p.m. 07/05/2026	3:07 p.m. 07/05/2026	23 minutes	-

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Section 2. Task Panel Description

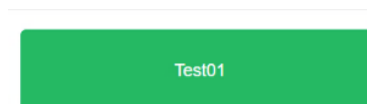


- | | |
|------------------------|-------------------------|
| 1. Create Task | 7. AGVs |
| 2. Live Map | 8. Daily Schedule |
| 3. Components | 9. AMR Tasks |
| 4. Management | 10. Running AMR Tasks |
| 5. Transaction | 11. Completed AMR tasks |
| 6. Device Connectivity | |

Create Tasks

There are few ways to initiate the tasks through the DFleet system. Among them are using the task button and using the create task menu.

1. Create Tasks Using Task Buttons



2. Create Tasks using "Create Task" menu

Help

Create Task

Name*

Task template*

Test01

Start after

Confirm

Cancel

Live Map

Live Map displays the location of all AMRs and updates their location in real time. The map can be panned or zoomed to show a particular section of the map and switch tabs to select different active maps. The button at the top right allows the user to switch between the 2D and 3D views of the Live Map.



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Description of colour of AMR icon in DFleet

No.	AMR Colour	Description
1.	Yellow	Running task with "Wait" function included into the task template. (Refer to 3.4.2 for "Wait" function)
2.	Green	AMR idle or running task
3.	Blue	Waiting for traffic controller
4.	Red	AMR safety triggered. Click on the AMR to display the safety message.

Live Map 2D Features

The settings tab on the Live Map allows the user to configure the elements to be displayed on the Live Map.

Live Map

[3D View](#)

Traffic CPU ZONE A ZONE B Zone C ZONE E ZONE F ZONE H 

Show labels for AGVs

Nothing selected

Show labels for stations

Nothing selected

Show only AGVs in the following states

Running Task Runner App

☒ Display map annotation

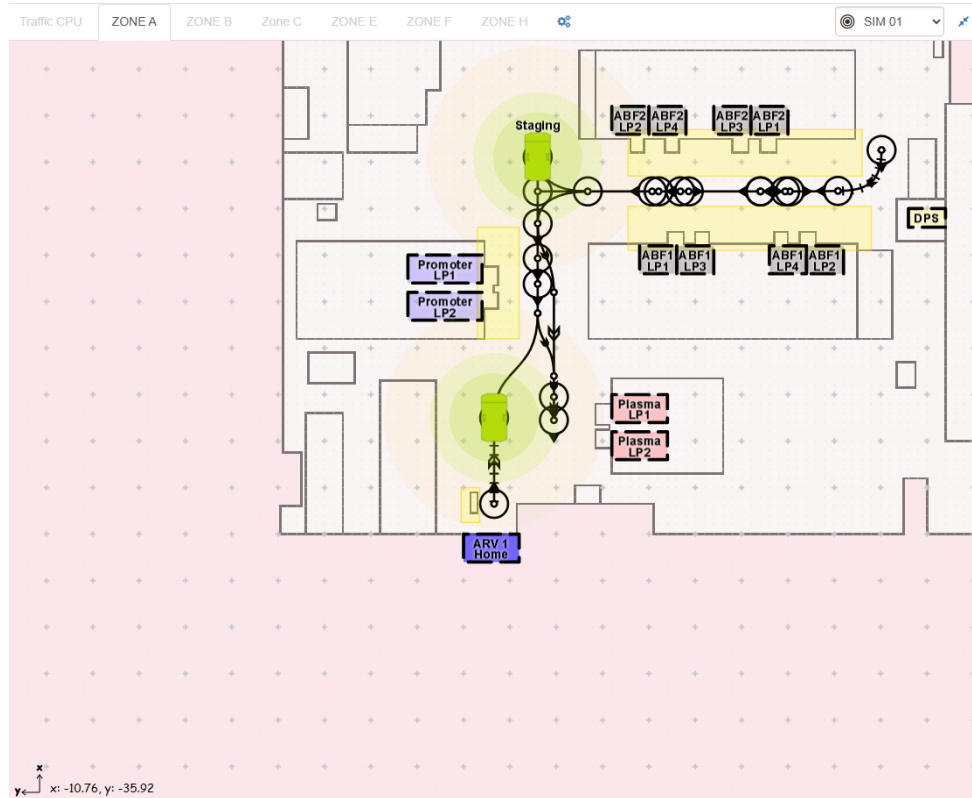
☐ Display traffic reservation

Apply

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Live Map Settings	
Show labels for AGVs	Dropdown to select AGVs to display, with name labels.
Show labels for stations	Dropdown to select stations to display, with name labels.
Show only AGVs in the following states	Display AGVs only when: - Running Task Runner App - Running Other App - Not Ready
Display map annotation	Display map annotation edited in the Map Editor.
Display traffic reservation	Display traffic reservation.

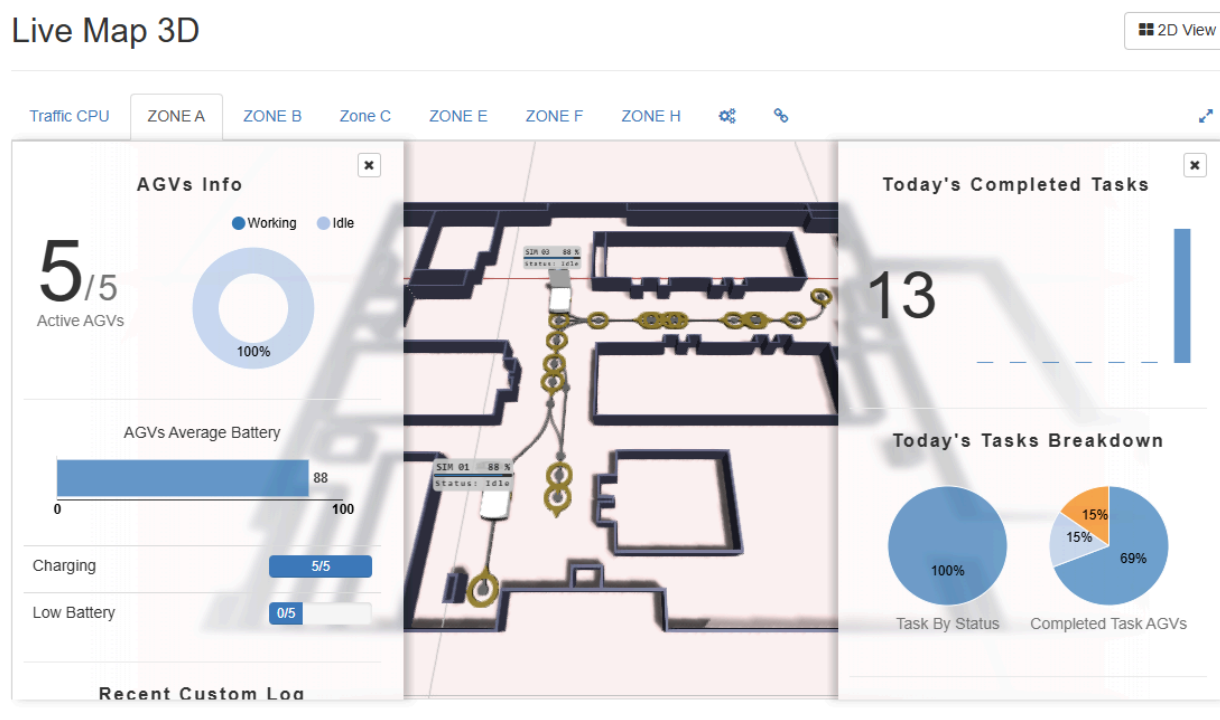
The toolbar at the top right consists of a **Focus To AGV** dropdown and a **Fullscreen** button. The **Focus To AGV** dropdown allows the user to lock the Live Map view so that it always centers on the selected AGV. The fullscreen toggle button will enter or exit fullscreen mode when clicked.



Live Map 3D Features

The Live Map in 3D view comes by default with the AMR, stations, walls, and other 3D map annotations, along with dashboard information.

Live Map 3D



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The settings tab on the Live Map 3D allows the user to configure the elements to be displayed on the Live Map.

Live Map Settings	
Show only AGVs in the following states	Display AGVs only when: - Running Task Runner App - Running Other App - Not Ready
Display map path	Display map path in 3D map.
Display map wall	Display map wall in 3D map.

The **Link to Current View** button, located to the right of the settings button, is used to update the dashboard information, while the **Fullscreen** toggle button at the top right will enter or exit fullscreen mode when clicked.

Components

Display all available components and provide a live view of component status. Components need to be configured in the component page which can be accessed by going through Settings > Components.

Components (Live View)

Menu	#	Component	View
Task	1	Audio	
Live Map	2	Device Monitoring	
Components	3	Lift	
Management	4	Register	
Transaction	5	Remote Io	
	6	Script	
	7	Secsgem	
	8	Sim Agv	
	9	Ups	

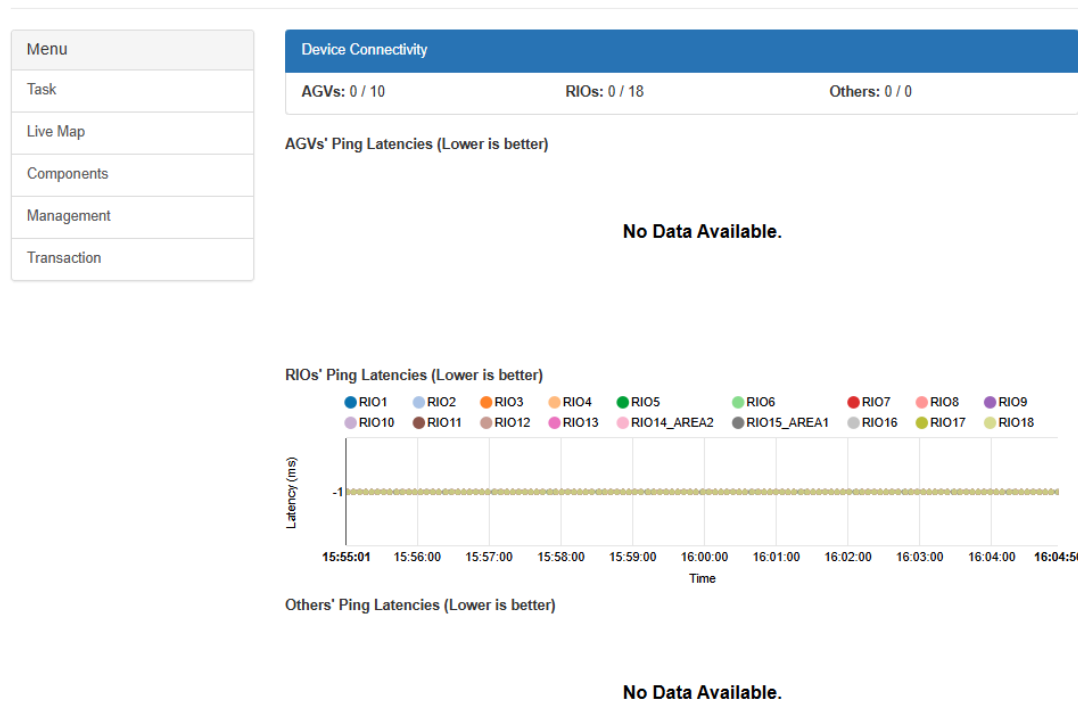
Component Live View Panel

Audio (Live View)

Menu	#	Name	Status	Action
Dashboard	1	AGV Alarm		
Live Map	2	Channel 1		  
Components	3	Channel 2		  
Management	4	Channel 3		  
Transaction	5	Channel 4		  
	6	Channel 5		  

Audio Live View

Device Monitoring (Live View)



Device Monitoring Live View

Lift (Live View)

Menu	#	Name	Status
Dashboard	1	Lift 1	Type: Task Template Lift Reservation: False
Live Map			Reserve Unreserve
Components			

Lift Live View

Register (Live View)

Menu	FA	FB	FC	FP
Dashboard				
Live Map				
Components				
Management				
Transaction				

Register Name	Register Value	Remarks	Operation
FA0	0	Trigger barcode scanner for Area 1 (0 = no pallet detected, 1 = pallet detected)	
FA1	0	Trigger barcode scanner for Area 2 (0 = no pallet detected, 1 = pallet detected)	
FA2	0	Test Load Unload	
FA3	0	Button Trigger for Area 2 (0 = Button not trigger, 1 = Button trigger)	
FA4	0		
FA5	0	Pallet Destination Overwrite(0 = Recovery Task, 1 = Destination Overwrite)	
FA6	0		
FA7	0		
FA8	0		

Register Live View

Remote IO (Live View)

Menu	#	Name	Status	Remarks
Dashboard	1	RIO1	Model: WISE-4051 🔗 Connection failed I7 I6 I5 I4 I3 I2 I1 I0	I0 I1 I2 I3
Live Map				Empty(A7) Condenser(A7) Basepan(A7) Compressor 2(A7)
Components				I4 I5 I6 I7
Management				Compressor 1(A7) Tube Assembly(A7) P-Fan(A7) Small part(A7)
Transaction	2	RIO2	Model: WISE-4051 🔗 Connection failed I7 I6 I5 I4 I3 I2 I1 I0	I0 I1 I2 I3
				Home 13(A6) Condenser(A6) Compressor 2(A6) Compressor 1(A6)
				I4 I5 I6 I7
				P-Fan(A6) Basepan(A6) Tube Assembly(A6) Condenser(A4)

Remote IO Live View

Script (Live View)

Menu	Warning: Stopping script too frequently might crash the system.			
Dashboard	Scripts			
Live Map	#	Name	State	Action
Components	1	HelloWorldScript	RUNNING	pid 30109, uptime 0:00:01
Management	▶ Start All ■ Stop All			
Transaction	Scheduled Scripts			
	#	Name	Last Run	Action
	No data available.			

Scripts Live View

SecsGem (Live View)

Menu Task Live Map Components Management Transaction	Communication State	DISABLED
	Control State	HOST OFF-LINE
	Initial Online Control State	REMOTE Set Local
	Terminal Message Acknowledge 	
	-	
	Message to host Send 	

[Download secsgem log](#)

Secs/Gem Live View

Management

The Management Panel allows users to create management tasks from a list of available task templates created beforehand through the Management Panel. The panel also provides a status overview of all created tasks at the bottom. To create a task, simply click on the desired task from the list.


Panel
Settings ▾
Users ▾
Report ▾
Search
Help
🔊
June 30, 2023, 10:49:03 AM
admin ▾

Management Panel

Menu Dashboard Live Map Components Management Transaction	TestAGV		
	Task Status		
	Task Name	Status	Operation
	TestAGV	active	ⓘ

Management panel

Create management task "TestAGV" ✕

Create management task

Transaction

The Transaction Panel displays the status overview of running transactions, completed transactions and active carriers. It also allows users to add or stop transactions.

Transaction Panel

Add Transaction

Running Transaction

Name	Transfer	AGV	Action	Status	Owner
▶ SimSim Group	ddd: a1 → e2 cccc: e2 → b2	SimSim	Goto E	-	-
Manual 72	ddf: d1 → SIM 02 (port1)	-	-	Queue	admin

Completed Transaction

Name	Transfer	AGV	Status	Error	Creation Time	Start Time	Transfer Time	End Time	Owner
Manual 73	ddd: b2 → a1	SimSim	Completed	-	3:10 PM	3:10 PM	3:10 PM	3:11 PM	admin
Manual 71	ss: b2 → SimSim (port2)	SimSim	Completed	-	3:10 PM	3:10 PM	3:10 PM	3:10 PM	admin
Manual 70	dfdfdf: e1 → a1	SimSim	Cancelled	Task Aborted	10:24 AM	10:24 AM	-	10:24 AM	admin
Manual 69	dfdfdf: b1 → a1	SimSim	Aborted	Task Aborted	10:22 AM	10:23 AM	10:23 AM	10:23 AM	admin
Manual 68	ddd: b1 → a1	SimSim	Aborted	Task Aborted	9:05 AM	9:05 AM	-	-	admin

[View all completed transactions.](#)

Active Carrier

Name	Port	On Agv	Reporter	
ddd	SimSim (port1)	✓	carrier_manager	+ ✎ ✕
ss	SimSim (port2)	✓	carrier_manager	✎ ✕

[View all carriers.](#)

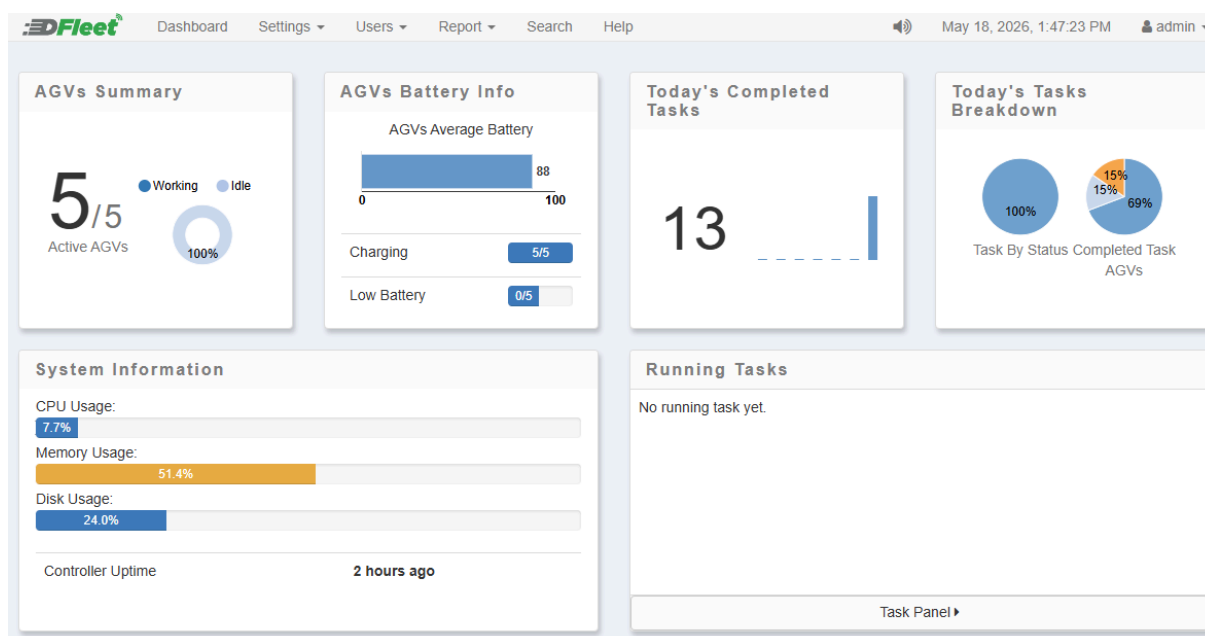
A carrier is considered in active state when it is either currently on AMR port or when the carrier's last update is not more than 6 hours ago. Under the Active Carrier section, users are allowed to manually change the carriers' locations as well as register new carriers. By clicking the "View all carriers" link, it can route to the Carrier page where users can download the carriers' log, which records all the editing actions of the carriers.

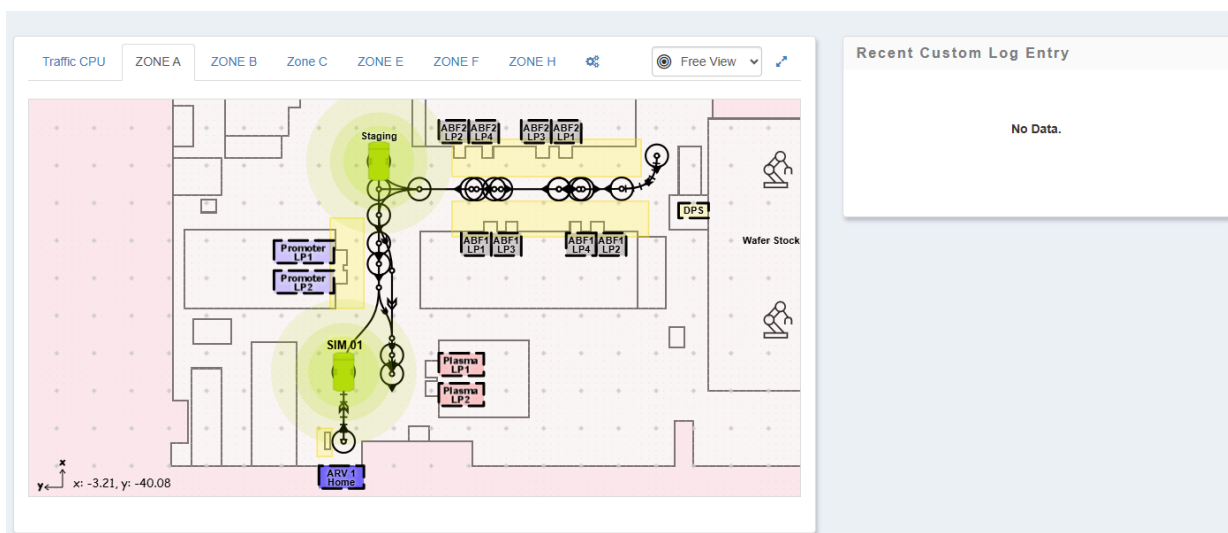
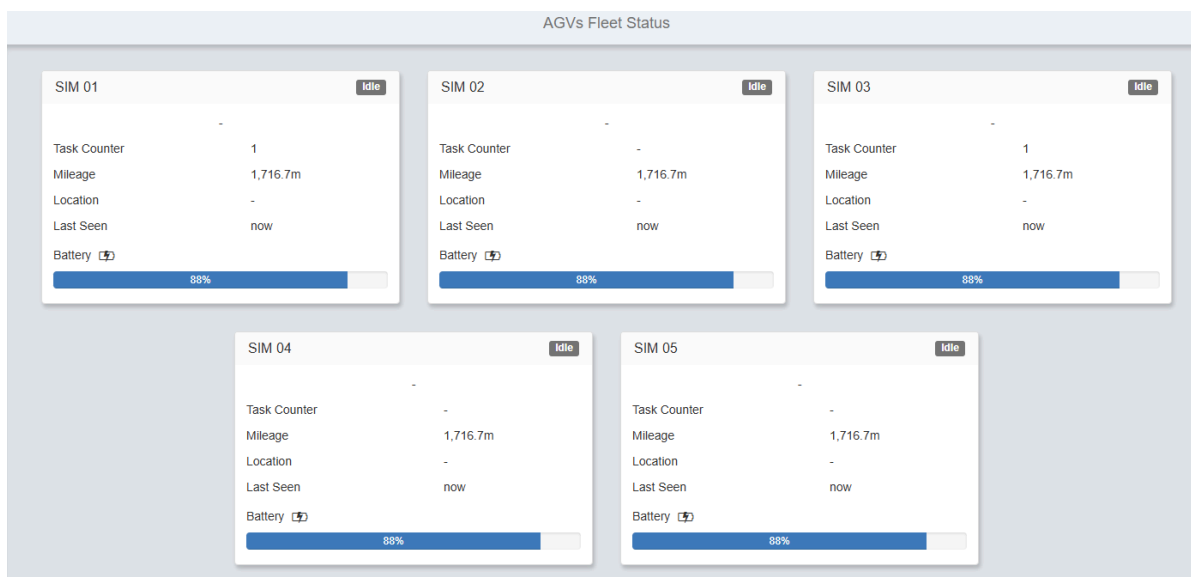
Dashboard

The Dashboard is the main overview page of DFleet for monitoring purposes. The info provided includes real-time status and logging.

The Dashboard provides the following information:

1. **AGVs Summary** — Shows the current number of active AGVs and their percentage relative to the total fleet.
2. **AGVs Battery Info** — Displays the average battery level, individual AGV battery levels, and charging status for each AGV.
3. **Today's Completed Tasks** — Shows the total number of tasks completed today.
4. **Today's Tasks Breakdown** — Provides a breakdown of tasks by status and highlights the AGVs that completed them.
5. **System Information** — Displays DFleet server hardware resource usage, such as CPU, memory, and storage.
6. **Running Tasks** — Shows a live list of tasks currently being executed by the AGVs.
7. **AGVs Fleet Status** — Shows all AGVs information.
8. **Live Map** — Similar to the Live Map page in 2D view.
9. **Recent Custom Log Entry** — Shows the latest custom log.

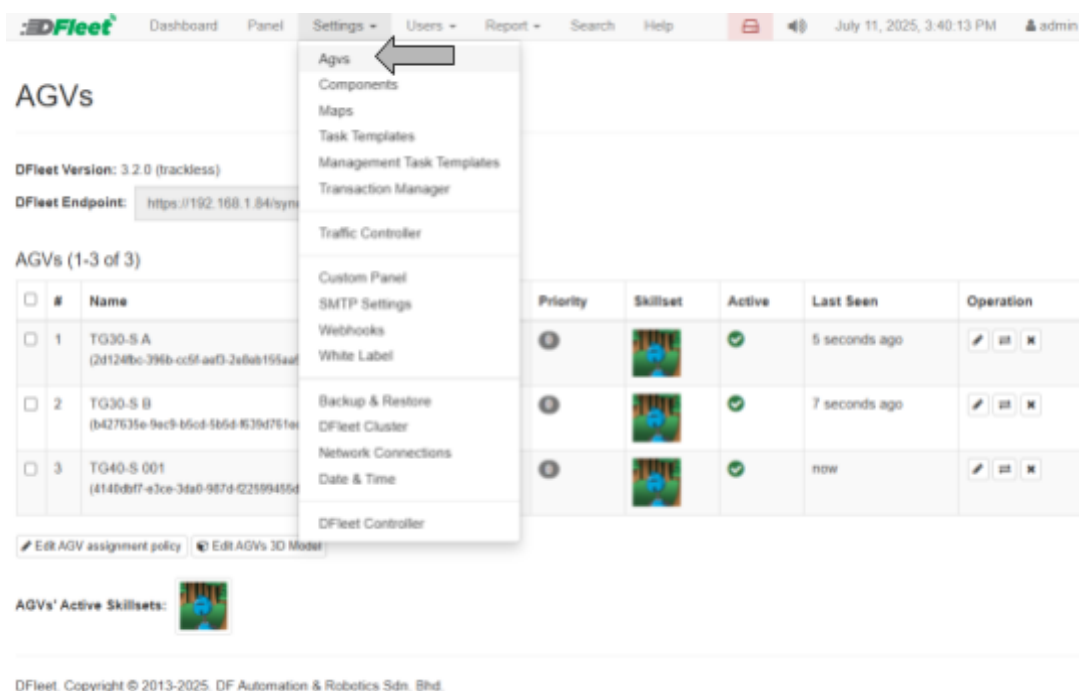




Configuration

Section 1. AGVs

In the AGV tab, we can know the description of the AMR. We can also know about the skillset, and its operation to cancel the operation or to check whether the AMR is active or not through this tab also. We can also set the priority of the AMRs.



DFleet Version: 3.2.0 (trackless)

DFleet Endpoint: <https://192.168.1.84/sym>

AGVs (1-3 of 3)

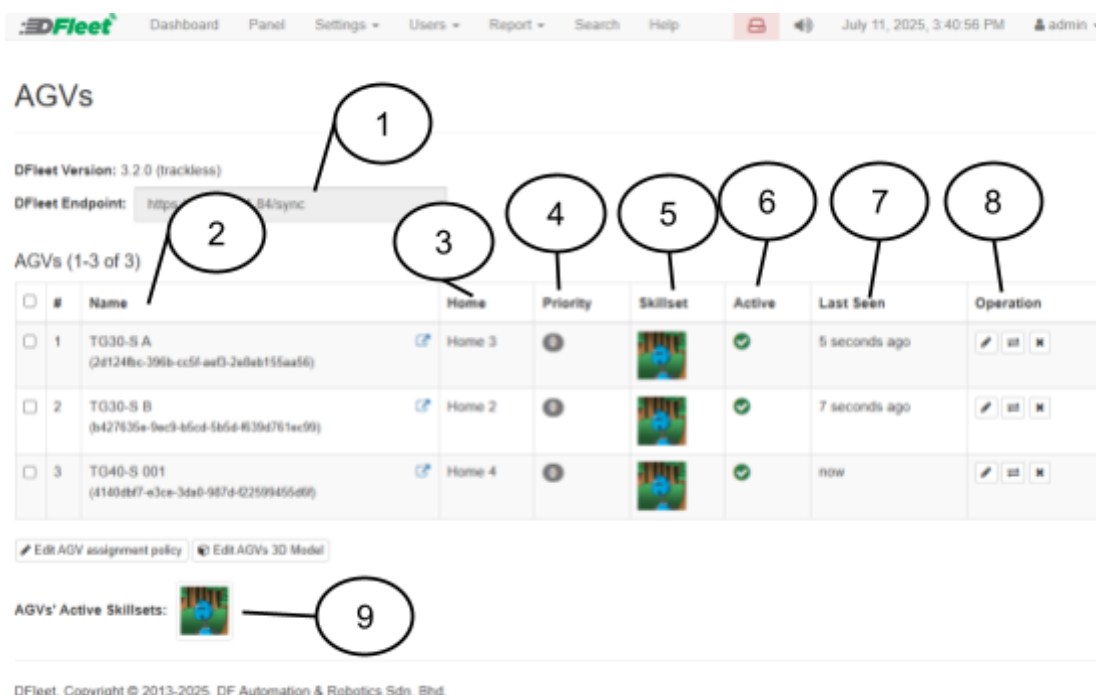
#	Name	Priority	Skillset	Active	Last Seen	Operation
1	TG30-S A (2d1248c-396b-cc5f-ae5-2e6ab155aaf)	3			5 seconds ago	  
2	TG30-S B (b427635e-9ac9-b5cd-5b54-639d751e)	3			7 seconds ago	  
3	TG40-S 001 (414dbd7-e3ce-3da0-987d-422599455d)	3			now	  

[Edit AGV assignment policy](#) [Edit AGVs 3D Model](#)

AGVs' Active Skillsets: 

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Navigate to AGVs page



AGVs page

No.	Component	Description
1.	DFleet Endpoint	DFleet endpoint (Used to connect AMR with DFleet)
2.	Name	AMR name
3.	Home	Home (Home of AMR)
4.	Priority	Priority (Making AMR to accomplish highest priority tasks first)
5.	Skillset	Skillset (AMR skillset varies depending on the skillset icon)
6.	Active	Active (AMR's Activeness)
7.	Last Seen	Last Seen (Last seen of AMR)
8.	Operation	Operation (Used to edit, swap or delete AMR from DFleet)
9.		Skillset icons (Used to display the AMR skill set as not all AMRs have the same skillset)

Applying AMR Skillset

By default, the AMR's skillset is applied automatically. However, if the AMR's skillset changes, you must manually update it. For example, if AMR2 Zone A below is upgraded and its skillset changes, you need to apply the new skillset.

AGVs

DFleet Version: 3.2.0 (trackless)

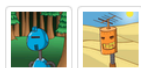
DFleet Endpoint: <https://dfleet-p2.demo.dfautomation.com/syn>

AGVs (1-3 of 3)

☐	#	Name	Home	Priority	Skillset	Active	Last Seen	Operation
☐	1	AMR1 Zone A (e65f3b8e-5988-14b7-70c8-98c4ec83ef7f)	circle_home4	0		✓	3 weeks ago	Edit Reset Delete
☐	2	SIM 01 (00000000-0000-0000-0001-000000000000)	circle_home1	0		✓	4 days ago	Edit Reset Delete
☐	5	AMR2 Zone A (b39c9f0b-db0d-708e-83aa-87f9a9e55823)	circle_home1	0	Active Actual	✗	3 weeks ago	Edit Reset Delete

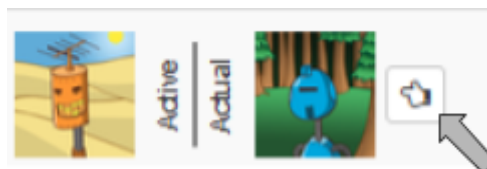
[Edit AGV assignment policy](#) [Edit AGVs 3D Model](#)

AGVs' Active Skillsets:



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Applying the AMR's actual skillset is simple. Just click the hand icon in the skillset column.



[< AGVs](#)

Apply skillset for AGV "AMR2 Zone A?"

Are you sure you want to apply skillset for AGV "AMR2 Zone A"?

[Apply Skillset](#)

[Cancel](#)

Once applied, the active skillsets in the bottom left corner of the page will be updated.

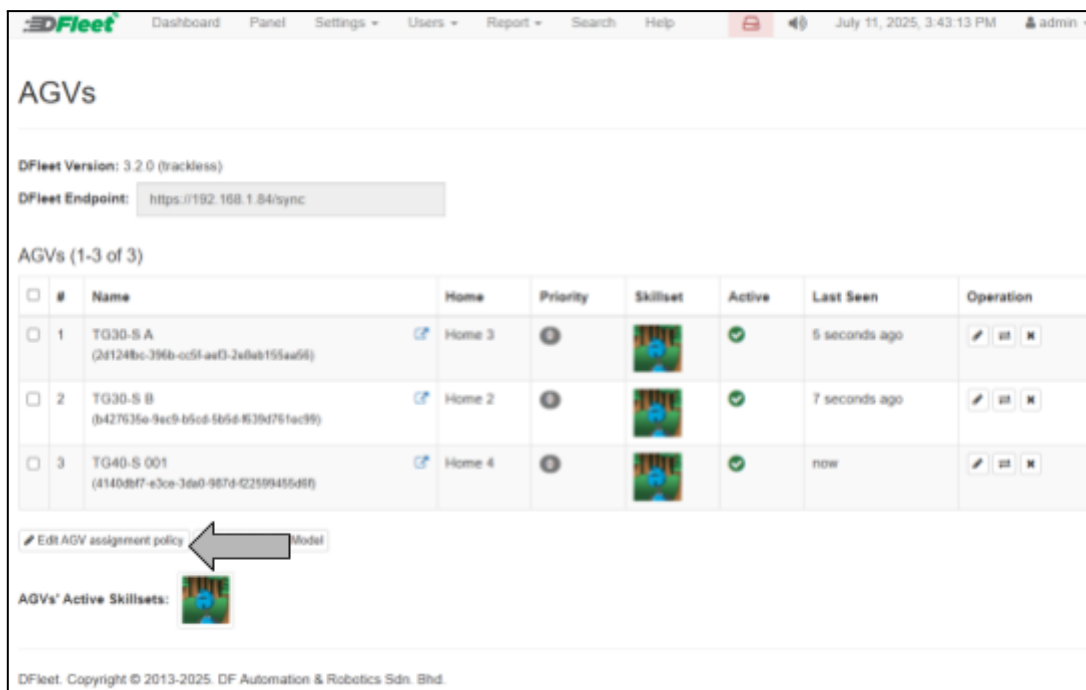


Notice:

Starting from version 3.1, DFleet is able to handle AMRs with different skillsets..

AGV Assignment policy

For viewing the assignment policy of AMR, you need to select the edit AGV assignment policy. It is located at the bottom of the AMR list.



DFleet Version: 3.2.0 (trackless)

DFleet Endpoint: <https://192.168.1.84/sync>

AGVs (1-3 of 3)

#	Name	Home	Priority	Skillset	Active	Last Seen	Operation
1	TG30-S A (2d124bc-396b-cc5f-ae03-2e8ab155aa54)	Home 3	3			5 seconds ago	  
2	TG30-S B (b427635e-9ec9-b5c6-5b56-8539d761ac99)	Home 2	3			7 seconds ago	  
3	TG40-S 001 (4140bf7-e3ce-3da0-987d-02259945546f)	Home 4	3			now	  

[Edit AGV assignment policy](#) 

AGVs' Active Skillsets: 

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Navigate to AGV Assignment Policy

[← AGVs](#)

AGV Assignment Policy

Enabled Criteria (From most to least significant)	
+	Universal Agv Priority
+	Highest ▾ Battery Percentage (Tolerance: 5 %)
+	Least ▾ Recently Used (Tolerance: 5 seconds)

Disabled Criteria	
+	Lowest ▾ Task Counter (Tolerance: 1 tasks)
+	Task Specific Agv Priority
+	Shortest ▾ Travel Distance (Tolerance: 1 m)
+	Lowest ▾ Mileage (Tolerance: 10 m)

[Submit](#)

Assignment Policy page

Here you can enable or disable the criteria for the AMR. Simply just drag and drop it to arrange it.

Enabled Criteria (From most to least significant)	
+	Universal Agv Priority
+	Highest ▾ Battery Percentage (Tolerance: 5 %)
+	Least ▾ Recently Used (Tolerance: 5 seconds)

Disabled Criteria	
+	Lowest ▾ Task Counter (Tolerance: 1 tasks)
+	Task Specific Agv Priority
+	Shortest ▾ Travel Distance (Tolerance: 1 m)
+	Lowest ▾ Mileage (Tolerance: 10 m)

Drag and drop criteria

The sorting also can be made from highest to lowest or vice versa. These are some of the criteria:

No.	Assignment Criteria	Description
1.	Recently used (Least / Most)	Task assignment based on most / least recently used AMR.
2.	Battery Percentage (Highest / Lowest)	Task assignment based on AMR with the highest / lowest battery percentage.
3.	Task Counter (Highest / Lowest)	Task assignment based on AMR with the highest / lowest task counter.
4.	Task Specific Agv Priority	Task assignment based on the priority configured in the "Allowed AGVs" column. (Refer to Task Templates section for AMR priority)
5.	Universal Agv Priority	Task assignment based on AMR priority defined in AMR setting. (Refer to AGV Configuration Page section for AMR priority)
6.	Travel Distance (Shortest/ Longest)	Task assignment based on travel distance to destination hint defined in task template. (Refer to Task Templates section for destination hint)
7.	Mileage (Lowest / Highest)	Task assignment based on lowest / highest mileage of the AMR.

Enabled Criteria (From most to least significant)

+	Universal Agv Priority
+	Highest Battery Percentage (Tolerance: 5 %)
+	Highest Recently Used (Tolerance: 5 seconds)

Set sorting to highest or lowest



Notice:

Sorting can only be configured after the criteria is enabled.

The tolerance for each assignment policy is configurable. For example, in the scenario below, if the AMRs have the same Universal AGV Priority, the system will next compare their battery percentages. If the difference in battery levels among the AMRs is less than 5%, it will proceed to select the AMR that was least recently used.

[< AGVs](#)

AGV Assignment Policy

Enabled Criteria (From most to least significant)

+	Universal Agv Priority
+	Highest Battery Percentage (Tolerance: 5 %)
+	Least Recently Used (Tolerance: 5 seconds)

Example of Assignment Policy Configuration

Pairing with AMR

There are a few steps to be taken to pair the AMR with the DFleet system.

Step 1:

Login to the DFleet system as Supervisor and enter the AGVs tab.

AGVs

DFleet Version: 2.2.0 (tracked)

DFleet Endpoint:

AGVs (1-1 of 1)

#	Name	Home	Priority	Skillset
1	Keon (8005e787-b603-4393-9bcb-7872e81e625a)	A	0	

[Edit AGV assignment policy](#)

AGVs' Active Skillset:



Step 2:

Copy the DFleet Endpoint link. It is needed to activate the AMR later.

AGVs

DFleet Version: 2.2.0 (tracked)

DFleet Endpoint:

Step 3:

Login to the AGV as a supervisor and enter the AGV tab.

AGV

AGV Name*

UUID

Executor Mode* ☒ Standalone

☐ Dfleet

[Set executor mode](#)

AGV Home

[Specify AGV's home station](#)

Step 4:

Change the executor mode from Standalone to Dfleet.

Executor Mode*

☐ Standalone

☒ Dfleet

Set executor mode

Step 5:

After changing to Dfleet mode, you will be able to see the DFleet Endpoint link box. Paste the DFleet Endpoint link here.

DFleet Endpoint:

Step 6:

Ensure that the AMR is in the Activated mode. If it is not activated, you will need to activate it. (Steps for activating are shown below)

0		
0		 

Step 7:

Click on the “Edit” button (pencil icon) to open the AMR configuration page.

	1 day ago	 
---	-----------	---

Step 8:

On the AMR configuration page, change the AMR to “Active” state.

Edit AGV "Leon"



AGV Name: Leon

AGV Home*: A
Specify AGV's home station

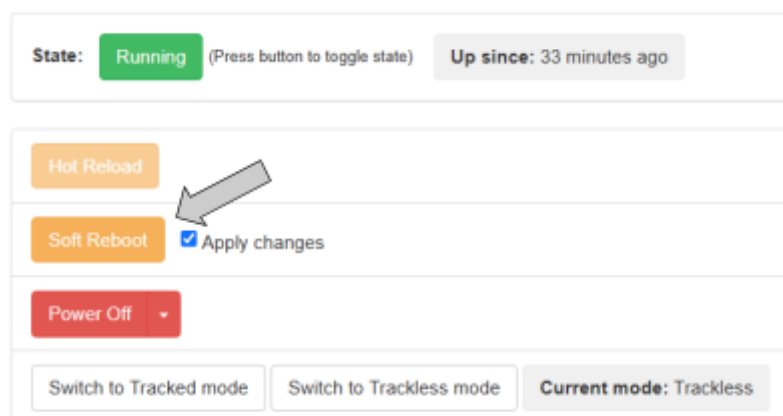
Priority*: 0
The AGV with a greater priority number will be chosen first.

☒ Active

Step 9:

Soft-reboot both the AMR and DFleet to apply the changes.

DFleet Controller



State: **Running** (Press button to toggle state) Up since: 33 minutes ago

Hot Reload

Soft Reboot ☒ Apply changes

Power Off

Switch to Tracked mode Switch to Trackless mode Current mode: Trackless

NavWiz



Start Controller

Soft Reboot

Power Off

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NavWiz 3.0.0-ibionk.20240812.121823-nightlyc18cc2e

Edit AGVs 3D Model

Each active AGV by default is using the Zalpha robot 3D model. Users can override the 3D models used by the active AGVs using this configuration.

 Dashboard Panel Settings ▾ Users ▾ Report ▾ Search Help

[< AGVs](#)

Edit AGVs 3D Model

TG30A 3D Model	-----	▼	
TG30B 3D Model	-----	▼	
TG40 3D Model	Tugger4	▼	
Zetha3 A 3D Model	zettha 3 AI	▼	

[Submit](#)

[Upload 3D](#) [Manage 3D](#)

AGV Configuration Page

Each AMR connected to DFleet will have its own AGV configuration page. The page provides configuration on how DFleet handles the AMR.

General Settings

 Panel Settings ▾ Users ▾ Report ▾ Search Help June 30, 2023, 11:10:27 AM admin ▾

[< AGVs](#)

Edit AGV "Lowbed R&d"

AGV Name	Lowbed R&d
AGV Home*	A1-PreReturn ▼ Specify AGV's home station
Priority*	0 The AGV with a greater priority number will be chosen first.
☐ Active	

AGV Home

The home option allows users to define a specific home station for each AMR. This allows for more general task definition later on. For example, users may define a task to command AMR to move to home. This task can be used on any AMR and each AMR executing the task will move to its own home station defined here.

Priority

This value is used by AGV assignment policy to determine which AMR will be selected when the “*Universal Agv Priority*” is enforced.

Active

This will enable or disable the AMR to pair with DFleet.

Dimension Profiles

The Dimension Profiles section allows users to configure the SVG (Scalable Vector Graphics) for both the AMR body and payload.

Users have the option to configure the SVG of the AMR body using the provided built-in templates and variations.

Dimension Profiles

Body

Payload 1

Payload 2

Payload 3

Payload 4

Payload 5

Template

Zalpha

Variation

Extended

Length

1.03

m

Width

0.55

m

Center of Rotation

0.51

m

Safety Margin

Top	0	m	Bottom	0	m
Left	0	m	Right	0	m

x: -0.7, y: -0.2

Furthermore, users can configure up to 5 payload profiles, enabling them to define different dimensions for various payload types. Later on, users can switch between Dimension Profiles using the “Select Dimension Profile” Task Template action in Task Template.

Dimension Profiles

Body

Payload 1

Payload 2

Payload 3

Payload 4

Payload 5

Template

Pallet

Length

1

m

Width

1

m

Center (Y-axis)

0

m

Center (X-axis)

0

m

Safety Margin

Top	0	m	Bottom	0	m
Left	0	m	Right	0	m

x: 1.3, y: 0.3



Info:

Configuring dimension profiles and safety margins for AMRs enables effective spatial grouping, enhancing traffic control management and collision avoidance. Proper configuration is highly recommended.

Executor Settings

Executor Settings

Allowed Motions

- ☒ Omni
- ☒ Forward
- ☒ Reverse
- ☒ Rotate Left
- ☒ Rotate Right
- ☒ U-turn Left, Rotate Left at Destination
- ☒ U-turn Right, Rotate Right at Destination
- ☒ Dynamic Path Planning

The types of manoeuvres allowed for this AGV.

Allowed Motions

Enable or disable certain motions to be executed by the AMR. This allows traffic controllers to plan suitable paths with restricted motion defined here.

Task Triggers

Task Triggers	Type	Configuration	Task Template
	AGV Idle	Timeout 3 Trigger task when AGV is idle for more than this duration (in seconds). ☐ Auto-abort when new task is available. Do not trigger when: ☐ AGV is charging. ☐ AGV is at home. AGV is at station: ☐ AGV's battery is below the minimum level to execute a new task. ☐ The trigger has run once and the trigger condition remains active.	

Battery Low

Threshold

0

Trigger task when AGV's battery below threshold (in percent).

☐ Auto-abort when new task is available.

Do not trigger when:

☐ AGV is charging.

☐ AGV is at home.

AGV is at station:

☐ AGV's battery is below the minimum level to execute a new task.

☐ The trigger has run once and the trigger condition remains active.

Task trigger is a DFleet feature to allow for automatic task assignment when certain conditions are met. There are two types of task triggers that are AGV Idle Task Trigger and Battery Low Task Trigger. Using the feature, users are able to define automatic tasks such as automatic return to home station for charging when battery low or when no new task was assigned to AMR.

AGV Idle Task Trigger	
Description	DFleet will automatically assign "idle task" to AMR that is idle for a certain amount of time.
Timeout	The duration in seconds before DFleet assigning idle tasks to AMR.
Auto-abort when new task is available.	If enabled, the assigned idle task will be aborted automatically when a new task is available.
AGV is charging	If enabled, the idle task will not be assigned if the AMR is currently idle for charging.
AGV is at home	If enabled, the idle task will not be assigned if the AMR is currently idle at home station.
AGV is at station	If enabled, the idle task will not be assigned if the AMR is idle at one of the selected stations.
AGV's battery is below the minimum level to execute a new task.	If enabled, the idle task will not be assigned if the AMR battery is below its minimum battery level configured inside the AGV configuration page.
The trigger has run once and the trigger condition remains active	If enabled, the idle task will not be assigned if the AMR has executed the idle task previously and no other task has been assigned to the AMR after that.
Task Template	The task template assigned to AMR when the idle task triggered. If the task template is not selected here, the idle task trigger will be disabled.

Battery Low Task Trigger	
Description	DFleet will automatically assign “battery low task” to AMR that are low in battery.
Threshold	The threshold battery value in percentage where DFleet will trigger battery low task if AMR battery is lower than this value.
Auto-abort when new task is available.	If enabled, the assigned battery low task will be aborted automatically when a new task is available.
AGV is charging	If enabled, the battery low task will not be assigned if the AMR is currently charging.
AGV is at home	If enabled, the battery low task will not be assigned if the AMR is currently at home station.
AGV is at station	If enabled, the battery low task will not be assigned if the AMR is at one of the selected stations.
AGV's battery is below the minimum level to execute a new task.	If enabled, the battery low task will not be assigned if the AMR battery is below its minimum battery level configured inside the AGV configuration page.
The trigger has run once and the trigger condition remains active	If enabled, the battery low task will not be assigned if the AMR has executed the battery low task previously and no other task has been assigned to the AMR after that.
Task Template	The task template assigned to AMR when the battery low task triggered. If the task template is not selected here, the battery low task trigger will be disabled.

Initialization Task Template*

Nothing selected

Task template for initialization of task runner.

Initialization at a Station*

☐ Not allowed
 ☒ Allowed at all stations
 ☐ Allowed at specific stations:

Nothing selected

Whether to allow initialization at a station.

Pre Initialization Task Template

Nothing selected

Task template for pre-initialization of task runner. Leaving it empty will disable the function.

Default Initialization Task Template

Task template for default initialization of task runner. Leaving it empty will disable the function.

Default Initialization Timeout*

5

Perform default initialization if there is no user input for more than this duration (in seconds).

☐ Default to starting task runner in paused state.

Default App

Default app to be run. Leaving it empty will disable the function.

Default App Timeout*

5

Start default app if there is no user input for more than this duration (in seconds).

Minimum Battery Level*

30

Minimum AGV battery level (%) for executing a new task.

Initialization Task Template

Enable users to select user defined task templates to be listed when initializing the AMR when running task runner. Must select at least one.

Initialization at a Station

Enable users to select user defined stations to be listed as starting position when initializing the AMR on task runner.

Pre-Initialization Task Template

Enable users to select user defined task template to be listed as pre-initialization task that can be executed before initializing the AMR when running task runner.

Default Initialization Task Template

Enable the user to select default initialization where the initialization task will be executed automatically when count down timeout. If the initialization task template is not selected, the function will be disabled.

Default Initialization Timeout

Countdown in seconds before default initialization task is executed when starting task runner.

Default to Starting Task Runner in Paused State

If enabled, task runner will be in pause state immediately after initialization.

Default App

When the default application is selected, the default application will start automatically after the controller starts when count down timeout. If no default application is selected the function will be disabled.

Default App Timeout

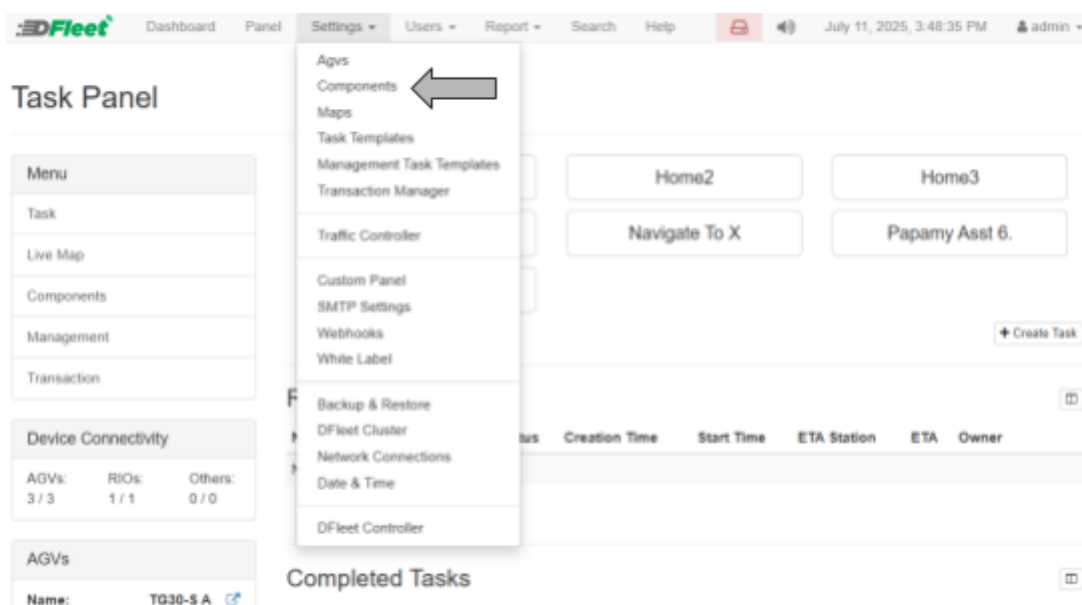
Countdown in seconds after the controller starts where the default application will be started when timeout.

Minimum Battery Level

Battery level in percentage where AMR will not be assigned with a new task if its battery falls below the level.

Section 2. Component

Component page lists out all components available for AMR. This is the place to configure each component. Currently available components consist of Audio, Lift, Registers, Remote I/O, Scheduler, Script, SecsGem, Sim AGV and Ups.



Navigate to Components

Components

#	Component	Operation
1	Audio	
2	Device Monitoring	
3	Lift	
4	Register	
5	Remote Io	
6	Scheduler	
7	Script	
8	Secsgem	
9	Sim Agv	
10	Ups	

Components page

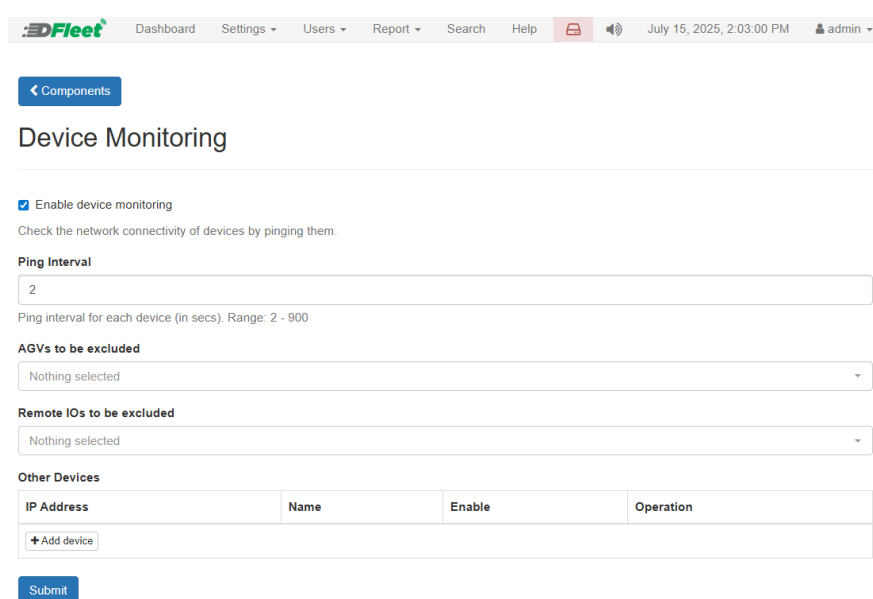
To start editing, click the pencil icon to access the respective component.

Audio

The audio component configuration page allows the user to configure the initial state of the audio channel and also configure the music playlist.

Device Monitoring

The Device Monitoring component configuration page allows users to enable or disable device monitoring. When enabled, the system will check the network connectivity of the devices by pinging them. Users can configure the ping interval and specify which AMRs and Remote IOs should be excluded from monitoring.



The screenshot shows the 'Device Monitoring' configuration page. At the top is a navigation bar with links: Dashboard, Settings, Users, Report, Search, Help, and a user profile 'admin'. Below the navigation bar is a 'Components' button. The main heading is 'Device Monitoring'. There is a checkbox 'Enable device monitoring' which is checked. Below it is a note: 'Check the network connectivity of devices by pinging them.' The 'Ping Interval' is set to '2' in a text input field, with a note below it: 'Ping interval for each device (in secs). Range: 2 - 900'. There are two dropdown menus: 'AGVs to be excluded' and 'Remote IOs to be excluded', both currently showing 'Nothing selected'. Below these is a table titled 'Other Devices' with columns: IP Address, Name, Enable, and Operation. There is an 'Add device' button with a plus icon. At the bottom is a 'Submit' button.

Lift

The lift component configuration page allows the user to configure the actions to be performed for all the lift operations such as:

- Reserve
- Unreserve
- Check Reservation
- Call Level
- Wait Level
- Clear Level
- Read Level

Register

The register page lists down all available registers. Here you can also configure remarks for specific registers.

Register

Registers		Remarks
#	Name	
1	FA0	Register Testing 01
2	FA1	Register Testing 02
3	FA2	Register Testing 03
4	FA3	
5	FA4	
6	FA5	

No.	Component	Description
1.	Name	List down all registered names.
2.	Remarks	Editable remarks for respective registers. Only show up under live view.



Caution:

Register FA0-FA99, FB0-FB99, and FC0-FC99 are temporary registers. Value stored would be erased after DFleet is powered off or soft reboot.

Registers FP0-FP99 are persistent registers. Value stored would remain even after AMR powered off.

Remote I/O

Configure remote I/O connection and its remarks. Support a few models of ADAM and WISE digital I/O module.

Remote IO

Communication Timeout 1

3

Timeout for remote IO connection (in secs). Range: 1.0 - 20.0

Devices

#	Name	Model	IP Address	Remarks
1	RIO1	WISE-4060	10.77.214.1	<pre><table style="width: 100%" border="1"> <tr> <td text align = "center"> I3 </td> <td text align = "center"> I2 </td> <td text align = "center"> I1 </td> <td text align = "center"> I0 </td> </tr></pre>
2	RIO2	WISE-4060	10.77.214.2	<pre><table style="width: 100%" border="1"> <tr></pre>

No.	Component	Description
1.	Communication Timeout	Set timeout for remote I/O. Unit is in seconds.
2.	Name	Designated name of remote I/O.
3.	Model	Pick from a list of I/O modules. Currently support: - ADAM-6050 - ADAM-6052 - ADAM-6250 - ADAM-6256 - WISE-4050 - WISE-4051 - WISE-4056 - TURCK-FEN20-16DXP - OMRON-NX-EIC202 - SIMULATOR
4.	IP Address	IP address of Remote I/O
5.	Remarks	Remarks need to be written in text or HTML format.

Scheduler

Schedulers

Schedulers (1-3 of 3)

#	Name	Type	Block Task	Active	Owner	Operation
1	Example 01	Disable Callbutton	✓	✓	admin	  
2	Example 02	Add Task	✓	✓	admin	  
3	Example 03	Add Management Task	✓	✓	admin	  

+ "Disable Callbutton" scheduler

+ "Add Task" scheduler

+ "Add Management Task" scheduler

 Calendar

Scheduler component page provides option for users to schedule disable call button, add task and add management task. The scheduler provides options to schedule for a single event or recurring event. The scheduler page lists all available schedulers with options to edit, clone or delete the scheduler.



Notice:

Users are advised to configure the scheduler after soft reboot. Unlike other configurations, scheduler configurations work with active data that are enabled after soft reboot. Changes in scheduler will be active immediately, no soft reboot required for scheduler configuration.



Caution:

Scheduler error usually appears after soft reboot due to it working on active data. Schedulers that are in error state will be disabled. Scheduler errors are shown exclusively on the scheduler page to avoid confusion.

Disable Call Button Scheduler

[< Schedulers](#)

Add new scheduler [Mode: Disable Callbutton]

Name*

Description

☒ Block task
 Block task assignment from overlapping with this event.

Start*

Duration*

Duration of each occurrence in "hh:mm:ss" format.

Occurrences

Disable call button scheduler allows users to schedule a time period where disable call button is enforced. When the disable call button is enforced, all users with call button permission will not be able to add a new task.

Parameter	Description
Name	Name of the scheduler. The name must be unique and will be used to identify the schedule.
Description	Description of the scheduler for future reference.
Block task	If enabled, Tasks that will overlap with this event will not be assigned to AMR.
Start	The start time of the first event.
Duration	Duration of the event.
Occurrences	Occurrences of the event or empty for a one-time event.
	Yearly Schedule event that reoccurs yearly on the same date defined in start time. Enable "has expired date" to stop the event after this date.
	Monthly Schedule event that recurs monthly on the same date defined in start time. Enable "has expired date" to stop the event after this date.
	Weekly Schedule event that reoccurs weekly on the same day and time defined in start time. Enable "has expired date" to stop the event after this date.
	Daily Schedule event that reoccurs daily on the same time defined in start time. Users can select which day of the week this event will trigger. (e.g., user can define schedule that only happened on weekend) Enable "has expired date" to stop the event after this date.
	Hourly Schedule event that reoccurs hourly on the same minute defined in start time.

Add Task Scheduler

Add task scheduler allows users to schedule events where tasks can be automatically added to the task queue. The configuration parameters are similar to the disable call button scheduler listed in table above with an additional list of tasks described in table below.

**Info:**

Compared to disable call button scheduler, add task scheduler duration parameter is optional. When set to 0, the duration is automatically selected from the longest task duration in the tasks list. Setting the value will override this automatic behaviour.

Parameter	Description
Task Template	Task template that will be added when the event is triggered.
Params	Parameters required for the selected task template.
Agv	Optional AMR that will be assigned for the task. Leaving it empty will allow for AMR selection based on priority.
Operation	Operations button. Users may remove, clone or rearrange the task listed.
Add task	Add a new entry inside the task list.

Add Management Task Scheduler

Add new scheduler [Mode: Add Management Task]

Name*

Description

☒ Block task
 Block task assignation from overlapping with this event.

Start*

Duration*

Duration of each occurrence in "hh:mm:ss" format.

Occurrences

Management Tasks

Management Task Template	Params	Operation
+ Add management task		

Submit

Add management task scheduler allows users to schedule events where management tasks can be automatically added to the task queue. The configuration parameters are similar to the disable call button scheduler listed in the previous table with an additional list of tasks described in the table below.

Parameter	Description
Management Task Template	Management Task template that will be added when the event is triggered.
Params	Parameters required for the selected management task template.
Operation	Operations button. Users may remove, clone or rearrange the management task listed.
Add management task	Add a new entry inside the management task list.

Calendar

Schedulers Calendar

Year Month Day						
< July 2023 >						
Sunday	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday
25	26	27	28	29	30	1
2	3 Tasks 2	4 Tasks 1	5 Tasks 1	6 Tasks 1	7 Tasks 1	8
9	10 Tasks 2	11 Tasks 1	12 Tasks 1	13 Tasks 1	14 Tasks 1	15
16	17 Tasks 2	18 Tasks 1	19 Tasks 1	20 Tasks 1	21 Tasks 1	22
23	24 Tasks 3	25 Tasks 1	26 Tasks 1	27 Tasks 1	28 Tasks 1	29
30	31 Tasks 2	1	2	3	4	5

Calendar views in the scheduler component provide ways for users to monitor and interact with schedulers' events. Users have options to reschedule or cancel an event. This is useful for situations where only a single event needs to be canceled or rescheduled.

Script

Script

Scripts

#	Name	Script File	Option	Operation
1	HelloWorldScript	Currently: helloWorld.py Change: Choose File No file chosen	☐ Autostart ☐ Autorestart	✕
+ Add script				

Scheduled Scripts

#	Name	Script File	Option	Operation
+ Add script				

On the Script configuration page, users have the ability to run custom Python scripts. Additionally, users can schedule scripts to run at specific times using the scheduled scripts functionality. This feature provides flexibility and customization options for performing specific tasks or automating processes within the DFleet system.

SecsGem

SecsGem

☒ Enable Equipment Server

Submit

This page is designated for users to enable or disable SecsGem server communications.

Sim AGV


Dashboard Settings Users Report Search Help

Components

Sim AGV

Number of Sim AGVs*

5

If this number is reduced, the extraneous Sim AGVs will be removed.

Speedup Factor*

1

The ratio of simulated running speed to actual speed.

☐ Make Sim AGVs automatically emulate the active skillset assigned to each of them. Otherwise, this can also be performed manually from the Sim AGVs live view page.

Submit

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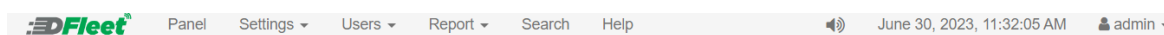
Sim AGV components provide virtual AGV to perform simulation of DFleet operation. Set the number of simulator AGV inside the Sim AGV component. The speedup factor can be used to adjust the ratio between the simulated running speed and the actual speed. By checking the box at the bottom, the Sim AGVs will automatically emulate the active skillset assigned to each of them.

Then, the AGV profile will be shown under AGVs configuration. Below is the example of a simulator AGV inside AGVs configuration.

☐	#	Name	Home	Priority	Skillset	Active	Last Seen	Operation
☐	1	SIM 01 (000000000-0000-0000-0001-0000000000000)	🔗	0			12 minutes ago	✎ ⚙️ ✖️

A simulator AGV can be controlled from the Sim Agv live view. Navigate to Task Panel > Live component > Sim Agv to interact with the Sim AGV. There are three available apps for the simulator AGV:

- Task runner
- Action test
- Homing



Sim AGV (Live View)

Menu	#	Name	Status
Dashboard	1	SIM 01	Status: Not Ready Available Apps (click to run): Task Runner Action Test Homing

Task Runner

The task runner is used to execute tasks for the simulator AGV. All the available task templates will be shown and the user could click on the task template to create a new task for the simulator AGV. The simulator AGV will then perform the task as configured in the task template.

To start executing the task, users are required to select the position of the simulator AGV. There are four options for users to choose from. Below shows the four available options to select the simulator AGV position.

App: Task Runner

Choose an option below to initialize AGV location.

AGV is parked at its home.

AGV is parked at previous Task Runner stopped.

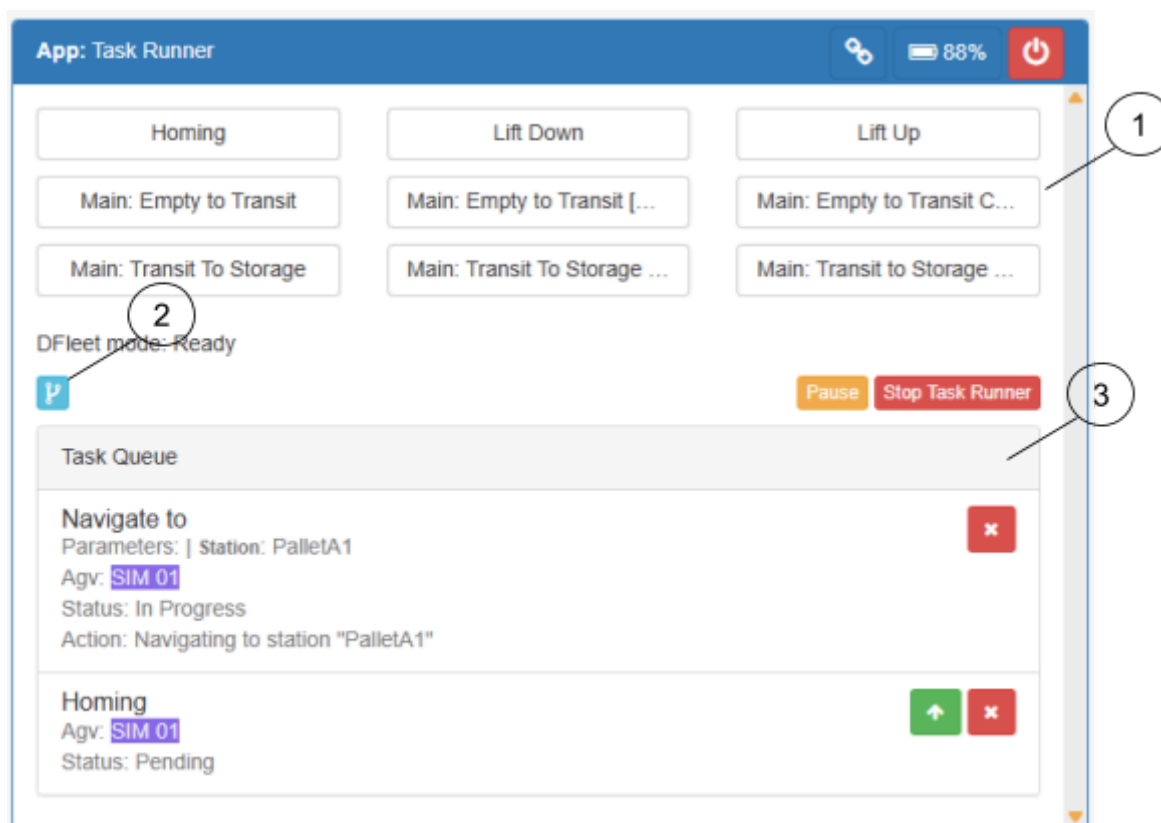
AGV is parked at a station. ▼

Start task runner in paused state ☐

! Cancel

Options	Description
AGV is parked at its home	Option for users to choose when the AMR is already parked on its home station.
Forward dock to charger (tracked)	Option for users to choose when a forward docking is required for AMR to move forward (on a track) towards its home station.
Reverse dock to charger (tracked)	Option for user to choose when a reverse docking is required for AMR to move reverse (on a track) towards its home station.
AGV is parked at previous Task Runner stopped.	Option for users to choose when an AMR doesn't move after stopping Task Runner.
Agv is parked at a station	Option for the user to choose when the AMR is already parked on the desired station.

The simulator AGV will be able to execute tasks after the location has been selected. After initialization the task runner page will show the operational panel as shown below.



No.	Component	Description
1.	Task List	List of tasks allowed for this simulator AGV.
2.	Action Live View	Live view of current task being executed.
3.	Task Queue	List of active tasks in queue, user can cancel, abort or prioritize tasks in this view.

Action Test

Under the action test option, users could execute selected actions to the simulator AGV.

App: Action Test
88%

Select action:
Navigate To
Start
Stop
Stop App

Station
A
Choose a destination station.

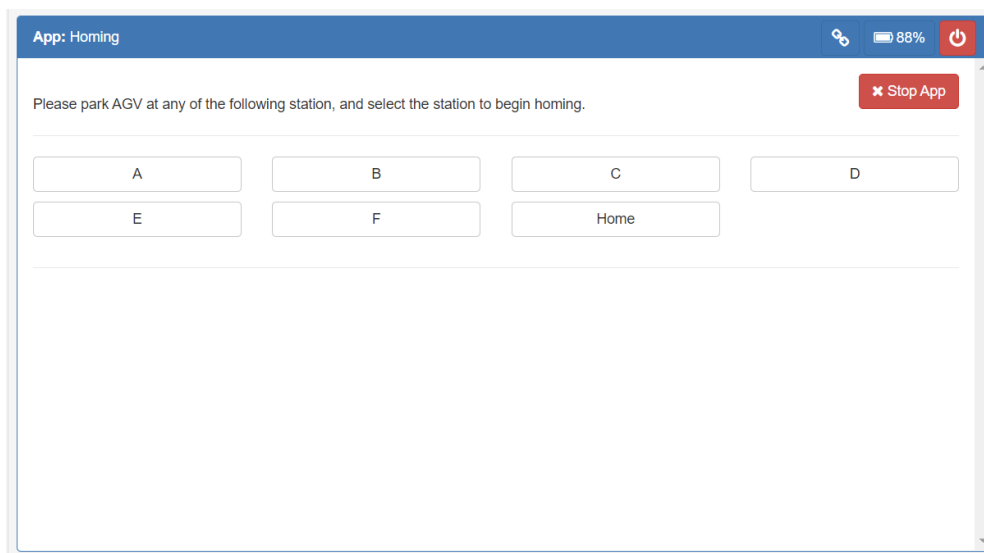
Align Station Type
-1
Min: -1 ; Max: 1
Use front(0) or rear(1) line sensor, or trackless localization(-1) for station alignment.

Next Motion
0
Min: 0 ; Max: 4
Next motion after final junction (0=Idle, 1=Non-Stop, 2=Rotate Left, 3=Rotate Right, 4=Non-Stop Bezier)

Next Speed
0
Min: 0.0 ; Max: 3.0

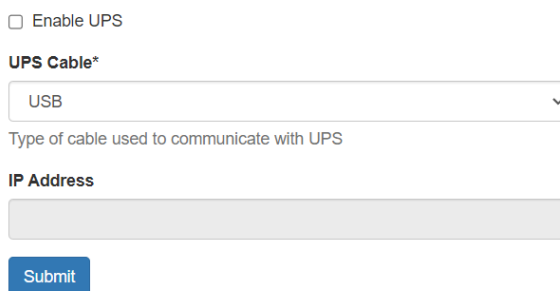
Homing

Under homing application the simulator AGV will go back to its home from the selected station.



UPS

UPS

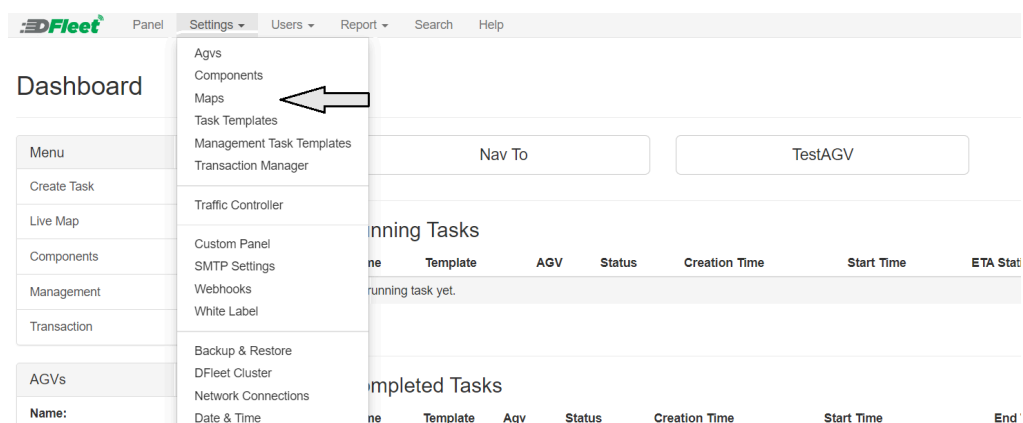


UPS components provide connection to Uninterrupted Power Supply, UPS. It allows DFleet to display connected UPS status and provide management skills to signal power off to the UPS. Table below lists available options in the UPS configuration page.

Parameters	Description
Enable UPS	To enable the UPS component.
UPS Cable	Select how the UPS device connected to DFleet. Options available are USB connection or NET for connection through the local network.
IP Address	IP Address of the UPS if UPS Cable is configured to use NET mode.

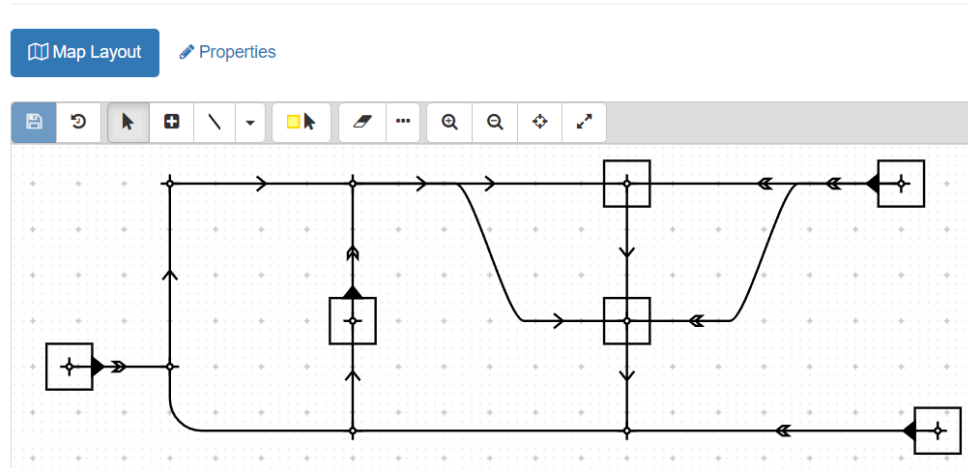
Section 3. Maps

The key function of the Map under DFleet System is to let the system understand all the track/path configuration for the AMR on site.



The following is an example of AMR map:

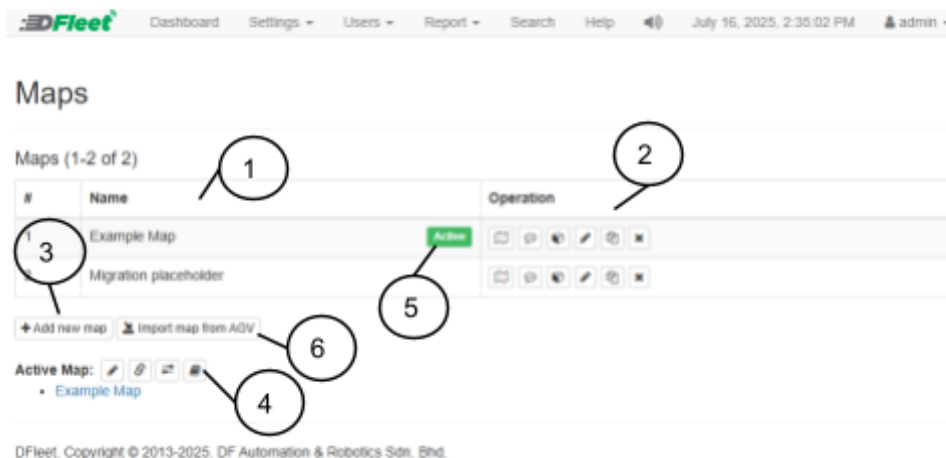
Map "Example 1"



Under the map, users will need to configure the station and path according to the track setup of the AMR on site. Main purpose of the mapping system is to let the system understand the map configuration and enable AMR to move to the destination station on the map.

Track AMR Map

List down all available maps already created inside the map layout page. Also provides a few side operations.

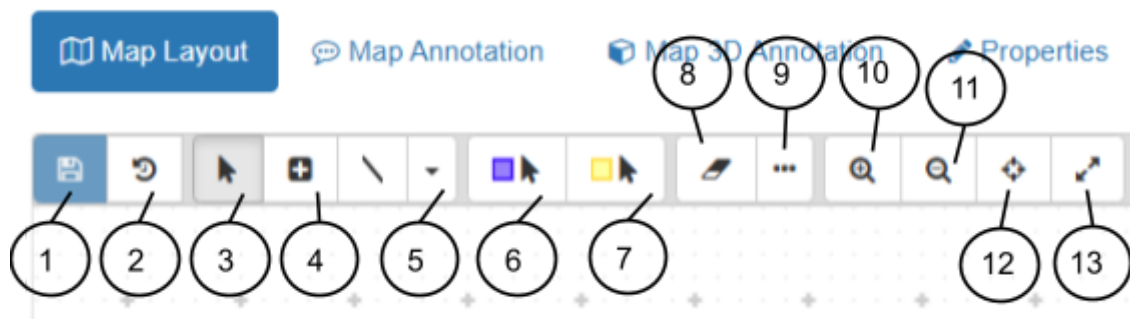


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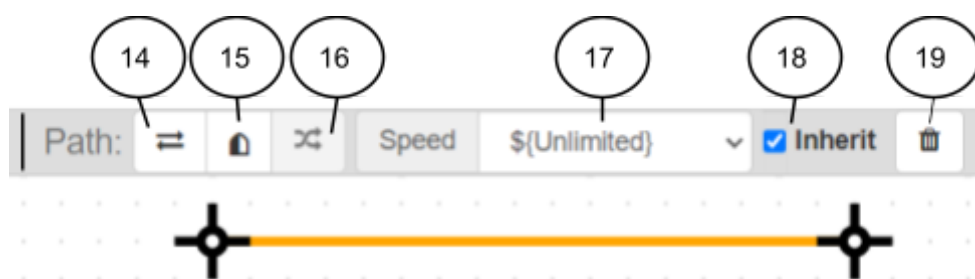
No.	Component	Description	
1.	Name	Map name which can be configured in the map editor page	
2.	Operation		Open map layout.
			Open map annotation.
			Open 3D map annotation
			Open map properties.
			Clone map.
			Delete map.
3.		Open a new map editor page.	
4.	Active Map		Open active map editor page.
			Open map teleportation editor page.
			Open map transition trigger editor page.
			Open map parameters editor page
5.		Indicate current active map	
6.		Open import map from AGV page.	

Map Config Panel

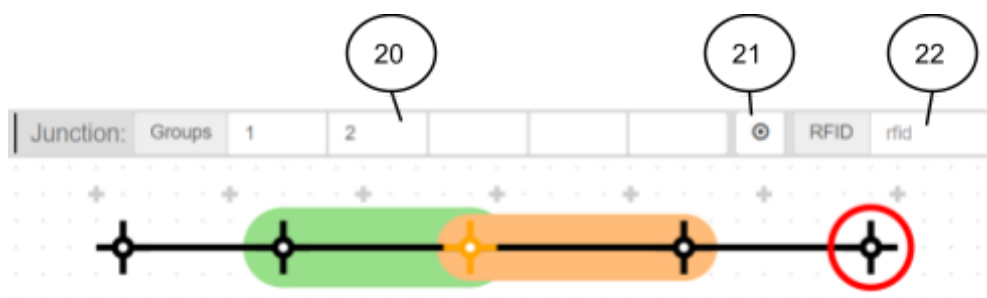
This is where the map is created or edited. Layouts are equipped with a few different functions on its toolbar which will be explained in the table below.



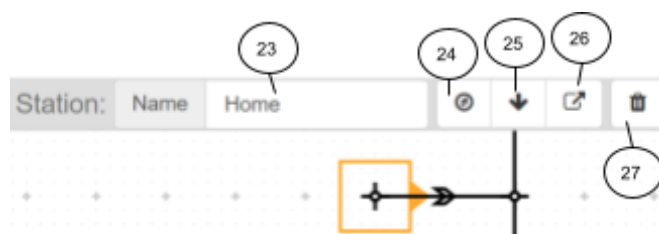
Map Layout Toolbar



Path

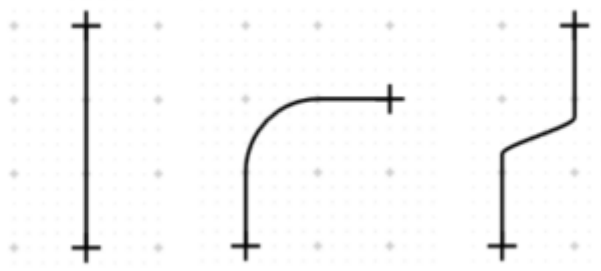


Junction

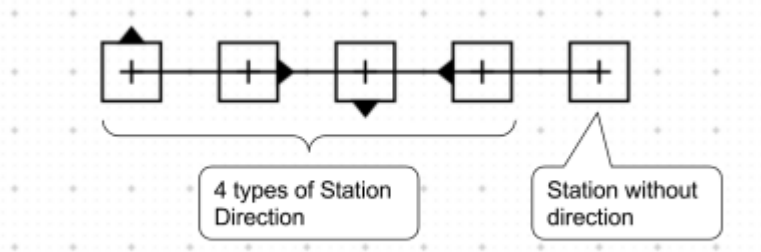
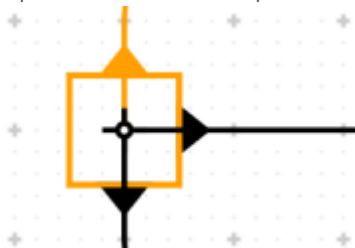


Station

No.	Component	Description
1.		Function for users to save the edited map. A Soft Reboot is required after the changes on the map for the changes to be effective.

2.		This function allows the user to restore back to the previous changeset.
3.		This is a function to change the current tool to Select, that allows the user to drag the map for different area views, and to select the Path, Junction or Station for configuration.
4.		Function to change the current tool to Station placement that allows the user to place a station on the map. Take note that stations must be placed on either end of a path. Multiple station placement on one junction is also supported. After selecting a Station using the Select tools, addition Station menu will appear
5.		Function to change the current tool to Path placement that allows the user to place a path on the map. Paths are the basic element in creating the map. There are 3 types of path available for the map, Straight Path, Right-angled Path, and S-curved Path.
		 Straight Path Right-angle Path S-curved Path
		After selecting a Path using the Select tools, an additional Path menu will appear.
		 Straight path tool.
		 Right angle path tool.
		S-curve path tool.
6.		Change the current tool to "Location Hint Zone", enabling the user to define the location hint for the areas.
7.		Change the current tool to "No-rotate Zone Placement", enabling the user to define specific areas where the AMR should avoid rotating to prevent collisions.
8.		Quickly delete everything on the map. Do take note that all unsaved changes will be lost if this function is used.
9.		 Undo last change on map.
		 Redo last change on map.
10.		Zoom in on the map editing window.
11.		Zoom out on the map editing window.
12.		Function to fit all content into the window size. System will adjust the view to show all the content on the map editing window.
13.		This function is to enable the map editor to show in full screen mode.

14.		Change Flow allows the user to set a flow for the AMR movement. Once the flow has been set in the Map, DFleet will only allow the movement of AMR in the configured flow direction. This is useful to configure one way directions for the path.										
15.		Change forward/reverse motion of the path, forcing the AMR to forward or reverse when passing through this path, depending on the type of forward/reverse configuration below. <table><tr><th>Forward/Reverse motion</th><th>Description</th></tr><tr><td> Free </td><td>The navigation direction (forward or reverse) solely depends on whether it is forward navigation or reverse navigation.</td></tr><tr><td> Force AMR heading </td><td>Enforces the AMR to follow the specified heading, determining the forward or reverse navigation based on the incoming direction of the AMR.</td></tr><tr><td> Force forward </td><td>Enforces the AMR to navigate forward when passing through this path.</td></tr><tr><td> Force reverse </td><td>Enforces the AMR to navigate in reverse when passing through this path.</td></tr></table> For paths connected to the home station, they will be automatically enforced to the correct motion during navigation.	Forward/Reverse motion	Description	Free	The navigation direction (forward or reverse) solely depends on whether it is forward navigation or reverse navigation.	Force AMR heading	Enforces the AMR to follow the specified heading, determining the forward or reverse navigation based on the incoming direction of the AMR.	Force forward	Enforces the AMR to navigate forward when passing through this path.	Force reverse	Enforces the AMR to navigate in reverse when passing through this path.
Forward/Reverse motion	Description											
Free	The navigation direction (forward or reverse) solely depends on whether it is forward navigation or reverse navigation.											
Force AMR heading	Enforces the AMR to follow the specified heading, determining the forward or reverse navigation based on the incoming direction of the AMR.											
Force forward	Enforces the AMR to navigate forward when passing through this path.											
Force reverse	Enforces the AMR to navigate in reverse when passing through this path.											
16.		Change Bending is a function to change the bending direction of the Right-angled Path and S-curved Path. This function is not available for Straight Path. Before After Before After										
17.	Speed Limit	Allow users to set a speed limit for the selected path. Users can choose from predefined Medium Fast, Medium and Slow or specify the speed in meter/second units.										
18.	Inherit	Tick to inherit values from map parameters.										
19.		Delete currently selected Path. Take note that the deletion of a path is not allowed if there is a station placed on either end of the path. To delete the path, the station must be first deleted.										
20.	Group	Change junction group. Each junction can have up to 5 different junction groups.										
21.		Toggle swapping junction or via junction.										

22.	RFID	Set junction RFID.
23.	Name	Name configuration for the station. It will be used as the reference to the particular Station under Task Template and AMR Home Station configuration.
24.		Station heading, it will change AMR heading according to the station heading despite which direction the AMR navigates from. Take note in the map view that all stations will/will not have a small arrow showing the station direction. Clicking this Change Direction button will toggle the direction of the Station.  For a Station without direction, the navigation of AMR to this station will end at the direction depending on where it is navigating from.
25.		A "switch" to select different station placed under the same point as currently selected station 
26.		Shortcut button to Search Page that searches all objects that used this station as reference.
27.		Function to delete the currently selected Station



Info:

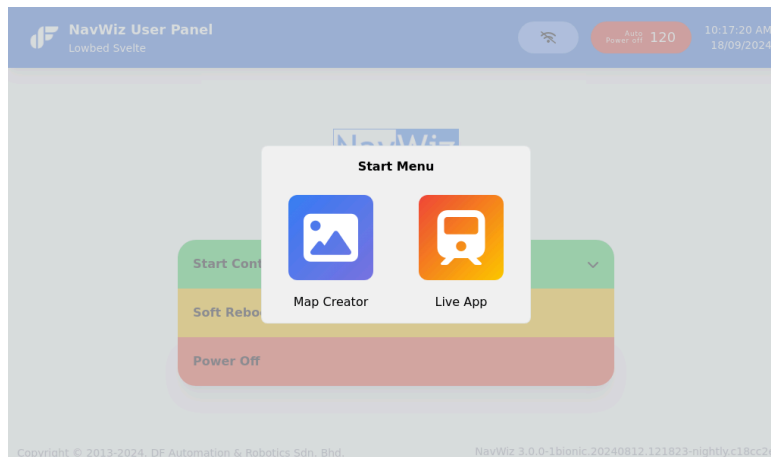
To select multiple objects, click and drag the mouse while holding down the "Ctrl" key.

To clone selected objects (ensure that paths are selected), hover the mouse over the junction, then click and drag the mouse while holding down the "Shift" key.

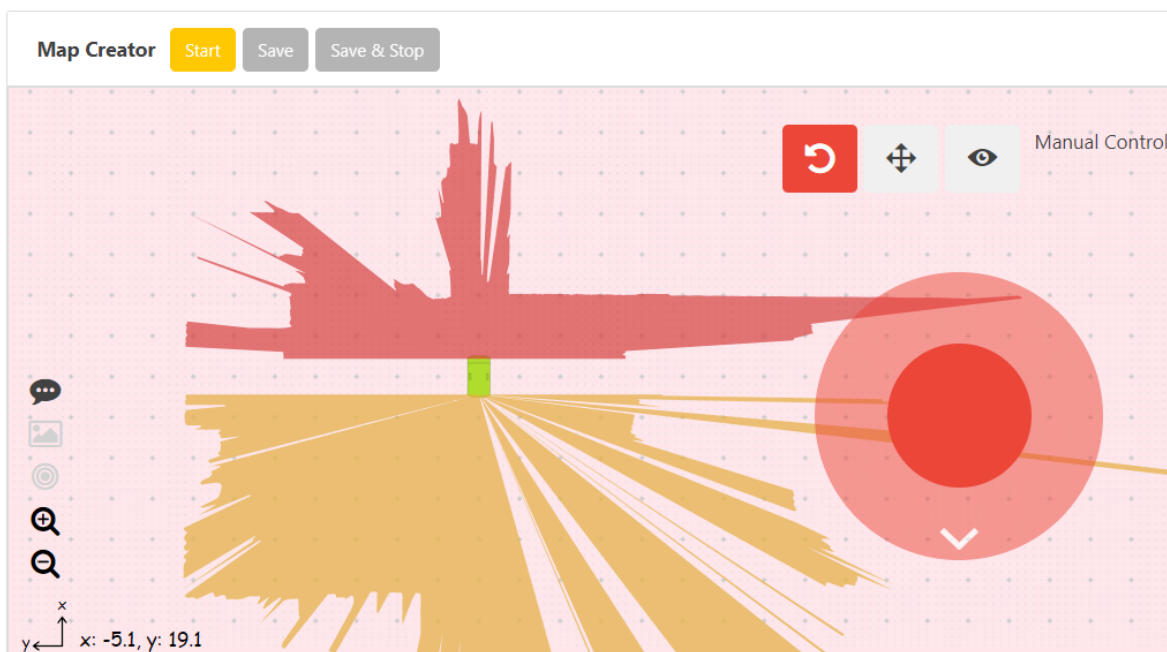
Trackless AMR Map

Creating a map for a trackless AMR is a more involved process. First, a raw map must be generated using the AMR's laser sensor. This step can be performed with the 'Map Creator' tool. To activate Map Creator, click the arrow next to 'Start Controller' and select 'Map Creator'.

In short, users must use the AGV NavWiz system to plot a trackless AMR map, as Map Creator is not available within the DFleet system.

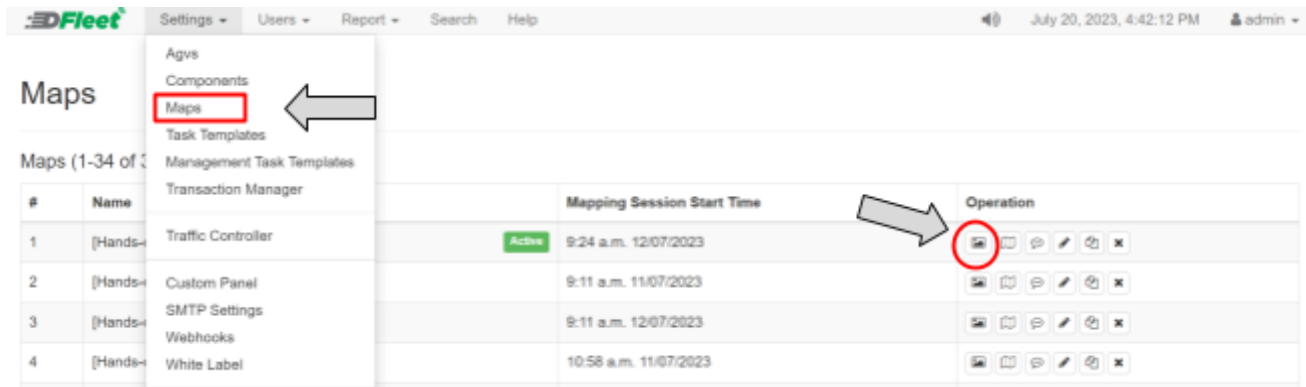


Press “Start” and control the AMR with the controller and navigate to let AMR plot the map. You will be able to see the outline of the layout after the map is successfully plotted. After that, press “Save & Stop” to stop Map Creator.



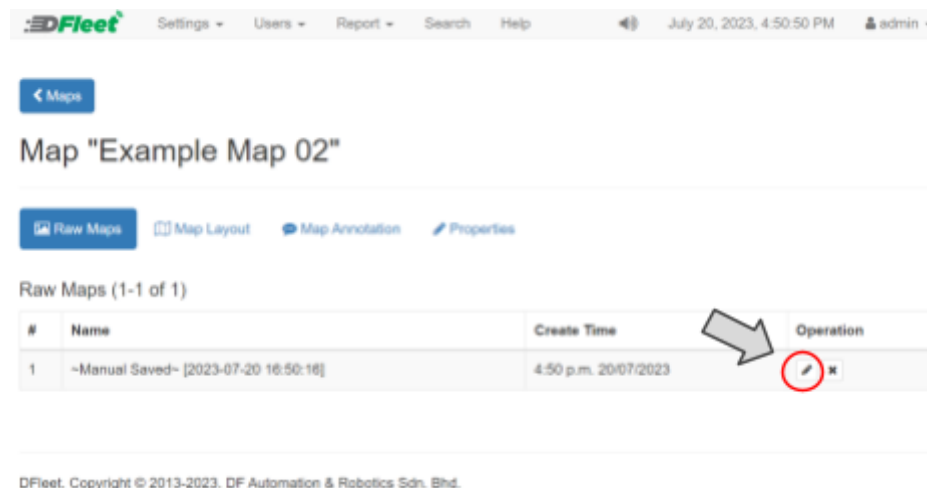
Raw Map Editor

The raw map editor is available in DFleet. To access this page, navigate to Setting > Maps. Under operation column, click on View Raw Maps Icon and then click on the edit button under operation column on the raw map to edit.



#	Name	Mapping Session Start Time	Operation
1	[Hands-on]	9:24 a.m. 12/07/2023	[View Raw Maps] [Map Layout] [Map Annotation] [Properties]
2	[Hands-on]	9:11 a.m. 11/07/2023	[View Raw Maps] [Map Layout] [Map Annotation] [Properties]
3	[Hands-on]	9:11 a.m. 12/07/2023	[View Raw Maps] [Map Layout] [Map Annotation] [Properties]
4	[Hands-on]	10:58 a.m. 11/07/2023	[View Raw Maps] [Map Layout] [Map Annotation] [Properties]

Navigate to maps > Click on View Raw Map



Map "Example Map 02"

Raw Maps (1-1 of 1)

#	Name	Create Time	Operation
1	~Manual Saved~ [2023-07-20 16:50:16]	4:50 p.m. 20/07/2023	[Edit] [Delete]

Click on Edit

The raw map editor page will be shown and editing on the raw map can be performed.

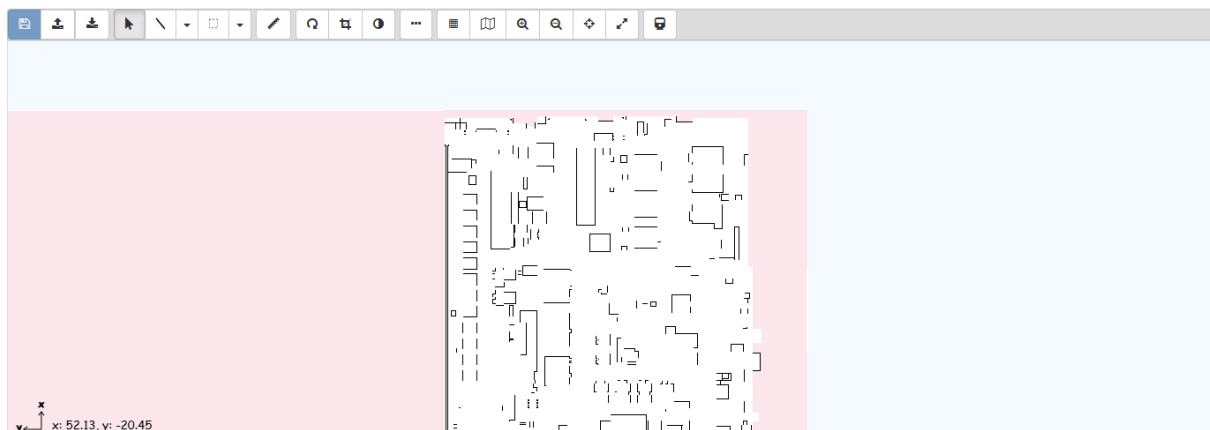
Edit raw map "-Manual Saved- [2026-02-24 15:48:20]"

New name

(unnamed)

☐ Clear name

☐ Replace the original raw map instead of saving a new copy.



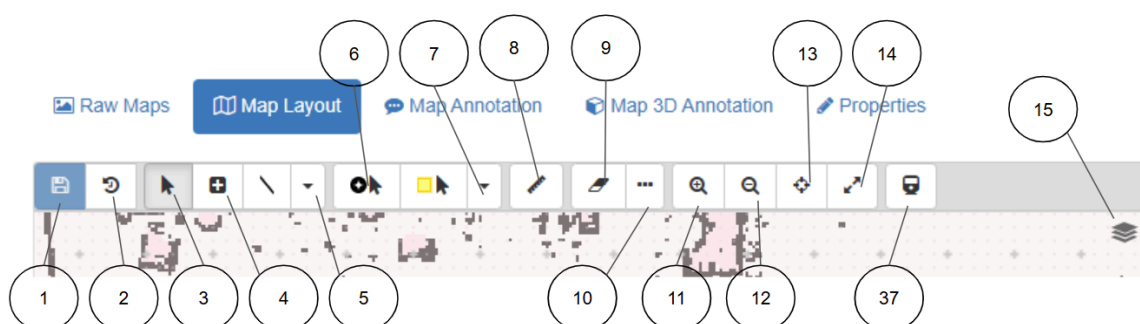
No.	Component		Features
1.	New name		Edited map name input.
2.	Raw map name		Raw map current name
3.	Clear name		Tick to clear map name when saved as new map.
4.	Replace the original raw map...		Tick to replace the original raw map instead of saving a new one.
5.			Function for users to save the edited map. If the raw map is saved as a new map, the users need to apply the new raw map under the map layout editor to use it.
6.			Function to upload raw map.
7.			Function to download raw map.
8.			Function to change current tool to Select, that allows users to drag the map for different area views.
9.			Draw black line (Alt + 1).
			Draw black hollow rectangle (Alt + 2).
10.			Draw white rectangle (Alt + 3).
			Draw white rectangle behind black pixels (Alt + 4).
			Erase line or rectangle drawn using editor (Alt + 5).

11.		Ruler tool to measure distances on canvas (Alt + 0).																							
12.		Rotate the whole raw map - The rotation may reduce the quality of the map. consider remapping with proper alignment for better raw map. - There is an automatic darken feature running after the rotation to solve black wall becoming gray issue.																							
13.		Open resize canvas tools drawer. <table> <tr> <td rowspan="3">Resize canvas</td><td></td><td>Canvas x dimension in meters.</td></tr> <tr> <td></td><td>Canvas y dimension in meters.</td></tr> <tr> <td>Resolution</td><td>Canvas resolution in meter/pixel.</td></tr> <tr> <td rowspan="2">Canvas offset</td><td></td><td>X offset in meters.</td></tr> <tr> <td></td><td>Y offset in meters.</td></tr> <tr> <td rowspan="2">Raw map offset</td><td></td><td>X offset within canvas in meters.</td></tr> <tr> <td></td><td>Y offset within canvas in meters.</td></tr> <tr> <td colspan="2"></td><td>Apply configuration to current canvas and close canvas resize tools drawer.</td></tr> <tr> <td colspan="2"></td><td>Close canvas resizes tools drawer without saving the new configuration.</td></tr> </table>	Resize canvas		Canvas x dimension in meters.		Canvas y dimension in meters.	Resolution	Canvas resolution in meter/pixel.	Canvas offset		X offset in meters.		Y offset in meters.	Raw map offset		X offset within canvas in meters.		Y offset within canvas in meters.			Apply configuration to current canvas and close canvas resize tools drawer.			Close canvas resizes tools drawer without saving the new configuration.
Resize canvas		Canvas x dimension in meters.																							
		Canvas y dimension in meters.																							
	Resolution	Canvas resolution in meter/pixel.																							
Canvas offset		X offset in meters.																							
		Y offset in meters.																							
Raw map offset		X offset within canvas in meters.																							
		Y offset within canvas in meters.																							
		Apply configuration to current canvas and close canvas resize tools drawer.																							
		Close canvas resizes tools drawer without saving the new configuration.																							
14.		Undo last change on map.																							
15.		Redo last change on map.																							
16.		Function to show or hide grid (Alt + G).																							
17.		Convert pixels with color that fall below the configured threshold to become true black. AMCL only considers true black pixels as a reference obstacle.																							
18.		Toggle map layout.																							
18.		Zoom in on the editor canvas.																							
19.		Zoom out of the editor canvas.																							

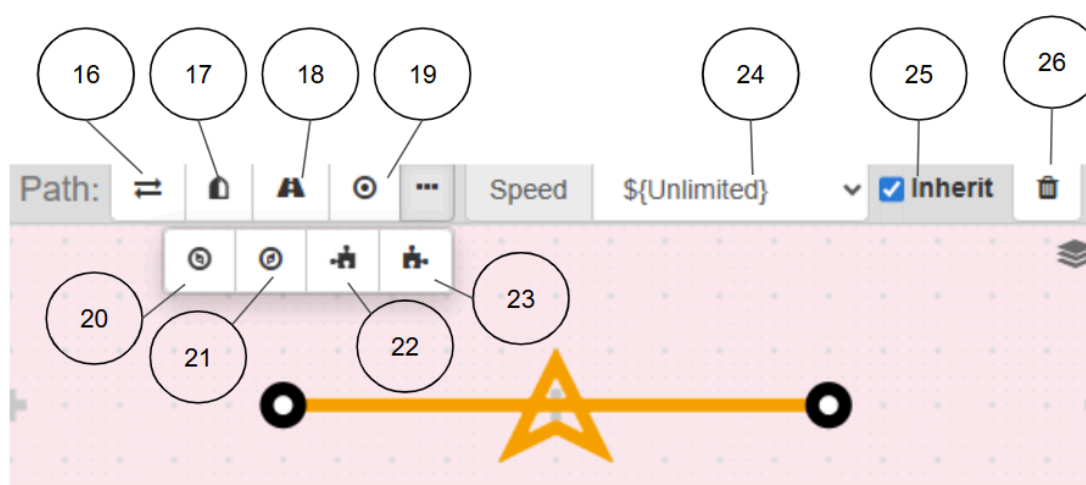
20.		Function to fit all content into the window size. The system will adjust the view to show all the content on the map editing window.
21.		This function is to enable the map editor to show in full screen mode.
22.		Select the AGV to connect to the Live App.

Map Layout Editor

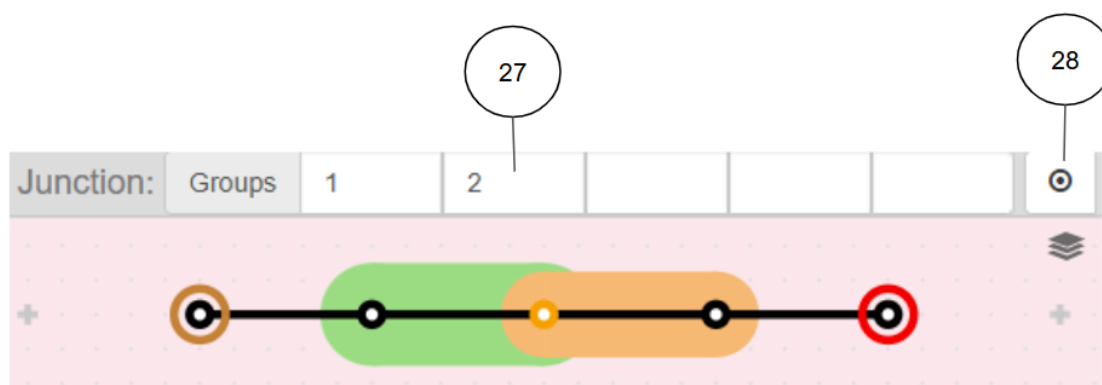
This is where the map is created or edited. Layouts are equipped with a few different functions on its toolbar which will be explained in the table below.



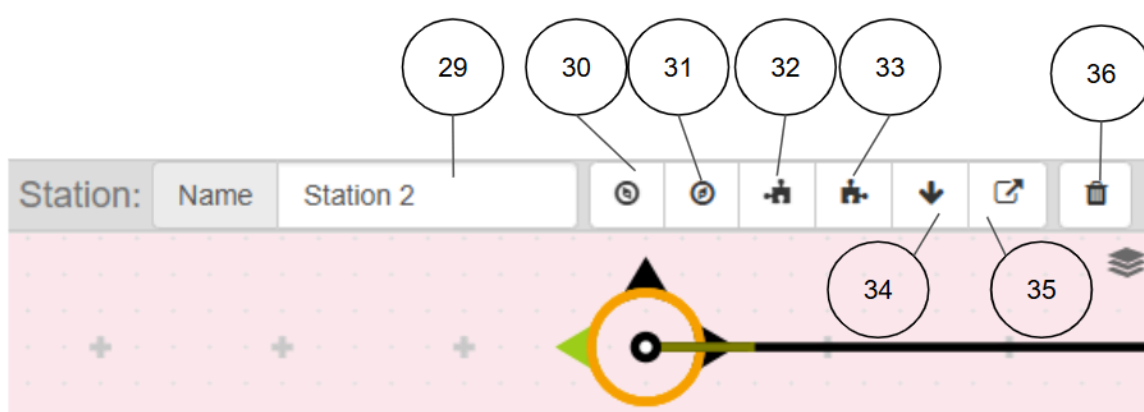
Map Layout Toolbar



Path

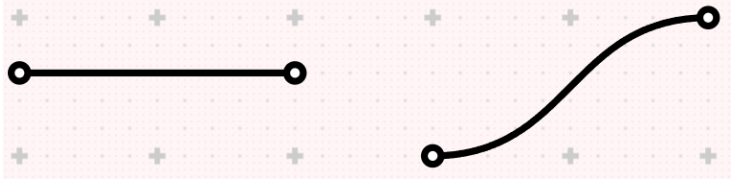


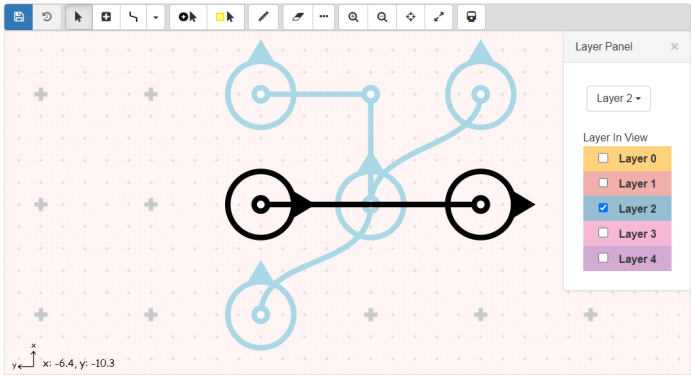
Junction



Station

No.	Component	Description
1.		Function for users to save the edited map. A Soft Reboot is required after the changes on the map for the changes to be effective.
2.		This function allows the user to restore back to the previous change set.
3.		This is a function to change the current tool to Select, that allows the user to drag the map for different area views, and to select the Path, Junction or Station for configuration.
4.		Function to change the current tool to Station placement that allows the user to place a station on the map. Take note that stations must be placed on either end of a path. Multiple station placement on one junction is also supported. After selecting a Station using the Select tools, addition Station menu will appear

5.		Change the current tool to "Path Placement" enabling the user to place a path on the map. Paths are the basic elements in creating the map and can be of two types: Straight Path, Bezier Path.  Straight Path Bezier Path After selecting a Path using the "Select" tool, an additional Path menu will appear.	
			Straight path tool.
			Bezier path tool.
6.		Change the current tool to "Landmark Placement", enabling the user to place landmarks (reflectors) for better localization performance and accuracy.	
7.			Change the current tool to "Location Hint Zone", enabling the user to define the location hint for the areas..
			Change the current tool to "No-rotate Zone Placement", enabling the user to define specific areas where the AMR should avoid rotating to prevent collisions.
			Change the current tool to "Forbidden Zone Placement", enabling the user to define specific zones where the AMR is not allowed to enter when using Dynamic Path Navigation. Take note that this feature is only available if the version of DFleet supports Dynamic Path Planning.
8.		Measure distance on the map.	
9.		Quickly delete everything on the map. Do take note that all unsaved changes will be lost if this function is used.	
10.			Undo last change on map.
			Redo last change on map.
11.		Zoom in on the map editing window.	
12.		Zoom out on the map editing window.	
13.		Function to fit all content into the window size. System will adjust the view to show all the content on the map editing window.	
14.		This function is to enable the map editor to show in full screen mode.	

15.		Expand the layer panel menu, enabling the user to configure the paths, stations and junctions into different layers. The paths, stations and junctions in active layers will be displayed in black colour, while the ones in other layers will be shown in light blue. Additionally, the items that are not currently in view on the map cannot be selected. Take note that the layers do not impact navigation behaviour, they are purely for visualization purposes. 										
16.		Change Flow allows the user to set a flow for the AMR movement. Once the flow has been set in the Map, DFleet will only allow the movement of AMR in the configured flow direction. This is useful to configure one way directions for the path.										
17.		Change forward/reverse motion of the path, forcing the AMR to forward or reverse when passing through this path, depending on the type of forward/reverse configuration below. <table><tr><th>Forward/Reverse motion</th><th>Description</th></tr><tr><td> Free </td><td>The navigation direction (forward or reverse) solely depends on whether it is forward navigation or reverse navigation.</td></tr><tr><td> Force AMR heading </td><td>Enforces the AMR to follow the specified heading, determining the forward or reverse navigation based on the incoming direction of the AMR.</td></tr><tr><td> Force forward </td><td>Enforces the AMR to navigate forward when passing through this path.</td></tr><tr><td> Force reverse </td><td>Enforces the AMR to navigate in reverse when passing through this path.</td></tr></table> For paths connected to the home station, they will be automatically enforced to the correct motion during navigation.	Forward/Reverse motion	Description	Free 	The navigation direction (forward or reverse) solely depends on whether it is forward navigation or reverse navigation.	Force AMR heading  	Enforces the AMR to follow the specified heading, determining the forward or reverse navigation based on the incoming direction of the AMR.	Force forward 	Enforces the AMR to navigate forward when passing through this path.	Force reverse 	Enforces the AMR to navigate in reverse when passing through this path.
Forward/Reverse motion	Description											
Free 	The navigation direction (forward or reverse) solely depends on whether it is forward navigation or reverse navigation.											
Force AMR heading  	Enforces the AMR to follow the specified heading, determining the forward or reverse navigation based on the incoming direction of the AMR.											
Force forward 	Enforces the AMR to navigate forward when passing through this path.											
Force reverse 	Enforces the AMR to navigate in reverse when passing through this path.											
18.		Switch Trackless / Tracked allow the user to switch from Trackless to Tracked navigation or vice versa. Note that if Tracked is selected, there must be a magnetic track placed on the layout.										

19.		Toggle Omni Path heading. The robot's direction follows the heading along the path if the robot supports Omni Drive.
20.		Rotate the direction of the Omni Path Heading in Counter-clockwise direction . Each click on this function will initiate a 10 degree counterclockwise turn.
21.		Rotate the direction of the Omni Path Heading. Each click on this function will initiate a 10 degree clockwise turn.
22.		Rotate the direction of the Omni Path Heading in Counter-clockwise direction. Each click on this function will initiate a 90 degree Counterclockwise turn.
23.		Rotate the direction of the Omni Path Heading in Clockwise direction. Each click on this function will initiate a 90 degree Clockwise turn.
24.	Speed Limit	Allow users to set a speed limit for the selected path. Users can choose from predefined Medium Fast, Medium and Slow or specify the speed in meter/second units.
25.	Inherit	Tick to inherit values from map parameters.
26.		Delete currently selected Path. Take note that the deletion of a path is not allowed if there is a station placed on either end of the path. To delete the path, the station must be first deleted.
27.	Group	Change junction group.
28.		Toggle swapping junction or via junction.
29.	Name	Name configuration for the station. It will be used as the reference to the particular Station under Task Template and AMR Home Station configuration.
30.		Rotate Heading (Counter-clockwise) is a function to change the direction of the station. This function will rotate the direction of the Station in Counter-clockwise direction. Each click on this function will initiate a 10 degree counterclockwise turn.
31.		Rotate Heading (Clockwise) is a function to change the direction of the station. This function will rotate the direction of the Station in Clockwise direction. Each click on this function will initiate a 10 degree counterclockwise turn.
32.		Rotate Heading 90 degree (Counter-clockwise) is a function to change the direction of the station. This function will rotate the direction of the Station in Counter -clockwise direction. Each click on this function will initiate a 90 degree Counterclockwise turn.
33.		Rotate Heading 90 degree (Clockwise) is a function to change the direction of the station. This function will rotate the direction of the Station in Clockwise direction. Each click on this function will initiate a 90 degree Clockwise turn.
34.		A "switch" to select different stations placed under the same point as currently selected station.
35.		Shortcut button that links to the search page that searches all the references that used this station.
36.		Function to delete the currently selected Station.
37.		Select the AGV to connect to the Live App.



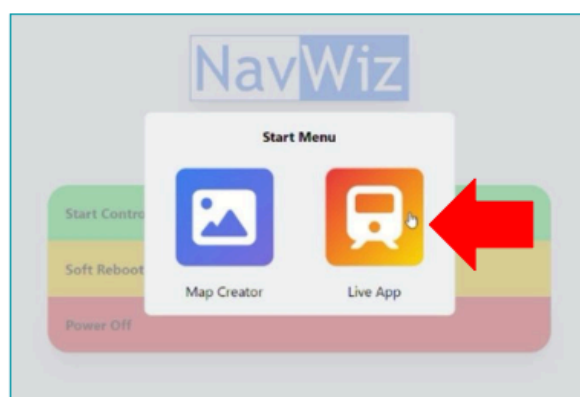
Info:

To select multiple objects, click and drag the mouse while holding down the "Ctrl" key.

To clone selected objects (ensure that paths are selected), hover the mouse over the junction, then click and drag the mouse while holding down the "Shift" key.

Live App

The Live App is a feature that utilizes real-time data to simplify map editing, including raw map, layout, and landmark editing. The AGV NavWiz must start the Live App in order to use the Live App map editing feature in DFleet.

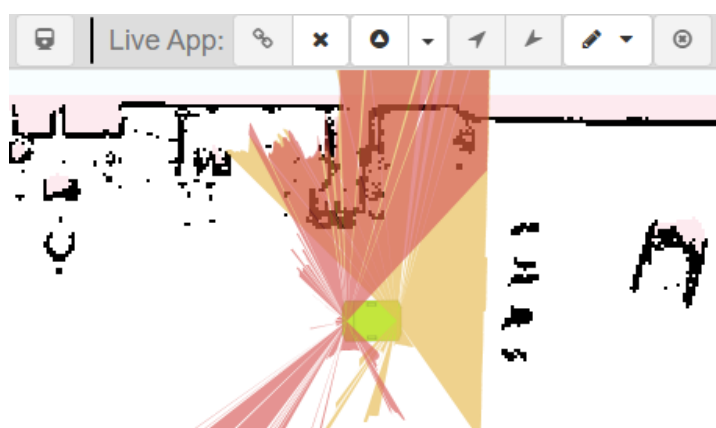


Raw Map Editor Live App

If the Live App is running on the AGV side, the Raw Map Editor can select an available AGV to connect to and display data from that AGV.



There are few function buttons available once connected.

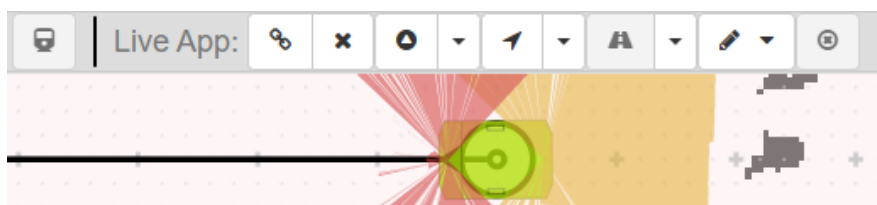


No.	Component	Description
1.		Display connection status.
2.		Disable Live App
3.		Reset the AGV position on the raw map either using the cursor or the manual arrow controls. For Reset Position, use the cursor to select a position to reset the AGV's position. For manual reset position, the AGV position is controlled using manual arrow movements. 

4.		Go To Position will move the actual robot to the desired position in forward motion. Use the cursor to select the desired position to go to, then press Shift + Spacebar to move the actual robot. Only enabled after resetting the AGV position.
5.		Reverse Go To Position, this will move the actual robot to the desired position in a reverse motion. Use the cursor to select the desired position to go to, then press Shift + Spacebar to move the actual robot. Only enabled after resetting the AGV position.
6.		Apply detected walls.
7.		Abort action such as Go To Position / Reset Position

Map Layout Editor Live App

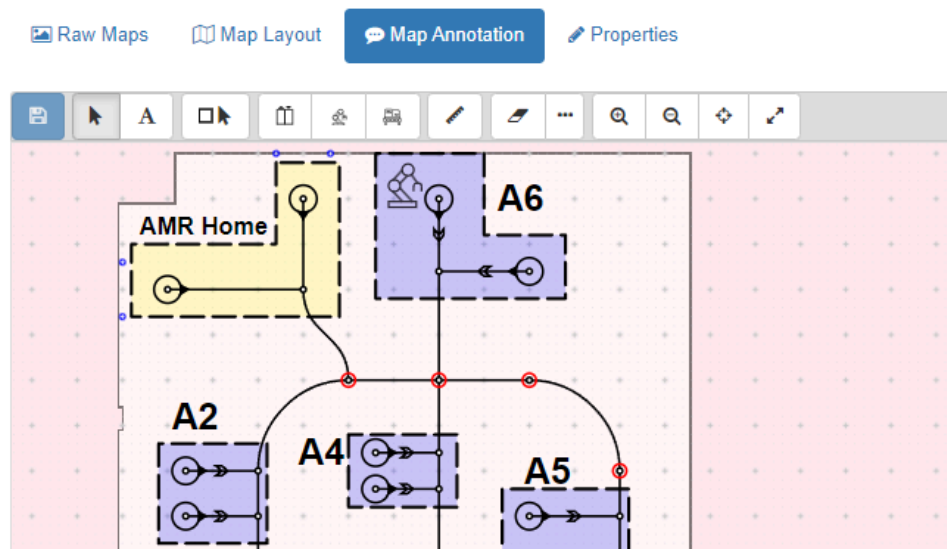
The Live App in the Map Layout Editor can be connected in a similar way to the Raw Map Editor, with some different functions available.



No.	Component	Description
1.		Display connection status.
2.		Disable Live App
3.		Reset Position, Manual Reset Position and Reset tracker to station. For Reset Position and Reset tracker, use the cursor to select a position to reset the AGV's position. For manual reset position, the AGV position is controlled using manual arrow movements.
4.		Go To Position and Reverse Go To Position, will move the actual robot to the desired position. Use the cursor to select the desired position to go to, then press Shift + Spacebar to move the actual robot. Only enabled after resetting the AGV position.
5.		Navigate To and Reverse Navigate To a station, will move the actual robot to the desired station. Use the cursor to select the desired station to go to, then press Shift + Spacebar to move the actual robot. Only enabled after resetting the AGV position tracker.
6.		Apply detected reflector.
7.		Abort the running action such as Go To Position, Navigate To and Reset Position.

Map Annotation

Map "Example Map 01"



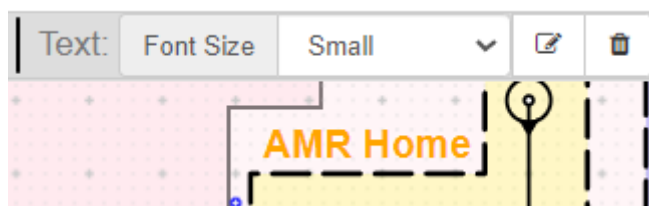
The Map Annotation Editor page enables users to create annotations on the map. These annotations are solely for visual reference and do not impact the functionality of the AMR system. They serve to provide additional information and enhance the user's understanding of the map. The table below outlines and explains the various functions available.

Common Features

No.	Features	Descriptions
1.		Save any edits made to the map annotation. In order for the changes to take effect, Soft Reboot is required.
2.		Change the current tool to "Select," enabling users to drag the map for different area views. This tool also enables users to drag the text annotation and icon annotation around.
3.		Change the current tool to "Text Annotation Placement", allowing the user to place text annotation. The default text annotation is "Add text".
4.		Change the current tool to "Polygon Annotation Placement", allowing the user to place polygon annotation.
5.		Change the current tool to "Lift Icon Annotation Placement", allowing the user to place lift icon annotation.
6.		Change the current tool to "Robot Arm Icon Annotation Placement", allowing the user to place robot arm icon annotation.
7.		Change the current tool to "Conveyor Icon Annotation Placement", allowing the user to place conveyor icon annotation.

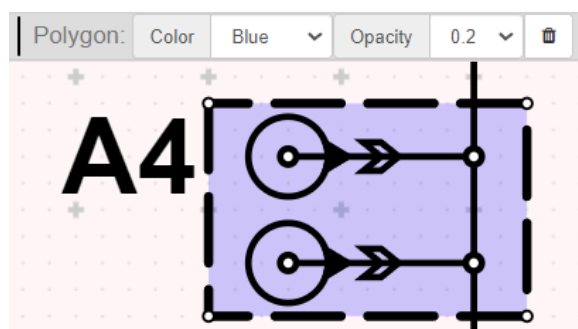
8.		Change the current tool to “Pallet Icon Annotation Placement”, allowing the user to place pallet icon annotation .
9.		Quickly delete all map annotations. Do take note that all unsaved changes will be lost if this function is used.
10.		Undo the last change on the map annotation.
		Redo the last change on the map annotation.
11.		Zoom in on the map annotation editing window.
12.		Zoom out on the map annotation editing window.
13.		Adjust the view to show all the content on the map annotation editing window, ensuring that everything is visible within the window size.
14.		Enable full-screen mode for the map annotation editor.

Text Annotation Properties Toolbar - Appear when a text annotation is selected.



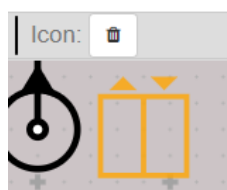
No.	Features	Descriptions
1.	Font Size	Font size of the text annotation. Available options: ● Extra Small ● Small ● Medium ● Large ● Extra Large
2.		Click to edit the content of the text annotation. Multi-line content is allowed.
3.		Delete currently selected text annotation.

Polygon Annotation Properties Toolbar - Appear when a polygon annotation is selected.



No.	Features	Descriptions
1.	Color	Color of the polygon annotation. Available options: • Red • Green • Blue • Yellow • Black • White
2.	Opacity	Opacity of the polygon annotation. Available options: • 0 • 0.2 • 0.4 • 0.6 • 0.8 • 1
3.		Delete currently selected polygon annotation.

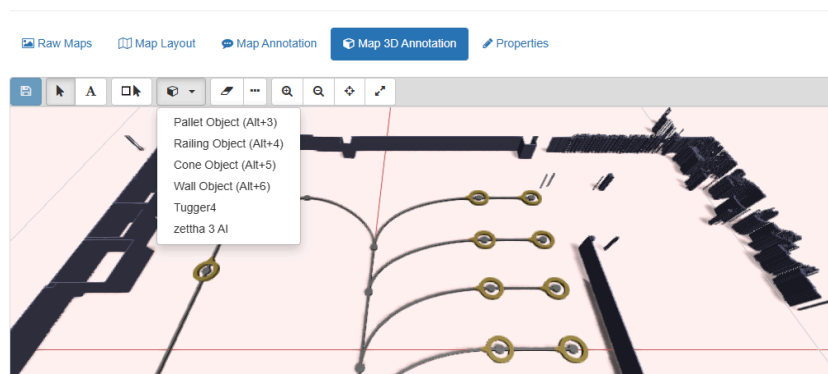
Icon Annotation Properties Toolbar - Appear when an icon annotation is selected.



No.	Features	Descriptions
1.		Delete currently selected icon annotation.

Map 3D Annotation

3D Map "example"



The Map 3D Annotation Editor page enables users to create 3D annotations on the 3D map. Similar to 2D annotations, these annotations are solely for visual reference and do not impact the functionality of the AMR system. They serve to provide additional information and enhance the user's understanding of the map. The table below outlines and explains the various functions available.

Common Features

No.	Features	Descriptions
1.		Save any edits made to the map 3d annotation. In order for the changes to take effect, Soft Reboot is required.
2.		Change the current tool to "Select," enabling users to drag the map for different area views and select the annotation objects for configuration. This tool also enables users to drag the text annotation and object annotation around.
3.		Change the current tool to "Text Annotation Placement", allowing the user to place text annotation. The default text annotation is "Add new text".
4.		Change the current tool to "Polygon Annotation Placement", allowing the user to place polygon annotation.
5.		Change the current tool to "Object Annotation Placement", allowing the user to select Pallet Object, Railing Object, Cone Object, Wall Object and custom 3D assets (if any) into the map.
6.		Quickly delete all map annotations. Do take note that all unsaved changes will be lost if this function is used.
7.		Undo the last change on the map annotation.
		Redo the last change on the map annotation.
8.		Zoom in on the map annotation editing window.

9.		Zoom out on the map annotation editing window.
10.		Adjust the view to show all the content on the map annotation editing window, ensuring that everything is visible within the window size.
11.		Enable full-screen mode for the map annotation editor.

Properties

It is possible to edit the name of a map in the Properties tab.

Map "TestMap01"

The screenshot shows the 'Properties' tab for a map named 'TestMap01'. It includes a 'Name' input field with the value 'TestMap01' and an 'Update' button. Above the input field are two tabs: 'Map Layout' and 'Properties'.

Import Map from AGV

It is possible to directly import maps from any AMR currently synch with the DFleet. To do so, click the "Import map from AGV" button at the bottom left corner of the Maps page.

The screenshot shows the 'Maps (1-1 of 1)' section. It contains a table with one map entry: 'TestMap01'. Below the table are two buttons: '+ Add new map' and 'Import map from AGV'. A mouse cursor is pointing at the 'Import map from AGV' button. Below the buttons, it says 'Active Map: TestMap01'.

Once inside, wait for the page to refresh as it may take some time for the system to retrieve all available maps from all currently sync AMR.

Import map from AGV

The screenshot shows the 'Import map from AGV' page. It has a 'Refresh all' button at the top. Below it is a table with columns '#', 'Map', and 'Operation'. The table has two rows: 'AMR 001' and 'SIM 01'. The 'SIM 01' row shows a map named 'Sample Map' with an 'Import' button in the 'Operation' column. A mouse cursor is pointing at the 'Import' button.

From the list choose any map to import and click the icon under the operation column to start importing.

Sample Map
 Imported into map "Sample Map"

A notification would come out after import is done and map could now be access from the map main page

Active Map

Edit an active map by clicking the pencil icon on the bottom left corner of the AGV page.

Maps (1-2 of 2)

#	Name
1	Example Map 01
2	Sample Map

[+ Add new map](#)
[Import map from AGV](#)

Active Map:    

• Example Map 01

Once inside, tick the map to be set active and click submit. It is possible to activate more than one map at a time. This is usually applied to maps within a multi-storey building.

Edit active map

[Active Map](#)
[Map Teleportation](#)
[Map Transition Trigger](#)
[Map Parameters](#)

Active map*

☐ Example Map 01
☒ Sample Map

[Submit](#)



Info:

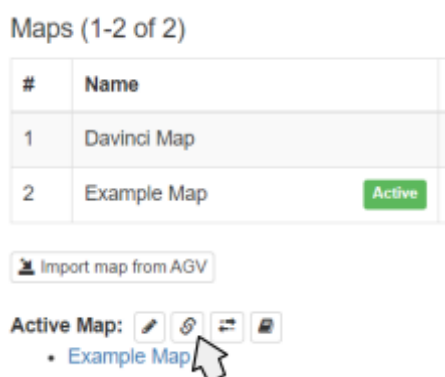
It is possible to activate more than one map at a time. This is mostly used when mapping multi-storied buildings.

Map Teleportation

Teleportation is essential for trackless AMRs that rely on switching between map layouts. This applies in situations where the AMR needs to transition through a curtain or use a lift to move to a different floor within a building.

Teleportation can also be applied between two stations within the same map. In earlier versions of DFleet, teleportation was often used to trigger specific actions automatically when the AMR passed through a designated station. However, with the introduction of the Map Transition Trigger feature, this usage of teleportation will become less common.

Access the map teleportation page by clicking the icon on the bottom left corner of the page.



Navigate to map teleportation page

Edit map teleportation

[Active Map](#)
[Map Teleportation](#)
[Map Transition Trigger](#)
[Map Parameters](#)

Teleports							
#	Start	End	Pre-Action	Action	Attributes	AGVs	Operation
1	A	B	---	Select Obstacle Profile Parameters: Profile 3	Align Station Type (before teleport) ☐ Non-stop Transition (before teleport) ☒ Auto-reset AGV Position (after teleport) ☐ Abortable Distance Cost (m) 100	(default) +	Up Down Delete

Map teleportation page

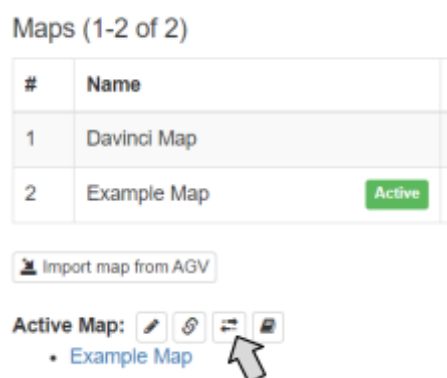
No.	Component	Description
1.	Start	Start position/station where teleport would start.
2.	End	Final position/station where teleport would end.
3.	Pre-Action	Task to execute right before entering the teleport station.
4.	Action	Task executed to complete teleportation.
		Align Station Type (before teleport) Select the align station method among the options below: • Trackless • Front Line Sensor • Rear Line Sensor
5.	Attributes	Non-stop Transition (before teleport) Untick to cancel non-stop motion before teleportation, causing the AMR to come to a complete stop before initiating the teleportation process.
		Auto-reset AGV (after teleport) Position Reset AMR position. Compulsory unless already reset in 'Action' template. Position Tracker Reset AMR tracker only after teleport. This is normally used if the teleportation happened without actual robot motion.
		Abortable Untick to make the teleport action non-abortable.
		Validate RFID (after teleport) To validate the RFID. (Only available in track mode)
		Distance Cost (m) Distance in map from Start to End position/station.
6.	AGVs	Different Pre-Action and Action for different AGVs
7.	Operation	 Move up the list.
		 Move down the list.
		 Clone teleport.
		 Delete teleport.
8.	Add Teleport	Add a new map teleport point.
9.	Submit	Submit change to be saved. Need to soft-reboot afterwards.

Map Transition Trigger

Transition Triggers are designed to automatically activate specific actions when the AMR passes through a designated station, or when the AMR changes its heading during navigation. Transition Triggers are configured between two stations located at the same junction.

It's important to note that Transition Triggers are only effective when navigation skills like "Navigate To" and "Reverse Navigate To" are in progress. Skills like "Rotate Left" or "Rotate Right" will not activate the transition actions.

Access the map transition trigger page by clicking the icon on the bottom left corner of the Maps page.

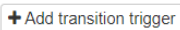


Edit map transition trigger

[Active Map](#)
[Map Teleportation](#)
[Map Transition Trigger](#)
[Map Parameters](#)

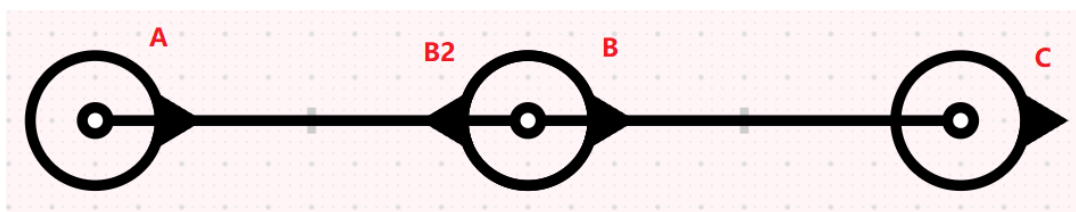
Transition Triggers									
#	Location	Incoming	Outgoing	Applicable Motion	Start Action (before rotate align)	End Action (after rotate align)	Attributes	AGVs	Operation
1	C	Conditions: Edit - Start Headings: - C	Conditions: Edit - End Headings: - C	☒ Navigate To ☒ Reverse Navigate To ☒ Ambi Navigate To	Select Obstacle Profile Parameters: Profile 1 ☐	Select Navigation Profile Parameters: Profile 1 ☐	☒ Always stop (before transition) Abortable: ☐ During "Start Action" ☐ Before rotate align ☐ After rotate align ☐ During "End Action"	(all AGVs) +	Up Down X

No.	Features	Descriptions
1.	Location	Start station (before transition).
2.	Incoming	Incoming conditions before transition. Available options: - All Conditions - Incoming Paths: - Departure - Swapping Departure - Linked station name - Start Headings: - Station heading
3.	Outgoing	Outgoing conditions after transition. Available options:

		• All Conditions • Outgoing Paths: ○ Departure ○ Swapping Departure ○ Linked station name • End Headings: ○ Station heading	
4.	Applicable Motion	Select the relevant applicable motions that activate this transition trigger. Available options: • Navigate To • Reverse Navigate To • Ambi Navigate To	
3.	Start Action (before rotate align)	Task to execute at start heading before transition.	
4.	End Action (after rotate align)	Task to execute at end heading after transition.	
5.	Attributes	Always stop (before transition)	Tick to cancel non-stop transition when passing through this junction.
		Abortable	Tick to make the transition trigger action abortable. Available options: • During 'Start Action' • Before rotate align • After rotate align • During 'End Action'
6.	AGVs	Specific AGV for this transition trigger.	
7.	Operation		Move up in the list.
			Move down in the list.
			Clone transition trigger.
			Delete transition trigger.
8.		Add a new transition trigger.	
9.	Submit	Submit changes. In order for the changes to take effect, Soft Reboot is required.	

Example Use Case 1

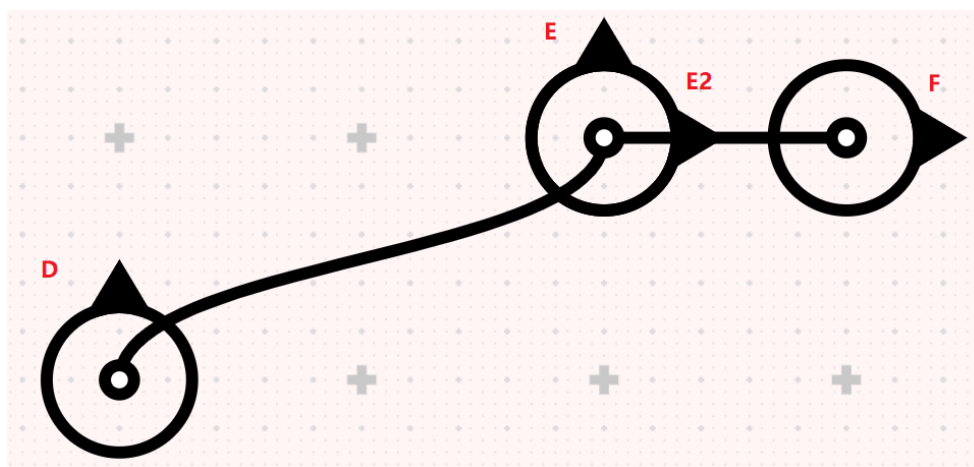
Switch to navigation profile 2 when forward passing through station B from left to right, and switch to navigation profile 1 when forward passing through station B2 from right to left.



#	Location	Incoming	Outgoing	Applicable Motion	Start Action (before rotate align)	End Action (after rotate align)	Attributes	AGVs	Operation
1	B	Conditions: - Incoming Paths: - A	Conditions: - End Headings: - B2	☒ Navigate To ☐ Reverse Navigate To ☐ Ambi Navigate To	Select Navigation Profile Parameters: Profile 2	---	☒ Always stop (before transition) Abortable: ☐ During "Start Action" ☐ Before rotate align ☐ After rotate align ☐ During "End Action"	(all AGVs)	+ - ↕
2	B2	Conditions: - Incoming Paths: - C	Conditions: - End Headings: - B2	☒ Navigate To ☐ Reverse Navigate To ☐ Ambi Navigate To	Select Navigation Profile Parameters: Profile 1	---	☒ Always stop (before transition) Abortable: ☐ During "Start Action" ☐ Before rotate align ☐ After rotate align ☐ During "End Action"	(all AGVs)	+ - ↕

Example Use Case 2

At station D, if the AMR executes "Navigate To Station F", it will rotate right 90 degrees at station E. If for some reason we want to force it to rotate left 275 degrees instead, we can achieve this through transition triggers.



Transition Triggers

#	Location	Incoming	Outgoing	Applicable Motion	Start Action (before rotate align)	End Action (after rotate align)	Attributes	AGVs	Operation
1	E	Conditions: - Incoming Paths: - D	Conditions: - End Headings: - E2	☒ Navigate To ☒ Reverse ☒ Ambli ☒ Navigate To	Rotate Left (trackless) Parameters: Angle: 270 Enable Sensor: ☒	----	☒ Always stop (before transition) Abortable: ☐ During "Start Action" ☐ Before rotate align ☐ After rotate align ☐ During "End Action"	(all AGVs) +	↑ ↓ ↺ ✕

Map Parameters

The Map Parameters page is used specifically for configuring custom speed parameters measured in meters per second (ms^{-1}). These parameters can be used to set the speed limit of the paths in the map editor.

Access the map parameters page by clicking the icon on the bottom left corner of the Maps page.

Maps (1-2 of 2)

#	Name	
1	Davinci Map	
2	Example Map	Active

Import map from AGV

Active Map:    
 • Example Map 

[Active Map](#)
[Map Teleportation](#)
[Map Transition Trigger](#)
[Map Parameters](#)

Map Parameters					
#	Name	Type	Description	Value	Operation
1	Unlimited	Double			(Reserved)
2	Medium Fast	Double		0.6	  
				AGV-specific Overrides: +	
3	Medium	Double		0.5	  
				AGV-specific Overrides: +	
4	Slow	Double		0.4	  
				AGV-specific Overrides: +	

+ Add map parameter

No.	Component	Description
1.	Name	Name of the map parameter.
2.	Type	Data type of the map parameter.
3.	Description	Description for the map parameter.
4.	Value	Value of the map parameter.
5.	Operation	 Move up in the list.
		 Move down in the list.
		 Delete the map parameter..

Section 4. Task Templates

Task Templates is a continuous step that can be carried out by the AMR. With a task template, the AMR is able to perform multiple works without continuous permission. We can also modify the tasks to match with the surroundings of the working workplace.

Task Templates

The screenshot shows the 'Task Templates' interface. It includes a 'FILTERS' dropdown (1), a list of task templates (2-3 of 3) with columns for #, Name, Active, Top-level, Allowed AGVs, Allowed Groups, Allowed Users, and Operation (3-8). Below the list are three buttons: 'Add new task template' (9), 'Edit task template global parameters' (10), and 'Edit task template variables' (11).

#	Name	Active	Top-level	Allowed AGVs	Allowed Groups	Allowed Users	Operation
1	TestAGV	✓	✓	SIM 01	Call Button (0:00:00) Supervisor (0:00:00) User (0:00:00)	demoUser (0:00:00)	[Edit] [Copy] [Delete] [More]
2	Open Door	✓	✗	-	-	-	[Edit] [Copy] [Delete]
3	testMarker	✗	✗	-	-	-	[Edit] [Copy] [Delete]

No.	Component	Description		
1.	Filters	Filters task template list.	Active	Yes
				No
		Top-level	Yes	
			No	
2.	Name	Task template name.		
3.	Active	Mark current status of task template.		Active
				Not active
4.	Top-level	Mark top-level configuration status		Top-level task
				Not top-level task
5.	Allowed AGVs	List all the AMRs that are allowed to participate in the task template, along with their priority configuration. The priority determines the priority in which AMRs are assigned to the task if multiple AMRs are available.		

6.	Allowed groups	List all user groups allowed to participate in the task template operation and the expected time for AMR to reach the station of the user. The format is hh:mm:ss.	
7.	Allowed users	List all users allowed to participate in the task template operation and the expected time for AMR to reach the station of the user. The format is hh:mm:ss.	
8.	Operation		Open task template editor page.
			Clone the task template.
			Open AMR allowed participation page.
			Open User allowed participation page.
			Delete the task template.
9.	 Add new task template	Add a new task template.	
10.	 Edit task template global parameters	Edit task template global parameters.	
11.	 Edit task template variables	Edit task template variables.	

Task Template Editor/Creator

Edit task template "[Basic] Navigate To"

Name*

Category

☒ Active
☒ Top-level template
☒ Abortable
☐ Suspend task initially

Params

Name	Type	Description	Default	Min	Max	Operation
station	Station					↑ ↓ ×

+ Add param

Outcomes *

Enter the outcomes separated by comma.

Structure

View [last saved](#) or [last cached](#) original structure.



Duration*

Expected time for AGV to complete the whole task template. The format is "hh:mm:ss".

Destination Hint

 ☐ Inherit

Station hint to determine AGV's distance during AGV assignment (optional)

AGV Assignment Min Distance

 m ☐ Inherit

Minimum distance of AGV to be allowed for task assignment. Set to zero to disable this function.

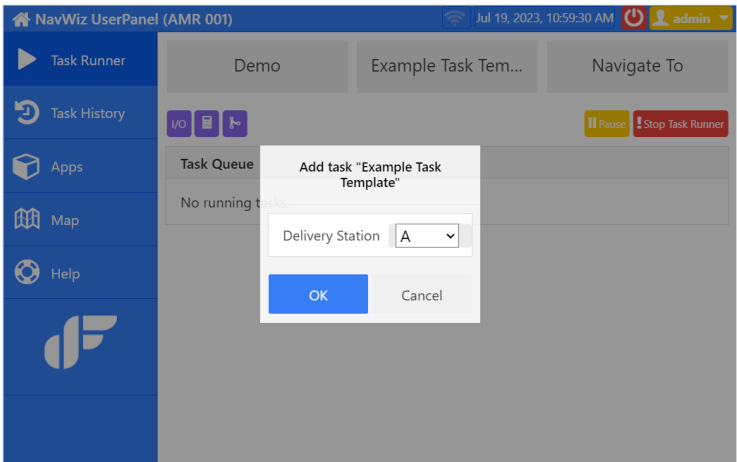
AGV Assignment Max Distance

 m ☐ Inherit

Maximum distance of AGV to be allowed for task assignment. Set to zero to disable this function.

No.	Component	Description
1.	Name	This is the name assignment function for the task template. The name can be some representative to ease the recognition of the task template, for example, "Delivery to Station A" or "Collection from Station C".
2.	Category	An optional field. By giving this task template a category, this task template can be found in this specific category. This eases the work of finding a task template.
3.	Active	A flag for users to activate or deactivate the current task template. Users can use this function to temporarily disable a task template without deleting it. A deactivated task template will be hidden from UserPanel preventing users from running or executing it.

4.	Top-level template	A flag configuration for the current task template. A top-level template is a task template that can be run by the user from UserPanel, and it can only end by an outcome, "End". Any task that is configured as a non-top-level template will be hidden from UserPanel.																					
5.	Abortable	This functions as a flag for the user to allow or prevent the current task template from being aborted. A non-abortable task template cannot be stopped or aborted through normal user actions.																					
6.	Suspend task initially	Task added would be initially suspended instead of immediately starting.																					
7.	Params	Parameters are configurations that enable users to add customizable variables to a task template. These parameters can then be used as inherited variables within the task template, enhancing the flexibility to accommodate various parameter values. Here is an example of how Params can be used: Params <table><thead><tr><th>Name</th><th>Type</th><th>Description</th><th>Default</th><th>Min</th><th>Max</th><th>Operation</th></tr></thead><tbody><tr><td>Delivery Station</td><td>Station</td><td></td><td></td><td></td><td></td><td>↑ ↓ ×</td></tr><tr><td colspan="7">+ Add param</td></tr></tbody></table> In the figure shown above, there is a parameter named "Delivery Station" and its type is Station. For the structure of the example task template, there is a "Navigate To" action, which requires a "Station" parameter. For the "Station" parameter within the "Navigate To" action, users have two options. They can either select a static station available on the map or utilize the Params feature by linking the "Delivery Station" parameter to the "Navigate To" action's parameter. Param: ↑ ↓ Station A ▾ ☐ Inherit Navigate To Parameters: Station: A Align Station Type: -1 Next Motion: 0 Next Speed: 0 Sense Line: 0 Done Configuring Static Station Param: ↑ ↓ Station \${Delivery Station} ▾ ☒ Inherit Navigate To Parameters: Station: \${Delivery Station} Align Station Type: -1 Next Motion: 0 Next Speed: 0 Sense Line: 0 Done Configuring Dynamic Station	Name	Type	Description	Default	Min	Max	Operation	Delivery Station	Station					↑ ↓ ×	+ Add param						
Name	Type	Description	Default	Min	Max	Operation																	
Delivery Station	Station					↑ ↓ ×																	
+ Add param																							

		When a task template with Params is executed, a window will prompt users to input the corresponding values for the Params, ensuring customizable and dynamic task execution. 		
8.	Params	Name	Params name to be used inside task template structure.	
		Type	Type of params to be created.	Bool
				Int
				Double
				Str
				Station
				Register
				DFleet register
		Description	Description of params to be created.	
		Default	Default value of params.	
		Min	Minimum value of params.	
		Max	Maximum value of params.	
		Operation		Move params up.
				Move params down.
				Delete params.
9.	Outcomes	Outcomes is a function to define outcomes or results of the task template execution. This allows the task template to have different results, that can be used as a decision maker for the task template flow. However, the Outcomes feature is only available for task templates that are being configured as non-Top-level Task Templates. This limitation is due to the reason that the Top-level Task Templates are only allowed to have one outcome which is "End".		

Outcomes *

Success,Fail

Enter the outcomes separated by comma.

Structure



Once the Top-level template is being unchecked in a task template, users will notice that the Outcomes Text input will be available for edit or modification.

Users can now insert all the possible outcomes for the task template into the textbox under Outcomes.

After adding the outcomes into the textbox, they will be shown under Structure. Example above shows that there are 2 outcomes, Success and Fail.

Outcomes *

Ok,Timeout

Enter the outcomes separated by comma.

Structure

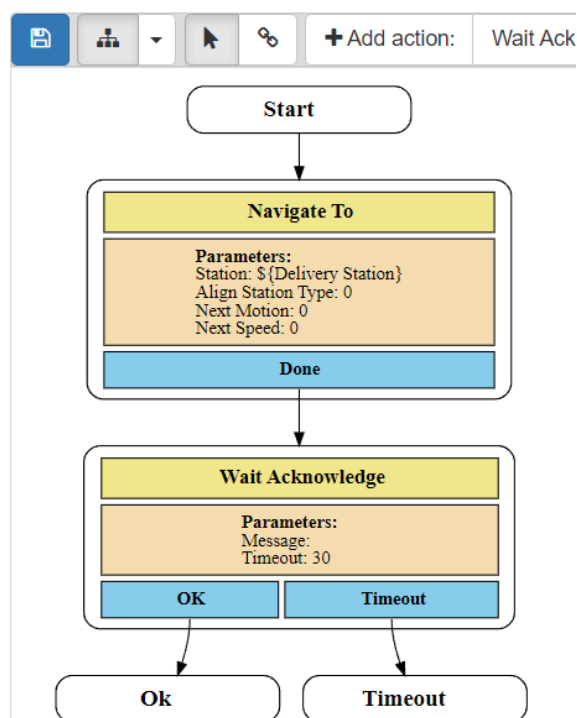
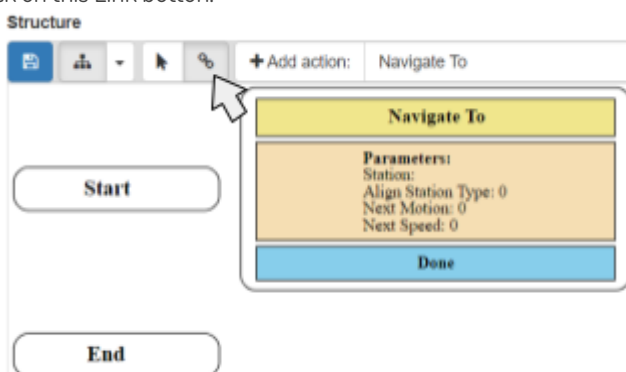
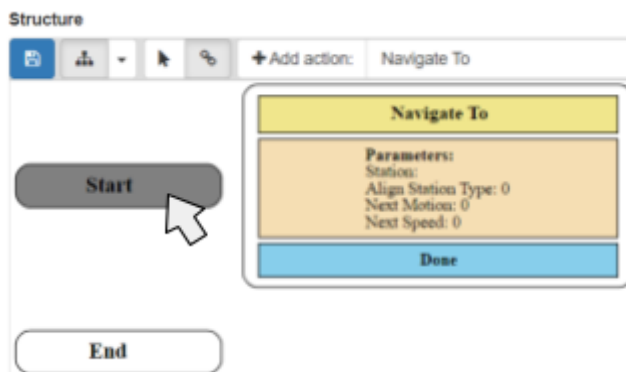
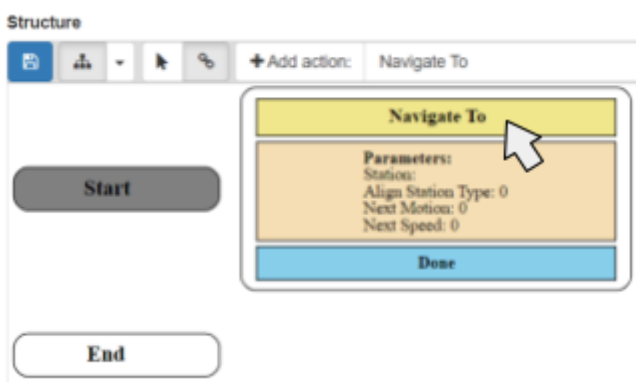
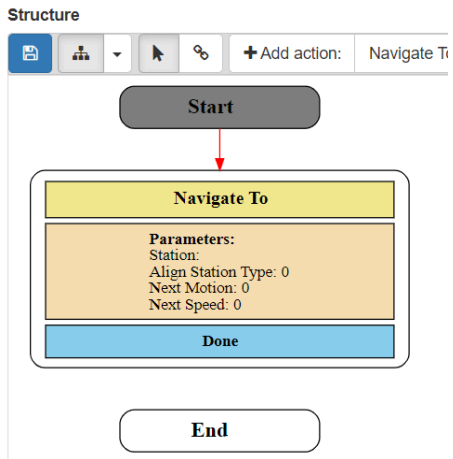


Figure above, shows a Task Template with 2 outcomes and under this task template, the first action is to navigate to a station, and second action is to perform Wait Acknowledge where there will be 2 results.

First result is OK (where the user presses ok while the Acknowledge message is shown) and second result is Timeout (where the timer timeout and no input from user side).

This task template can then be inserted into another task template to ease the task template creation.

10.	Last saved	Open view of the last saved original structure.		
11.	Last cached	Open view of last cached original structure.		
12.	Toolbar		Save edited task. A Soft Reboot is required after the changes for it to be effective.	
			 Click to change to auto layout, where the task template actions in the structure will be arranged automatically.	
			 Click to change to auto layout, where users have flexibility to drag the task template actions around in the structure.	
			Select is a function to change from the current tool to Select, that allows the user to drag the task template for different part views, and to select the action for configuration. If the current tool is set to this, users can also click and drag the outcomes to link task template actions.	
			Link is the function to link 2 actions under a task template. Steps to link 2 action: Step 1: Click on this Link button.	
				
			Step 2: Click on the first item to link.	
				
			Step 3: Click on the second item to link.	

			 Finally, a link is created. 	
			Add action	Add action into task template.
				Delete everything under the structure.
				Cut action tool.
				Copy action tool.
				Paste action tool.
				Undo last change.
				Redo last change.
				Zoom in on the structure window.
				Zoom out on the structure window.
				Fit Content fits all content into the window size. System will adjust the zoom level to show all content on the window.
				Show Fullscreen enables full screen viewing mode.

			Open note editor to allow users to add notes for each action. The note will pop out when a mouse cursor hover over the action. (Need to click and select an action first).
			Allow users to delete action or link. (Need to click and select an action or a link first).
13.	Duration	Expected time for AMR to complete the whole task template. The format is hh:mm:ss .	
14.	Destination hint	Station hint to determine the distance between AMR and Station which can be applied to "To Destination (Nearest / Farthest) AMR assignment policy. Check the inherit box to enable inherit function. Inherit function will take the parameter set at Params or Global Params as the value.	
15.	AGV Assignment Min Distance	Minimum distance of AMR to be allowed for task assignment. Set to zero to disable this function. Check the inherit box to enable inherit function. Inherit function will take the parameter set at Params or Global Params as the value.	
16.	AGV assignment max distance	Maximum distance of AMR to be allowed for task assignment. Set to zero to disable this function. Check the inherit box to enable inherit function. Inherit function will take the parameter set at Params or Global Params as the value.	

Task Template Groups and Users

The DFleet task template comes with access control for different users and user groups. There are some situations where the task templates need to be assigned to multiple users within the same user group, assigning the task template to the user group can eliminate the need to assign them individually for each user. The task template is assigned to a user if they are either listed as an allowed user or belong to a user group listed in the allowed groups.

Edit users of task template "Example Task Template"

Task template

Example Task Template

Allowed groups

☐ Guest
☒ Call Button
☐ User
☒ Supervisor

Allowed users

☐ CallButton
☐ Guest
☐ Supervisor
☐ User

Submit

After the configuration of the user access, a soft reboot is required for the changes to be effective.



Task Template Global Parameters

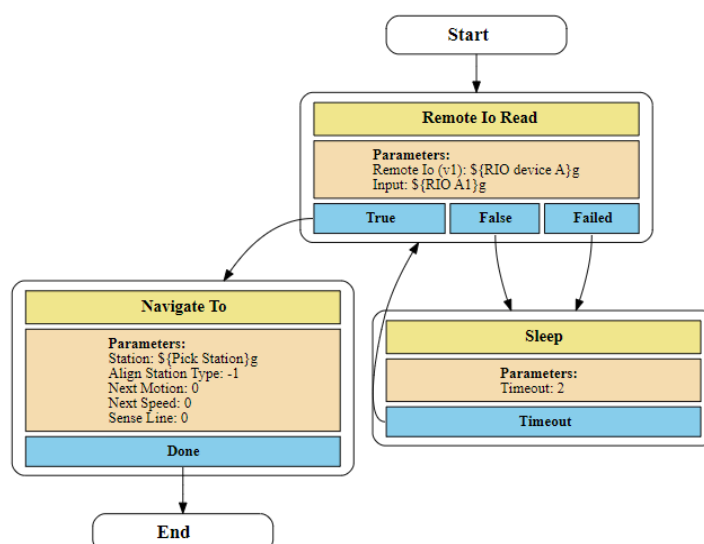
Task template global parameters are the constant values that can be accessed in all task templates. This makes users easily use the same constant value across different task templates, reducing the need for hard-coding task templates.

The global parameters can be of the following types:

- Bool
- Int
- Double
- Str
- Station
- Register
- DFleet Register
- Lookup Table

The following is an example that demonstrates the usage of task template global parameters.

Global Parameters					
#	Name	Type	Description	Value	Operation
1	agv_home	Station ▾			(Reserved)
2	RIO device A	Int ▾	Remote IO device at Area A	1	↑ ↓ ✕
3	RIO A1	Int ▾	Remote IO pin at station A1	0	↑ ↓ ✕
4	RIO A2	Int ▾	Remote IO pin at station A2	1	↑ ↓ ✕
5	Pick Station	Station ▾	Station to pick up load	A1 ▾	↑ ↓ ✕
6	Drop Station	Station ▾	Station to deliver load	A2 ▾	↑ ↓ ✕
+ Add global parameter					



Task Template Variables

Task template variables can be accessed in all task templates as well. However, unlike task template global parameters, the values of these task template variables can be changed in task templates through the following Task Template actions:

- Set Bool
- Set Integer
- Set String
- Set Double
- Set Station

The variables can be of the following types:

- Bool
- Int
- Double
- Str
- Station

Variables					
#	Name	Type	Description	Default	Operation
1	Pick Station	Station ▾	Station to pick up load	A1 ▾	⬆ ⬇ ✕
2	Drop Station	Station ▾	Station to deliver load	A2 ▾	⬆ ⬇ ✕
3	loaded	Bool ▾	Whether the AMR is loaded or not	☐	⬆ ⬇ ✕
+ Add variable					

**Notice:**

Take note that the values of the task template variables are stored on AMR instead of DFleet. Hence, please do not assume that the values of the Task Template variables are always the same all the time.

Skills of DFleet

There are some skills that are only available for DFleet.

1.	Play Music	Description	Play playlist music on selected audio channels.	
		Parameters	Channel (int)	Audio Channel No. (1 - 5)
				Default: 1, min: 1, max: 5
			Playlist (int)	Playlist No. (0 - 10)
				Default: 0, min: 1, max: 10
		Outcomes	Done	Action completed
2.	Stop Music	Description	Stop music on selected audio channels.	
		Parameters	Channel (int)	Audio Channel No. (1 - 5)
				Default: 1, min: 1, max: 5
		Outcomes	Done	Action completed
3.	Lift Reserve	Description	Lift Reserve is an action to reserve the lift for this AMR	
		Parameters	Lift (int)	Lift ID (1-10)
				Default: 1, min: 1, max: 10
		Outcomes	Done	Action completed
			Failed	Error occurred
4.	Lift Unreserve	Description	Lift Unreserve is an action to unreserve a previously reserve lift for this AMR	
		Parameters	Lift (int)	Lift ID (1-10)
				Default: 1, min: 1, max: 10
		Outcomes	Done: Action completed	
			Failed: Error occurred	
5.	Lift Check Reservation	Description	Lift Check Reservation is an action to check is the lift successfully reserved	
		Parameters	Lift (int)	Lift ID (1-10)
				Default: 1, min: 1, max: 10
		Outcomes	True	Lift is reserved
			False	Lift is not reserved
			Failed	Error occurred
6.	Lift Call Level	Description	Lift Call Level is an action to call the lift to specific floor	
		Parameters	Lift (int)	Lift ID (1-10)
				Default: 1, min: 1, max: 10

			Level (int)	Level which the lift should stop at
				Default: 0, min: -20, max: 100
		Outcomes	Done	Action completed
			Failed	Error occurred
7.	Lift Wait Level	Description	Lift Wait Level is an action to let the lift wait at the specific floor level	
		Parameters	Lift (int)	Lift ID (1-10)
				Default: 1, min: 1, max: 10
			Level (int)	Floor Level
				Default: 0, min: -20, max: 100
		Outcomes	Done	Action completed
			Failed	Error occurred
8.	Lift Clear Level	Description	Lift Clear Level is an action to stop lift from waiting at that level	
		Parameters	Lift (int)	Lift ID (1-10)
				Default: 1, min: 1, max: 10
			Level (int)	Floor level
				Default: 0, min: -20, max: 100
		Outcomes	Done	Action completed
			Failed	Error occurred
9.	Lift Read Level	Description	Lift Read Level is an action to read the current level of the lift position	
		Parameters	Lift (int)	Lift ID (1-10)
				Default: 1, min: 1, max: 10
			Level register (DFleet Register)	Register to store lift level
		Outcomes	Done	Action completed
			Failed	Error occurred
10.	Redis Delete Key	Description	Delete one redis key value	
		Parameters	Key	Redis key to delete
		Outcomes	Done	Action complete
			Fail	Error occurred
11.	Redis Set String Item	Description	Set one redis key to the given string value	
		Parameters	Key	Redis key
			Value	String value to set

		Outcomes	Done	Action complete
			Failed	Error occurred
12.	Redis Add String to List	Description	Add string item to a redis list	
		Parameter	Key	Redis key
			Value	String value to add inside list
			List Position	There are two options: 0 - head of list -1 - tail of list
		Outcomes	Done	Action complete
			Failed	Error occurred
13.	Redis Pop List	Description	Remove one string item from a redis list	
		Parameter	Key	Redis key
			List Position	There are two options: 0 - head of list -1 - tail of list
		Outcomes	Done	Action complete
			Failed	Error occurred
		Outcomes	Done	Action complete
14.	Redis Set Hash Field with String	Description	Set string value for one Hash field	
		Parameter	Key	Redis key
			Field	Redis hash field
			Value	String value to set
		Outcomes	Done	Action complete
			Failed	Error occurred
15.	Redis Delete Hash Field	Description	Delete one hash field and value from a redis	
		Parameter	Key	Redis key
			Field	Redis hash field
		Outcomes	Done	Action complete
			Failed	Error occurred
		Outcomes	Done	Action complete
16.	Redis Compare String Item with String	Description	Get redis string value and compare with the given string	
		Parameter	Key	Redis key
			String	String value to be compared with
		Outcomes	Equal	String item and string is the same
			Not Equal	String item and string is not the same
		Outcomes	Equal	String item and string is the same
			Not Equal	String item and string is not the same

			Failed	Not able to compare string item with string. Error occurred
17.	Redis Compare Two String Items	Description	Compare two string items between two redis key	
		Parameter	Key 1	Redis key 1
			Key 2	Redis key 2
		Outcomes	Equal	Redis key 1 and redis key 2 is the same
			Not Equal	Redis key 1 and redis key 2 is not the same
			Failed	Not able to compare. Error occurred.
18.	Redis Pop List and Compare String	Description	Remove one string item from redis list and compare it with the given string	
		Parameter	Key	Redis key
			String	String value to be compared with
			List Position	There are two options: • 0 - head of list • -1 - tail of list
		Outcomes	Equal	String item and string are the same.
			Not Equal	The string item and string are not the same.
			Failed	Not able to compare. Error occurred
19.	Redis Compare List Item and String	Description	Get string item from a redis list and compare it with the given string	
		Parameter	Key	Redis key
			String	String value to be compare with
			Index	Zero based index of the list item (Negative values start from the tail)
		Outcomes	Equal	List item and string is the same
			Not Equal	List item and string are not the same.
			Failed	Not able to compare. Error occurred
20.	Redis Compare Hash Field with String	Description	Get string item from one of the hash fields and compare it with the given string	
		Parameter	Key	Redis key
			Field	Redis hash field
			String	String value to be compare with
		Outcomes	Equal	String item in hash field and string is the same
			Not Equal	The string item in the hash field and string are not the same.
			Failed	Not able to compare. Error occurred

21.	Redis Compare Two Hash Fields	Description	Get two string items from two hash field and compare it with each other	
		Parameter	Key 1	Redis key 1
			Field 1	Redis hash field 1
			Key 2	Redis key 2
			Field 2	Redis hash field 2
		Outcomes	Equal	Two string from two hash field is the same
			Not Equal	Two string from two hash field is not the same
			Failed	Not able to compare. Error occurred
22.	Redis Get String Item	Description	Get Redis key value and set it to the variable.	
		Parameters	Key	Redis key
			Variable	String variable to set the value
		Outcomes	Done	Action completed
			Failed	Not able to set. Error occurred.
23.	Redis Pop List and Set Variable	Description	Redis pop head or tail of list item and set the value to the variable.	
		Parameters	Key	Redis key
			List Position	List position: [0] - head of list, [-1] - tail of list
			Variable	String variable to set the value.
		Outcomes	Done	Action completed
			Failed	Not able to set. Error occurred.
24.	Redis Get List Item	Description	Get the Redis key (list item) at an index and set the value to the variable.	
		Parameters	Key	Redis key
			Index	Zero based index of the list item. (Negative value start from tail) Default: -1
			Variable	String variable to set the value.
		Outcomes	Done	Action completed
			Failed	Not able to set. Error occurred.
25.	Redis Get Hash Field	Description	Get Redis hash field value for the key:field and set the value to the variable.	
		Parameters	Key	Redis key
			Field	Redis hash field.
			Variable	String variable to set the value.
		Outcomes	Done	Action completed
			Failed	Not able to set. Error occurred.

26.	Set Register Value	Description	Set Register Value is an action to write a value into a register	
		Parameters	Register (DFleet Register)	Register to be written
			Value (int)	Value to be assigned to the register
				Default: 0
		Outcomes	Done	Action completed
27.	Compare and Set Register	Description	An atomic action to compare register and only set a new value if the current value is equal to the expected value.	
		Parameters	Register (DFleet Register)	Register to be compared with expected value and set to new value.
			Expected Value (int)	The expected register current value.
			New Value (int)	The new value to be assigned.
		Outcomes	Equal	The register value is equal to expected value and the new value has been assigned to the register
			Not Equal	The register value is not equal to the expected value.
28.	Increment Register	Description	Increment Register is an action to increment the value of a Register. After this action, the value of the Register will be incremented according to the given increment value.	
		Parameters	Register (DFleet Register)	Register to be increment
			Value (int)	Increment value
				Default: 1
		Outcomes	Done	Action completed.
29.	Decrement Register	Description	Decrement Register is an action to decrement the value of a Register. After this action, the value of the Register will be decremented according to the given decrement value	
		Parameters	Register (DFleet Register)	Register to be decrement
			Value (int)	Decrement value
				Default: 1
		Outcomes	Done	Action completed.
30.	Add Register	Description	Add Register is an action to add two register values together. The result of this operation will be stored to the destination register.	
		Parameters	Destination (DFleet Register)	The register where the result will be stored.
			Source (DFleet Register)	The register where its value will be used to add to the destination register.

		Outcomes	Done	Action completed.
31.	Subtract Register	Description	Subtract Register is an action to subtract the destination register using the source register. The result of this operation will be stored in the destination register.	
		Parameters	Destination (DFleet Register)	The register where the result will be stored.
			Source (DFleet Register)	The register where its value will be used to subtract from the destination register.
		Outcomes	Done	Action completed.
32.	Multiply Register	Description	Multiply Register is an action to multiply two register values together. The result of this operation will be stored to the destination register.	
		Parameters	Destination (DFleet Register)	The register where the result will be stored.
			Source (DFleet Register)	The register where its value will be used to multiply with the destination register.
		Outcomes	Done	Action completed.
33.	Divide Register	Description	Divide Register is an action to divide the destination register with the source register. The quotient result of this operation will be stored to the destination register while the remainder will be stored to the remainder register.	
		Parameters	Destination (DFleet Register)	The register where the quotient result will be stored.
			Source (DFleet Register)	Divide the destination register with this register.
			Remainder (DFleet Register)	The register where the remainder value will be stored.
		Outcomes	Done	Action completed.
34.	Copy Register	Description	This action is to copy the register value from one to another register	
		Parameters	Destination (DFleet Register)	Destination register where the value is to be copied.
			Source (DFleet Register)	Register to be copied from.
		Outcomes	Done	Action Completed
35.	Copy Register From Agv	Description	Copy Register From AMR is an action to copy AMR register value to a register	
		Parameters	Destination (DFleet Register)	Destination register where the value is to be copied.
			Source Register	AMR register to be copied from.
		Outcomes	Done	Action completed

36.	Copy Register To Agv	Description	Copy Register To AMR is an action to copy register value to AGV register	
		Parameters	Destination Register	Destination AMR register where the value is to be copied.
			Source (DFleet Register)	Register to be copied from.
		Outcomes	Done	
37.	Remote Io Read	Description	This action will read the value from remote IO	
		Parameters	Remote io (int)	Remote io to read (1-20)
				Default: 1, min: 1, max: 20
			Input (int)	Remote io to read (1-20)
				Default: 1, min: 1, max: 20
		Outcomes	True	The value of remote IO is the same as input value
			False	The value of remote IO is not the same as input value
			Failed	Error occurred
38.	Remote Io Write	Description	This action will write a value to the remote io	
		Parameters	Remote io (int)	Remote io to write (1-20)
				Default: 1, min: 1, max: 20
			Output (int)	Remote io to write (1-20)
				Default: 1, min: 1, max: 20
			State (bool)	state of the output
				Default: False
		Outcomes	Done	Action completed
			Failed	Failed to complete
39.	Remote Io Read Register	Description	Read the register value from Remote Io	
		Parameters	Remote io (int)	Remote io to read (1-20)
				Default: 1, min: 1, max: 20
			Register (DFleet Register)	Register to read from
			Input mask (int)	Input mask
				Default: 255, min: 0, max: 65535
		Outcomes	Done	Action completed
			Failed	Error occurred
40.	Remote Io Write Register	Description	Write a register value through remote IO	

		Parameters	Remote io (int)	Remote io to write (1-20)
				Default: 1, min: 1, max: 20
			Register (DFleet Register)	Register to be written
			Output mask (int)	Output mask
		Outcomes		Default: 255, min: 0, max: 65535
			Done	Action completed
41.	Carrier Scanned at Port (Transaction)	Description		
				Register carrier at the port location and give signal to SecsGem component for carrier scanned event. To be called within a transaction loading or unloading task template.
		Parameters	Carrier Id	Carrier Id read by the AGV.
			Port	Carrier location when the read happened.
		Outcomes	Done	Action Completed
42.	Carrier Installed at Port (Transaction)	Description		Register carrier at the port location provided. To be called within a transaction loading or unloading task template.
		Parameters	Carrier Id	Carrier Id to be registered.
			Port	Port location of the carrier.
		Outcomes	Done	Action Completed
43.	Carrier Removed from Port (Transaction)	Description		Remove a carrier from port. To be called within a transaction loading or unloading task template.
		Parameters	Carrier Id	Carrier Id to be removed from port.
			Port	Previous known port of the carrier id.
		Outcomes	Done	Action Completed
44	Get Carrier at Port (Transaction)	Description		Check the registered carrier at the port
		Parameters	Port	Port location to check
			Variable	Assign found carrier to this variable
		Outcomes	Done	Action Completed
			Empty	No carrier found.
45	Install Dummy Carrier at Port (Transaction)	Description		Registers a placeholder carrier at a port that is hidden from the host.
		Parameters	Port	Installed port location.
		Outcomes	Done	Action Completed
46	Install Unknown Carrier at Port (Transaction)	Description		Registers a carrier with a synthesized name when the real ID is unknown.
		Parameters	Port	Installed port location.

			Prefix	Custom carrier identifier (appended after "UNKNOWN_").
		Outcomes	Done	Action Completed
47.	Carriers Removed from Port (Transaction)	Description	Remove all carriers from the specified port. To be called within a transaction loading or unloading task template.	
		Parameters	Port	The port from which all carriers will be removed.
		Outcomes	Done	Action Completed
48.	Abort Task - Carrier Handoff Failure (Transaction)	Description	Abort running task and set current transaction as carrier handoff failure. Transaction will then be in Paused state for user intervention. To be called within a transaction loading or unloading task template.	
		Outcomes	Done	Action Completed
49.	Abort Task - Carrier Read Failure (Transaction)	Description	Abort running task and set current transaction as carrier read failure. Transaction will be terminated. To be called within a transaction loading or unloading task template.	
		Parameters	Port	Port location of the carrier.
		Outcomes	Done	Action Completed
50.	Abort Task - Carrier Read Mismatch (Transaction)	Description	Abort running task and set current transaction as carrier read mismatch. Transaction will be terminated. To be called within a transaction loading or unloading task template.	
		Parameters	Carrier Id	The mismatched carrier ID.
			Port	Port location of the carrier.
		Outcomes	Done	Action Completed
51.	Abort Task - Destination Port Blocked (Transaction)	Description	Abort running task if destination port is blocked. Transaction will be terminated. To be called within a transaction loading or unloading task template.	
		Outcomes	Done	Action Completed
52.	Abort Task - No Empty Port (Transaction)	Description	Abort running task if no empty port. Transaction will be terminated. To be called within a transaction loading or unloading task template.	
		Outcomes	Done	Action Completed
53.	Enable All Swappings & Vias	Description	Enable all swapping & via junctions. Take note that: 1) The effect of this skill is accumulative, with the enabled/disabled states maintained until DFleet soft-reboot (apply changes). 2) Separate sets of states are kept for each AMR, i.e. the skill only affects the AMR which calls it.	
		Outcomes	Done	Action Completed
54.	Disable All Swappings & Vias	Description	Disable all swapping & via junctions. Take note that: 1) The effect of this skill is accumulative, with the enabled/disabled states maintained until DFleet soft-reboot (apply changes).	

			2) Separate sets of states are kept for each AMR, i.e. the skill only affects the AMR which calls it.	
		Outcomes	Done	Action Completed
55.	Enable Swappings / Vias	Description	Enable swapping/via for selected stations. Take note that: 1) The effect of this skill is accumulative, with the enabled/disabled states maintained until DFleet soft-reboot (apply changes). 2) Separate sets of states are kept for each AMR, i.e. the skill only affects the AMR which calls it.	
		Parameters	Swapping/Via 1	Station to be enabled swapping/via.
			Swapping/Via 2	Station to be enabled swapping/via.
			Swapping/Via 3	Station to be enabled swapping/via.
			Swapping/Via 4	Station to be enabled swapping/via.
			Swapping/Via 5	Station to be enabled swapping/via.
		Outcomes	Done	Action Completed
56.	Disable Swappings/Vias	Description	Disable swapping/via for selected stations. Take note that: 1) The effect of this skill is accumulative, with the enabled/disabled states maintained until DFleet soft-reboot (apply changes). 2) Separate sets of states are kept for each AMR, i.e. the skill only affects the AMR which calls it.	
		Parameters	Swapping/Via 1	Station to be enabled swapping/via.
			Swapping/Via 2	Station to be enabled swapping/via.
			Swapping/Via 3	Station to be enabled swapping/via.
			Swapping/Via 4	Station to be enabled swapping/via.
			Swapping/Via 5	Station to be enabled swapping/via.
		Outcomes	Done	Action Completed
57.	Check Traffic Group Reservation	Description	Check Traffic Group Reservation is an action to check whether a traffic group is reserved.	
		Parameters	Group 1	Group to be checked.
			Group 2	Group to be checked. (skipped if 0)
			Group 3	Group to be checked. (skipped if 0)
			Group 4	Group to be checked. (skipped if 0)
			Group 5	Group to be checked. (skipped if 0)

			All True	Tick to return "Reserved" only when all groups are reversed.
		Outcomes	Reserved	If All True is set, all the groups are reserved. Else, at least one of the groups is reserved.
			Unreserved	If All True is set, at least one of the groups is not reserved. Else, all groups are not reserved.
58.	Sleep	Description	Sleep is an action to perform a short delay on the AMR task template. There will be no on-screen message or indicator showing the countdown of this action.	
		Parameters	Timeout double	Sleep duration in seconds.
				Default: 1.0, min: 0, max: 9999
		Outcomes	Timeout	Action completed after the timeout
59.	Counter	Description	Counter is an action to perform counting for a register. The execution of this action will result in an increment of the value of the Register and a check on value compared to the limit. The counter will start with the initial value of the Register until the Limit. For example, if the initial value for the Register is 0, and the Limit is 3: First execution of Counter, Register value changed to 1, outcomes Counted Second execution of Counter, Register value changed to 2, outcomes Counted Third execution of Counter, Register value changed to 3, outcomes Counted Forth execution of Counter, Register value changed to 4, outcomes Limit Reached	
		Parameters	Register (DFleet Register)	Register to use the counter.
			Limit (int)	Number of counts.
				Default: 5, min: 0
		Outcomes	Counted	The register had been incremented, but value have not reached the limit
			Limit Reached	The register value reached limit
60.	Compare Registers	Description	Compare Register is an action to compare the value of 2 registers. The outcomes of this action will depend on the result of the comparison.	
		Parameters	Register 1 (DFleet Register)	Register 1 to compare
			Register 2 (DFleet Register)	Register 2 to compare
		Outcomes	Less Than	Register 1's value is less than Register 2's value
			Equal	Register 1's value is equal to Register 2's value
			Greater Than	Register 1's value is greater than Register 2's value

61.	Compare Register and Integer	Description	Compare Register and Integer is an action to perform a comparison of the value for a Register and an Integer.	
		Parameters	Register (DFleet Register)	Register for the compare
			Integer (int)	Integer value for the comparison. User can choose to insert a static number or enable Inherit to use the Params for task template
		Outcomes	Less Than	Register's value is less than the Integer's value
			Equal	Register's value is equal to the Integer's value
			Greater Than	Register's value is greater than Integer's value
62.	Compare Integers	Description	Compare Integers is an action to perform a compare of value for 2 Integers.	
		Parameters	Integer 1 (int)	The first integer for the comparison. User can choose to insert a static number or enable Inherit to use the Params for task template
			Integer 2 (int)	The second integer for the comparison. User can choose to insert a static number or enable Inherit to use the Params for task template
		Outcomes	Less Than	Integer 1's value is less than the Integer 2's value.
			Equal	Integer 1's value is equal to Integer 2's value
			Greater Than	Integer 1's value is greater than Integer 2's value
63.	Compare Doubles	Description	Compare Doubles is an action to perform a compare of value for 2 Double	
		Parameters	Double 1 double	The first double for the comparison. User can choose to insert a static double number or enable Inherit to use the Params for task template
			Double 2 double	The second double for the comparison. User can choose to insert a static double number or enable Inherit to use the Params for task template
		Outcomes	Less Than	Double 1's value is less than the Double 2's value
			Equal	Double 1's value is equal to Double 2's value
			Greater Than	Double 1's value is greater than Double 2's value
64.	Compare Stations	Description	Compare Stations is an action to perform a compare of 2 stations	
		Parameters	Station 1 Station	The first station for the comparison. User can choose from the list of available station or enable Inherit to use the Params for task template
			Station 2 Station	The second station for the comparison. Users can choose from the list of available stations or enable Inherit to use the Params for task template.
		Outcomes	Equal	Station 1 is equal to Station 2
			Not Equal	Station 1 is not equal to Station 2
65.	Compare Strings	Description	Compare Strings is an action to perform a compare for 2 string value	

		Parameters	String 1 (str)	The first string for the comparison. User can choose to insert a string or enable Inherit to use the Params for task template
			String 2 (str)	The second string for the comparison. User can choose to insert a string or enable Inherit to use the Params for task template
		Outcomes	Equal	String 1 is equal to String 2
			Not Equal	String 1 is not equal to String 2
66.	Compare Booleans	Description	Compare Booleans is an action to compare 2 Booleans values.	
		Parameters	Boolean 1 (bool)	The first boolean for the comparison. User can choose to set the boolean value or enable Inherit to use the Params for task template
			Boolean 2 (bool)	The second boolean for the comparison. User can choose to set the boolean value or enable Inherit to use the Params for task template
		Outcomes	Equal	Boolean 1's value is equal to Boolean 2's value
			Not Equal	Boolean 1's value is not equal to Boolean 2's value
67.	HTTP GET	Description	Perform HTTP GET request to target URL and compare the response with the expected response.	
		Parameters	Url (str)	Url of the GET request.
			Headers (str)	HTTP header of the request in JSON format.
			Params (str)	Parameters that will be sent as a query string in JSON format.
			Verify Ssl (bool)	Boolean to enable or disable SSL verification during execution of the skill.
			Response Verification (int)	Response Verification Method. 0 – The response matches exactly with the expected response string. 1 – The response matches with the expected response in JSON format. 2 – Partial of the response matches with the expected response in JSON format.
			Expected Response (str)	The expected response in raw or JSON format based on the response verification method.
		Outcomes	Success	The response of the HTTP GET request matches the expected response based on the verification method used.
			Response Mismatch	The response does not match the expected response based on the verification method used. Note:

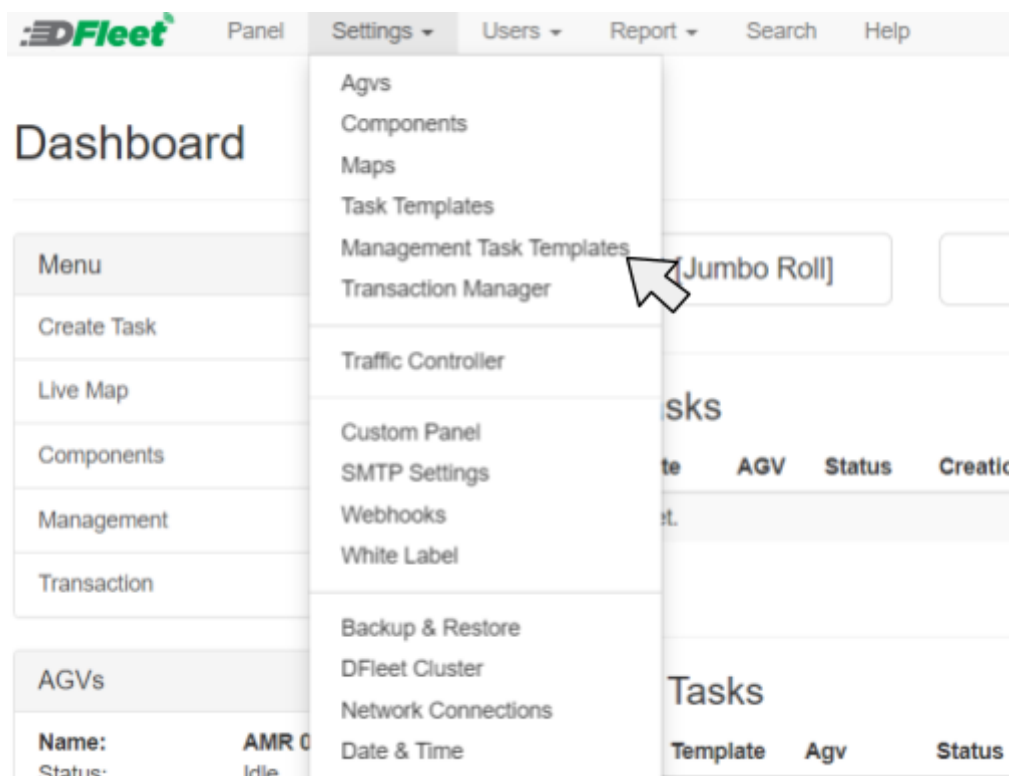
				Wrong JSON formatting of the expected response when using verification method 1 or 2 will also return this outcome.
			Error	The HTTP GET request failed.
68.	HTTP POST	Description	Perform HTTP POST request to target URL and compare the response with the expected response.	
		Parameters	Url (str)	Url of the POST request.
			Headers (str)	HTTP header of the request in JSON format.
			Params (str)	Parameters that will be sent as a query string in JSON format.
			Verify Ssl (bool)	Boolean to enable or disable SSL verification during execution of the skill.
			Data (str)	The body of the POST request in raw format.
			Response Verification (int)	Response Verification Method. 0 - The response matches exactly with the expected response string. 1 - The response matches with the expected response in JSON format. 2 - Partial of the response matches with the expected response in JSON format.
			Expected Response (str)	The expected response in raw or JSON format based on the response verification method.
		Outcomes	Success	The response of the HTTP POST request matches the expected response based on the verification method used.
			Response Mismatch	The response does not match the expected response based on the verification method used. Note: Wrong JSON formatting of the expected response when using verification method 1 or 2 will also return this outcome.
			Error	The HTTP POST request failed.
69.	HTTP PUT	Description	Perform HTTP PUT request to target URL and compare the response with the expected response.	
		Parameters	Url (str)	Url of the PUT request.
			Headers (str)	HTTP header of the request in JSON format.
			Params (str)	Parameters that will be sent as a query string in JSON format.
			Verify Ssl (bool)	Boolean to enable or disable SSL verification during execution of the skill.
			Data (str)	The body of the PUT request in raw format.
			Response Verification (int)	Response Verification Method:

				0 – The response matches exactly with the expected response string. 1 – The response matches with the expected response in JSON format. 2 – Partial of the response matches with the expected response in JSON format.
			Expected Response (str)	The expected response in raw or JSON format based on the response verification method.
		Outcomes	Success	The response of the HTTP PUT request matches the expected response based on the verification method used.
			Response Mismatch	The response does not match the expected response based on the verification method used. Note: Wrong JSON formatting of the expected response when using verification method 1 or 2 will also return this outcome.
			Error	The HTTP PUT request failed.
70.	HTTP DELETE	Description	Perform HTTP DELETE request to target URL and compare the response with the expected response.	
		Parameters	Url (str)	URL of the DELETE request.
			Headers (str)	HTTP header of the request in JSON format.
			Params (str)	Parameters that will be sent as a query string in JSON format.
			Verify Ssl (bool)	Boolean to enable or disable SSL verification during execution of the skill.
			Response Verification (int)	Response Verification Method: 0 – The response matches exactly with the expected response string. 1 – The response matches with the expected response in JSON format. 2 – Partial of the response matches with the expected response in JSON format.
			Expected Response (str)	The expected response in raw or JSON format based on the response verification method.
		Outcomes	Success	The response of the HTTP DELETE request matches the expected response based on the verification method used.
			Response Mismatch	The response does not match the expected response based on the verification method used.

				Note: Wrong JSON formatting of the expected response when using verification method 1 or 2 will also return this outcome.
			Error	The HTTP DELETE request failed.
71.	Ping	Description	Perform Ping request to the target hostname.	
		Parameters	Hostname (str)	The ping target hostname.
			Count (int)	Number of pings to be completed.
		Outcomes	Success	All ping successfully responded.
			Fail	One or more ping fails to respond.

Section 5. Management Task Templates

Management Task Templates are used to manage the AMRs when AMRs receive a trigger. In this page, a list of created management task templates will be displayed with its state (active and top-level). User is able to configure the created template by clicking the “pencil” button. Besides, the template can be cloned and deleted.



Navigate to Management Task Template page

Management Task Templates

FILTERS

Management Task Templates (1-1 of 1)

#	Name	Active	Top-level	Operation
1	Test AGV	✓	✓	✎ 🔄 ✕

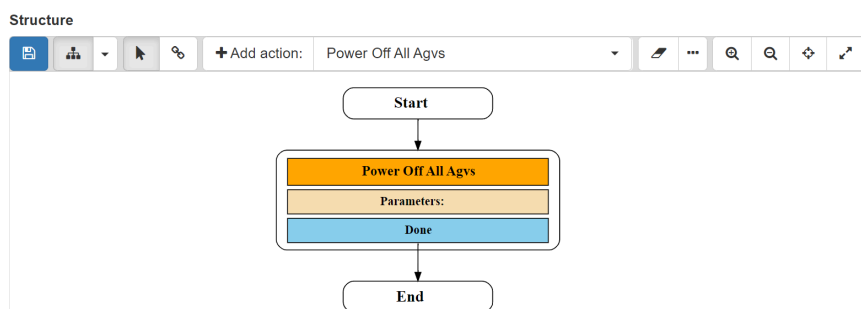
+ Add new management task template

⚡ Management task trigger

Management Task Template page

Creating Management Task Templates

By clicking the “Add new management task template” button located at the bottom of the page, users can create a new task template. The way of creating a management task template is exactly the same as the previous basic task template. The following figure shows an example on creating a management task template.



In this example, all AMR will power off when this task template is executed.

Skills Available in Management Task Templates

1.	Add Task	Description	Add a new task to the task queue. (Maximum active task is limited to 500)	
		Parameters	Note: The skill is unable to add tasks that expect parameters.	
			Task Template (TaskTemplate)	Target task template for the new task.
			Prioritize (bool)	To prioritize the task on the task queue.
2.	Add Task to Agv	Outcomes	Done	Action completed.
		Description	Add a new task with specified AMR to the task queue	
		Parameters	Note: The skill is unable to add tasks that expect parameters.	
			Task Template (TaskTemplate)	Target task template for the new task.
			Agv (Agv)	The AMR where the task is assigned to.
3.	Abort Task		Prioritize (bool)	To prioritize the task on the task queue.
		Outcomes	Done	Action completed.
		Description	This task will abort the task of the selected AMR.	
		Parameters	Agv (Agv)	AMR to abort the task.
		Outcomes	Done	Action completed.

4.	Abort All Task	Description	This task will abort all AMR tasks.	
		Parameters	None	
		Outcomes	Done	Action completed.
5.	Cancel All Task	Description	This task will cancel all running AMR task that is not in progress	
		Parameters	None	
		Outcomes	Done	Action completed
6.	Hard Reboot DFleet	Description	This action will hard reboot the DFleet system	
		Parameters	None	
		Outcomes	Done	Action completed
			Fail	Fail to execute action, trying to hard reboot DFleet within 60 seconds of up time.
7.	Power Off DFleet	Description	This action will power off the DFleet system	
		Parameters	None	
		Outcomes	Done	Action completed
			Fail	Fail to execute action, trying to power off DFleet within 60 seconds of up time.
8.	Log Message	Description	Log a message to custom log.	
		Parameters	Level (int)	Severity level. 1 - Info 2 - Warning 3 - Error
			Message (str)	The log messages.
		Outcomes	Done	Action completed.
9.	Send Email	Description	Send email to the recipients using the subject and message provided.	
		Parameters	To	The email of the recipient. Separate email addresses with a comma.
			Subject	The subject of the email message
			Message (str)	The content of the email message.
		Outcomes	Done	Action completed.
10.	Resume Agv	Description	This action will resume the operation of AMR	
		Parameters	Agv (Agv)	AMR to resume.
		Outcomes	Done	Action completed.
11.	Resume All Agvs	Description	This will resume the operation of all AMR	

		Parameters	None	
		Outcomes	Done	Action completed.
12.	Pause Agv	Description	This action will pause the operation of an AMR	
		Parameters	Agv (Agv)	AMR to pause.
		Outcomes	Done	Action completed.
13.	Pause All Agvs	Description	This action will pause the operation of all AMR.	
		Parameters	None	
		Outcomes	Done	Action completed.
14.	Start Agv	Description	This action will start the AMR	
		Parameters	Agv (Agv)	AMR to start.
		Outcomes	Done	Action completed.
15.	Start All Agvs	Description	This action will start all AMR	
		Parameters	None	
		Outcomes	Done	Action completed.
16.	Stop Agv	Description	This action will stop the AMR	
		Parameters	Agv (Agv)	AMR to stop.
		Outcomes	Done	Action completed.
17.	Stop All Agvs	Description	This action will stop all AMR	
		Parameters	None	
		Outcomes	Done	Action completed
18.	Reboot Agv	Description	This action will be able to perform hard or soft reboot for the AMR	
		Parameters		
			Mode (int)	Insert 0 for soft reboot Agv (Agv) AMR to reboot. and 1 for hard reboot
				Default: 0
		Outcomes	Done	Action completed
19.	Reboot All Agvs	Description	This action will be able to perform hard or soft reboot for all AMR	
		Parameters	Mode (int)	Insert 0 for soft reboot and 1 for hard reboot
				Default: 0
		Outcomes	Done	Action completed

20.	Power Off Agv	Description	This action will power off the AMR.	
		Parameters	Agv (Agv)	AMR to power off.
		Outcomes	Done	Action completed.
21.	Power Off All Agvs	Description	This action will power off all AMR.	
		Parameters	None	
		Outcomes	Done	Action completed.
22.	Check AGV Position	Description	Check whether the specified AGV is currently at the specified location..	
		Parameters	Agv	Agv to check position.
			Station	The station to check whether the AGV is at this location..
		Outcomes	Yes	The specified AGV is at the specified location.
			No	The specified AGV is not at the specified location.
23.	Check AGV Position Prefix	Description	Check whether the specified AGV is currently at any station that matches the given Station Prefix.	
		Parameters	Agv	Agv to check position.
			Station Prefix	The prefix to compare with the AGV's current station.
		Outcomes	Yes	The specified AGV is at a station that matches the Station Prefix.
			No	The specified AGV is not at a station that matches the Station Prefix.
24.	Check AGV Position Suffix	Description	Checks whether the specified AGV is currently at any station that matches the given Station Suffix.	
		Parameters	Agv	Agv to check position.
			Station Prefix	The suffix to compare with the AGV's current station.
		Outcomes	Yes	The specified AGV is at a station that matches the Station Suffix.
			No	The specified AGV is not at a station that matches the Station Suffix.
25.	Compare Station Distances to AGV	Description	Compare the distance from the AGV to Station 1 with the distance from the AGV to Station 2.	

		Parameters	Agv (Agv)	The AMR to compare distance from.
			Station 1	The first station to compare distance to.
			Station 2	The second station to compare distance to.
		Outcomes	Less Than	AMR is nearer to Station 1..
			Equal	The distances from the AMR to both stations are equal.
			Greater Than	AMR is nearer to Station 2..
			Unknown	The location of the AMR is unknown.
26.	Play Music	Description	Play playlist music on selected audio channels.	
		Parameters	Channel (int)	Audio Channel No. (1 - 5)
				Default: 1, Min: 1, Max: 5
			Playlist (int)	Playlist No. (0 - 10)
				Default: 0, Min: 0, Max: 10
		Outcomes	Done	Action completed.
27.	Stop Music	Description	Stop music on selected audio channels.	
		Parameters	Channel (int)	Audio Channel No. (1 - 5)
				Default: 1, min: 1, max: 5
		Outcomes	Done	Action completed.
28.	Redis Delete Key	Description	Delete one redis key value	
		Parameters	Key	Redis key to delete
		Outcomes	Done	Action complete
			Fail	Error occurred
29.	Redis Set String Item	Description	Set one redis key to the given string value	
		Parameters	Key	Redis key
			Value	String value to set
		Outcomes	Done	Action complete
			Failed	Error occurred
30.	Redis Add String to List	Description	Add string item to a redis list	
		Parameters	Key	Redis key

			Value	String value to add inside list
			List Position	There are two options: 0 - head of list -1 - tail of list
		Outcomes	Done	Action complete
			Failed	Error occurred
31.	Redis Pop List	Description	Remove one string item from a redis list	
		Parameters	Key	Redis key
			List Position	There are two options: 0 - head of list -1 - tail of list
		Outcomes	Done	Action complete
			Failed	Error occurred
32.	Redis Set Hash Field with String	Description	Set string value for one Hash field	
		Parameters	Key	Redis key
			Field	Redis hash field
			Value	String value to set
		Outcomes	Done	Action complete
			Failed	Error occurred
33.	Redis Delete Hash Field	Description	Delete one hash field and value from a redis	
		Parameters	Key	Redis key
			Field	Redis hash field
		Outcomes	Done	Action complete
			Failed	Error occurred
34.	Redis Compare String Item with String	Description	Get redis string value and compare with the given string	
		Parameters	Key	Redis key
			String	String value to be compared with
		Outcomes	Equal	String item and string is the same
			Not Equal	String item and string is not the same
			Failed	Not able to compare the string item with string. Error occurred
35.	Redis Compare Two String Items	Description	Compare two string items between two redis key	
		Parameters	Key 1	Redis key 1
			Key 2	Redis key 2

		Outcomes	Equal	Redis key 1 and redis key 2 is the same
			Not Equal	Redis key 1 and redis key 2 is not the same
			Failed	Not able to compare. Error occurred.
36.	Redis Pop List and Compare String	Description	Remove one string item from redis list and compare it with the given string	
		Parameters	Key	Redis key
			String	String value to be compared with
			List Position	There are two options: 0 - head of list -1 - tail of list
		Outcomes	Equal	String item and string are the same
			Not Equal	String item and string are not the same
			Failed	Not able to compare. Error occurred
37.	Redis Compare List Item and String	Description	Get string item from a redis list and compare it with the given string	
		Parameters	Key	Redis key
			String	String value
			Index	Zero based index of the list (Negative values start from the tail)
		Outcomes	Equal	List item and string is the same
			Not Equal	List item and string is not the same
			Failed	Not able to compare. Error occurred
38.	Redis Compare Hash Field with String	Description	Get string item from one of the hash fields and compare it with the given string	
		Parameters	Key	Redis key
			Field	Redis hash field
			String	String value to be compare with
		Outcomes	Equal	String item in hash field and string are the same
			Not Equal	String item in hash field and string are not the same
			Failed	Not able to compare.
39.	Redis Compare Two Hash Field	Description	Get two string items from two hash field and compare it with each other	
		Parameters	Key 1	Redis key 1
			Field 1	Redis hash field 1
			key 2	Redis key 2
			Field 2	Redis hash field 2

		Outcomes	Equal	Two string from two hash field are the same
			Not equal	Two string from two hash field are not the same
			Failed	Not able to compare. Error occurred
40.	Set Register Value	Description	Set Register is an action to write a value into a register.	
		Parameters	Register (DFleet Register)	Register to be written.
			Value (int)	Value to be assigned to the register
				Default: 0
		Outcomes	Done	Action completed.
41.	Compare and Set Register	Description	An atomic action to compare register and only set a new value if the current value is equal to the expected value.	
		Parameters	Register (DFleet Register)	Register to be compared with expected value and set to new value.
			Expected Value (int)	The expected register current value.
			New Value (int)	The new value to be assigned.
		Outcomes	Equal	The register value is equal to expected value and the new value has been assigned to the register
			Not Equal	The register value is not equal to the expected value.
42.	Increment Register	Description	Increment Register is an action to increment the value of a Register. After this action the value of the Register will be incremented according to the given increment value.	
		Parameters	Register (DFleet Register)	Register to be incremented.
			Value (int)	Increment value
				Default: 1
		Outcomes	Done	Action completed
43.	Decrement Register	Description	Decrement Register is an action to decrement the value of a Register. After this action the value of the Register will be decremented according to the given decrement value.	
		Parameters	Register (DFleet Register)	Register to be decrement
			Value (int)	Decrement value
				Default: 1

		Outcomes	Done	Action completed
44.	Add Register	Description	Add Register is an action to add two register values together. The result of this operation will be stored to the destination register.	
		Parameters	Destination (DFleet Register)	The register where the result will be stored.
			Source (DFleet Register)	The register where its value will be used to add to the destination register.
		Outcomes	Done	Action completed.
45.	Subtract Register	Description	Subtract Register is an action to subtract the destination register using the source register. The result of this operation will be stored in the destination register.	
		Parameters	Destination (DFleet Register)	The register where the result will be stored.
			Source (DFleet Register)	The register where its value will be used to subtract from the destination register.
		Outcomes	Done	Action completed.
46.	Multiply Register	Description	Multiply Register is an action to multiply two register values together. The result of this operation will be stored to the destination register.	
		Parameters	Destination (DFleet Register)	The register where the result will be stored.
			Source (DFleet Register)	The register where its value will be used to multiply with the destination register.
		Outcomes	Done	Action completed.
47.	Divide Register	Description	Divide Register is an action to divide the destination register with the source register. The quotient result of this operation will be stored to the destination register while the remainder will be stored to the remainder register.	
		Parameters	Destination (DFleet Register)	The register where the quotient result will be stored.
			Source (DFleet Register)	Divide the destination register with this register.
			Remainder (DFleet Register)	The register where the remainder value will be stored.
		Outcomes	Done	Action completed.
48.	Copy Register	Description	The action is to copy the register value from one to another register	

		Parameters	Destination (DFleet Register)	Destination register where the value is to be copied
			Source (DFleet Register)	Register to be copied from
		Outcomes	Done	Action completed
49.	Remote Io Read	Description	This action will read the value from remote IO	
		Parameters	Remote io (int)	Remote Io to read (1-20)
				Default: 1, min:1, max: 20
			Input (int)	Input to read (0-15)
				Default: 0, min:0, max:15
		Outcomes	True	The value of remote IO is the same as input value
			False	The value of remote IO is not the same as input value
			Failed	Error occurred
50.	Remote Io Write	Description	This action will write a value to the remote IO	
		Parameters	Remote io (int)	Remote Io to write (1-20)
				Default: 1, min: 1, max: 20
			Output (int)	Output to write (0-15)
				Default: 0, min: 0, max: 15
		Outcomes	State (bool)	State of the output
				Default: False
			Done	Action completed
51.	Remote Io Read Register	Description	Read the register value from Remote IO	
		Parameters	Remote io (int)	Remote Io to read (1-20)
				Default: 1, min: 1, max: 20
			Register (DFleet Register)	Register to read from
			Input mask (int)	Input mask
				Default: 255, min: 0, max: 65535
		Outcomes	Done	Action completed
			Failed	Error occurred
52.	Remote Io Write Register	Description	Write a register value through remote IO	

		Parameters	Remote Io (int)	Remote Io to write (1-20)
				Default: 1, min: 1, max: 20
			Register (DFleet Register)	Register to be written
			Output mask (int)	Output mask
				Default: 255, min: 0, max: 65535
		Outcomes	Done	Action completed
			Failed	Error occurred
53.	Enable All AGV Swappings & Vias	Description	Enable all swapping & via junctions. Take note that: 1) The effect of this skill is accumulative, with the enabled/disabled states maintained until DFleet soft-reboot (apply changes). 2) Separate sets of states are kept for each AMR, i.e. the skill only affects the AMR which calls it.	
		Parameter	Agv	AGV to enable all swappings and vias for.
		Outcomes	Done	Action Completed
54.	Disable All AGV Swappings & Vias	Description	Disable all swapping & via junctions. Take note that: 1) The effect of this skill is accumulative, with the enabled/disabled states maintained until DFleet soft-reboot (apply changes). 2) Separate sets of states are kept for each AMR, i.e. the skill only affects the AMR which calls it.	
		Parameter	Agv	AGV to disable all swappings and vias for.
		Outcomes	Done	Action Completed
55.	Enable AGV Swappings & Vias	Description	Enable swapping/via for selected stations for the specified AGV. Take note that: 1) The effect of this skill is accumulative, with the enabled/disabled states maintained until DFleet soft-reboot (apply changes).	
		Parameter	Agv	AGV to enable swappings and vias for.
			Swapping/Via 1	Station to be enabled swapping/via.
			Swapping/Via 2	Station to be enabled swapping/via.
			Swapping/Via 3	Station to be enabled swapping/via.
			Swapping/Via 4	Station to be enabled swapping/via.
			Swapping/Via 5	Station to be enabled swapping/via.
		Outcomes	Done	Action Completed
56.	Disable AGV Swappings & Vias	Description	Disable swapping/via for selected stations for the specified AGV. Take note that: 1) The effect of this skill is accumulative, with the enabled/disabled states maintained until DFleet soft-reboot (apply changes).	
		Parameter	Agv	AGV to disable swappings and vias for.

			Swapping/Via 1	Station to be disabled swapping/via.
			Swapping/Via 2	Station to be disabled swapping/via.
			Swapping/Via 3	Station to be disabled swapping/via.
			Swapping/Via 4	Station to be disabled swapping/via.
			Swapping/Via 5	Station to be disabled swapping/via.
		Outcomes	Done	Action Completed
57.	Check Traffic Group Reservation	Description	Check Traffic Group Reservation is an action to check whether a traffic group is reserved.	
		Parameter	Group 1	Group to be checked.
			Group 2	Group to be checked. (Skipped if 0)
			Group 3	Group to be checked. (Skipped if 0)
			Group 4	Group to be checked. (Skipped if 0)
			Group 5	Group to be checked. (Skipped if 0)
			All True	Tick to return "Reserved" only when all groups are reversed.
		Outcomes	Reserved	If All True is set, all the groups are reserved. Else, at least one of the groups is reserved.
			Unreserved	If All True is set, at least one of the groups is not reserved. Else, all groups are not reserved.
58.	Power Off UPS	Description	Power down UPS into hibernate mode.	
		Parameters	None	
		Outcomes	Done	Action completed.
			Fail	Fail to power down UPS.
59.	Sleep	Description	Sleep is an action to perform a short delay on the AMR task template. There will be no on-screen message or indicator showing the countdown of this action.	
		Parameters	Timeout (double)	Sleep duration in seconds
				Default: 1.0, min: 0, max: 9999
		Outcomes	Timeout	Action completed after the timeout
60.	Counter	Description	Counter is an action to perform counting for a register. The execution of this action will result in an increment of the value of the Register and a check on value compared to the limit. The counter will start with the initial value of the Register until the Limit. For example, if the initial value for the Register is 0, and the Limit is 3: First execution of Counter, Register value changed to 1, outcomes Counted	

			Second execution of Counter, Register value changed to 2, outcomes Counted Third execution of Counter, Register value changed to 3, outcomes Counted Forth execution of Counter, Register value changed to 4, outcomes Limit Reached	
		Parameters	Register (DFleet Register)	Register to use the counter.
			Limit (int)	Number of counts.
				Default: 5, min: 0
		Outcomes	Counted	The register had been incremented, but value have not reached the limit
			Limit Reached	The register value reached limit
61.	Compare Registers	Description	Compare Register is an action to compare the value of 2 registers. The outcomes of this action will depend on the result of the comparison.	
		Parameters	Register 1 (DFleet Register)	Register 1 to compare
			Register 2 (DFleet Register)	Register 2 to compare
		Outcomes	Less Than	Register 1's value is less than Register 2's value
			Equal	Register 1's value is equal to Register 2's value
			Greater Than	Register 1's value is greater than Register 2's value
62.	Compare Register and Integer	Description	Compare Register and Integer is an action to perform a compare of the value for a Register and an Integer	
		Parameters	Register (DFleet Register)	Register for the comparison
			Integer (int)	Integer value for the comparison. User can choose to insert a static number or enable Inherit to use the Params for task template
		Outcomes	Less Than	Register's value is less than the Integer's value
			Equal	Register's value is equal to the Integer's value
			Greater Than	Register's value is greater than Integer's value
63.	Compare Integers	Description:	Compare Integers is an action to perform a compare of value for 2 Integer	
		Parameters	Integer 1 (int)	The first integer for the comparison. User can choose to insert a static number or enable Inherit to use the Params for task template

			Integer 2 (int)	The second integer for the comparison. User can choose to insert a static number or enable Inherit to use the Params for task template
		Outcomes	Less Than	Integer 1's value is less than the Integer 2's value
			Equal	Integer 1's value is equal to Integer 2's value
			Greater Than	Integer 1's value is greater than Integer 2's value
64.	Compare Doubles	Description	Compare Doubles is an action to perform a compare of value for 2 Double	
		Parameters	Double 1 (double)	The first double for the comparison. User can choose to insert a static double number or enable Inherit to use the Params for task template
			Double 2 (double)	The second double for the comparison. User can choose to insert a static double number or enable Inherit to use the Params for task template
		Outcomes	Less Than	Double 1's value is less than the Double 2's value
			Equal	Double 1's value is equal to Double 2's value
			Greater Than	Double 1's value is greater than Double 2's value
65.	Compare Stations	Description	Compare Stations is an action to perform a comparison of 2 stations.	
		Parameters	Station 1 (Station)	The first station for the comparison. User can choose from the list of available station or enable Inherit to use the Params for task template
			Station 2 (Station)	The second station for the comparison. User can choose from the list of available station or enable Inherit to use the Params for task template
		Outcomes	Equal	Station 1 is equal to Station 2
			Not Equal	Station 1 is not equal to Station 2
66.	Compare Strings	Description	Compare Strings is an action to perform a compare for 2 string values.	
		Parameters	String 1 (str)	The first string for comparison. User can choose to insert a string or enable Inherit to use the Params for task template
			String 2 (str)	The second string for comparison. User can choose to insert a string or enable Inherit to use the Params for task template
		Outcomes	Equal	String 1 is equal to String 2

			Not Equal	String 1 is not equal to String 2
67.	Compare Booleans	Description	Compare Booleans is an action to compare 2 Booleans value	
		Parameters	Boolean 1 (bool)	The first boolean for the comparison. User can choose to set the boolean value or enable Inherit to use the Params for task template
			Boolean 2 (bool)	The second boolean for the comparison. User can choose to set the boolean value or enable Inherit to use the Params for task template
		Outcomes	Equal	Boolean 1's value is equal to Boolean 2's value
			Not Equal	Boolean 1's value is not equal to Boolean 2's value
68.	Compare Agvs	Description	Compare Agvs is an action to check if the two AMR is the same AMR.	
		Parameters	Agv 1 (Agv)	The first AMR to be compared with the other AMR.
			Agv 2 (Agv)	The second AMR to be compared with the other AMR.
		Outcomes	Equal	The first AMR is equal to the second AMR.
			Not Equal	The first AMR does not equal the second AMR.
69.	Compare Agv Status and String	Description	Compare AMR current status with a string (or a list of strings separated by comma). The comparison is case insensitive.	
		Parameters	Agv (Agv)	The target AMR for its status to be compared with the string.
			String (str)	The string where it will be compared to the AMR status. (case-insensitive)
		Outcomes	Equal	The AMR status is equal to the string provided.
			Not Equal	The AMR status does not match the string provided.
70.	HTTP GET	Description	Perform HTTP GET request to target URL and compare the response with the expected response.	
		Parameters	Url (str)	Url of the GET request.
			Headers (str)	HTTP header of the request in JSON format.
			Params (str)	Parameters that will be sent as a query string in JSON format.
			Response Verification (int)	Response Verification Method: 0 - The response matches exactly with the expected response string.

				1 – The response matches with the expected response in JSON format. 2 – Partial of the response matches with the expected response in JSON format.
			Expected Response (str)	The expected response in raw or JSON format based on the response verification method.
		Outcomes	Success	The response of the HTTP GET request matches the expected response based on the verification method used.
			Response Mismatch	The response does not match the expected response based on the verification method used. Note: Wrong JSON formatting of the expected response when using verification method 1 or 2 will also return this outcome.
			Error	The HTTP GET request failed.
71.	HTTP POST	Description	Perform HTTP POST request to target URL and compare the response with the expected response.	
		Parameters	Url (str)	Url of the POST request.
			Headers (str)	HTTP header of the request in JSON format.
			Params (str)	Parameters that will be sent as a query string in JSON format.
			Data (str)	The body of the POST request in raw format.
			Response Verification (int)	Response Verification Method. 0 – The response matches exactly with the expected response string. 1 – The response matches with the expected response in JSON format. 2 – Partial of the response matches with the expected response in JSON format.
			Expected Response (str)	The expected response in raw or JSON format based on the response verification method.
		Outcomes	Success	The response of the HTTP POST request matches the expected response based on the verification method used.

			Response Mismatch	The response does not match the expected response based on the verification method used. Note: Wrong JSON formatting of the expected response when using verification method 1 or 2 will also return this outcome.
			Error	The HTTP POST request failed.
72.	HTTP PUT	Description	Perform HTTP PUT request to target URL and compare the response with the expected response.	
		Parameters	Url (str)	Url of the PUT request.
			Headers (str)	HTTP header of the request in JSON format.
			Params (str)	Parameters that will be sent as a query string in JSON format.
			Data (str)	The body of the PUT request in raw format.
			Response Verification (int)	Response Verification Method: 0 – The response matches exactly with the expected response string. 1 – The response matches with the expected response in JSON format. 2 – Partial of the response matches with the expected response in JSON format.
			Expected Response (str)	The expected response in raw or JSON format based on the response verification method.
		Outcomes	Success	The response of the HTTP PUT request matches the expected response based on the verification method used.
			Response Mismatch	The response does not match the expected response based on the verification method used. Note: Wrong JSON formatting of the expected response when using verification method 1 or 2 will also return this outcome.
			Error	The HTTP PUT request failed.
73.	HTTP DELETE	Description	Perform HTTP DELETE request to target URL and compare the response with the expected response.	
		Parameters	Url (str)	URL of the DELETE request.
			Headers (str)	HTTP header of the request in JSON format.

			Params (str)	Parameters that will be sent as a query string in JSON format.
			Response Verification (int)	Response Verification Method: 0 – The response matches exactly with the expected response string. 1 – The response matches with the expected response in JSON format. 2 – Partial of the response matches with the expected response in JSON format.
			Expected Response (str)	The expected response in raw or JSON format based on the response verification method.
		Outcomes	Success	The response of the HTTP DELETE request matches the expected response based on the verification method used.
			Response Mismatch	The response does not match the expected response based on the verification method used. Note: Wrong JSON formatting of the expected response when using verification method 1 or 2 will also return this outcome.
			Error	The HTTP DELETE request failed.
74.	Ping	Description	Perform Ping request to the target hostname.	
		Parameters	Hostname (str)	The ping target hostname.
			Count (int)	Number of pings to be completed.
		Outcomes	Success	All ping successfully responded.
			Fail	One or more ping fails to respond.
75.	Power Off Remote Host	Description	Power Off Remote Host using SSH protocol.	
		Parameters	IP Address (str)	IP Address of the remote host.
			Username (str)	Username of the remote host.
			Password (str)	Password of the username.
		Outcomes	Done	Action completed.
			Fail	Fail to power off remote host due to trying to send power off command within 60 seconds of up time or fail to communicate with the remote host.



Creating Management Task Trigger

By clicking the button with the lightning symbol, the user will be able to add and control task triggers.

Management Task Templates (1-1 of 1)

#	Name	Active	Top-level	Operation
1	Test AGV	✓	✓	  

+ Add new management task template

⚡ Management task trigger 

There are 9 types of trigger:

Type

Executor Start

- Executor Start
- Agv Idle
- Agv Fleet Safety
- Agv Safety
- Remote IO Pin
- Remote IO
- UPS Status
- UPS Battery Charge Capacity
- Custom Log

When a trigger is received by the AMR, the management task template will be executed. Each trigger type has its own condition before executing the task.

Trigger Type	Trigger Condition
Executor Start	DFleet executor start.
Agv Idle	AMR is idle.
Agv Fleet Safety	Any or all AMR have safety messages.
Agv Safety	AMR has a safety message.
Remote I/O Pin	Remote I/O input pin change.
Remote I/O	Remote I/O binary input change.
UPS Status	Trigger task when the status of UPS changes.
UPS Battery Charge Capacity	Trigger task when the battery percentage reaches a specific level.
Custom Log	Trigger task when log message contains the search text.

Users can also configure the specific mode of trigger for each Task Trigger.

Trigger Type	Trigger Mode	
Executor Start	When DFleet starts running.	
AGV Idle	When AMR is idle.	
AGV Fleet Safety	-1	Once safety was triggered.
	0	Both states.
	1	Once recovered from the safety trigger.
AGV Safety	-1	Once safety was triggered.
	0	Both states.
	1	Once recovered from the safety trigger.
Remote I/O Pin	-1	On at any input pin state.
	0	On when the input pin is High.
	1	On when the input pin is Low.
Remote I/O	-1	On any state change of input binary.
	N _{trig}	On when input binary is equal to N _{trig} . Can apply a state mask to trigger. If masking is applied, the task would only trigger after checking the value of the specified bit.
UPS Status	-1	COMLOST
	0	ONBATT
	1	ONLINE
UPS Battery Charge Capacity	Set a value for minimum battery capacity before trigger	
Custom Log	1	INFO
	2	WARNING
	3	ERROR

Every time a task is triggered, a certain type of trigger would also provide a certain value of params to be exported into the triggered task template.

Trigger Type	Params Export	
Executor Start	-	
AGV Idle	Idle AMR (Agy)	AMR that is in idle mode
AGV Fleet Safety	Safety Count (Int)	Number of AMR that have safety message

	Safety Status (Bool)	True if any or all AMRs have safety messages.
	Navigation Failed (Bool)	True if any or all AMRs are triggered by navigation failure.
	Out of Line (Bool)	True if any or all AMRs are triggered by out of line error.
	Bumper Block (Bool)	True if any or all AMRs are triggered by bumper blocks.
	External Safety Trigger (Bool)	True if any or all AMRs are triggered by external safety triggers.
	Emergency Button Pressed (Bool)	True if any or all AMRs are triggered by an emergency button pressed.
	Charger Connected (Bool)	True if any or all AMRs are triggered by a charger connected.
	Laser Malfunction (Bool)	True if any or all AMRs are triggered by laser malfunction.
	Line Sensor Com Error (Bool)	True if any or all AMRs are triggered by line sensor communication error.
	Drive Overlimit Error (Bool)	True if any or all AMRs are triggered by drive overlimit error.
	Obstacle Block (Bool)	True if any or all AMRs are triggered by an obstacle block.
	Resume Pending (Bool)	True if any or all AMRs are waiting for manual safety resume.
AGV Safety	Safety AGV (Agv)	AMR that has a safety message.
	Safety Status (Bool)	True if AMRs are safety triggers.
	Navigation Failed (Bool)	True if AMRs are triggered by navigation failure.
	Out of Line (Bool)	True if AMRs are triggered by out of line error.
	Bumper Block (Bool)	True if AMRs are triggered by bumper blocks.
	External Safety Trigger (Bool)	True if AMRs are triggered by external safety triggers.
	Emergency Button Pressed (Bool)	True if AMRs are triggered by an emergency button pressed.
	Charger Connected (Bool)	True if AGVs are triggered by a charger connected.
	Laser Malfunction (Bool)	True if AMRs are triggered by laser malfunction.
	Line Sensor Com Error (Bool)	True if AMRs are triggered by line sensor communication error.
	Drive Overlimit Error (Bool)	True if AMRs are triggered by drive overlimit error.
	Obstacle Block (Bool)	True if AMRs are triggered by obstacle blocks.
	Resume Pending (Bool)	True if any or all AMRs are waiting for manual safety resume.
Remote I/O Pin	State (Bool)	Current state of remote I/O.
Remote I/O	State (Bool)	Current state of remote I/O.
UPS Status	-	
UPS Battery Charge Capacity	-	
Custom Log	-	

Below is an example of a management task trigger.

Name	Type	Configuration	Management Task Template	Params	Operation
TestTrigger	Executor Start ▾	-	Initialization ▾	-	↑ ↓ 📄 ✕
+ Add trigger					

Submit

Section 6. Transaction Manager

Transaction Manager

☐ Enable Transaction Manager

Task Templates	Type	Task Template	Parameters	AGVs
Goto Task Template		GOTO	station \${station}	(default) +
Loading Task Template		LOADING	carrier_id \${carrier_id} station_port \${source_port} destination_port \${destination_port}	(default) +
Unloading Task Template		UNLOADING	carrier_id \${carrier_id} station_port \${source_port} destination_port \${destination_port}	(default) +

Ports Settings

Agv Port Count	Port Count	Agvs
	4	(default) +
Number of port available on Agv. default: 1, range: [1, 100]		

Agv Backup Port Count	Backup Port Count	Agvs
	2	(default) +
Backup port available on Agv. Will no be used by the planner. default: 0, range: [0, 100]		

Ports Settings

Agv Port Count	Port Count	Agvs
	4	(default) +
Number of port available on Agv. default: 1, range: [1, 100]		

Agv Backup Port Count	Backup Port Count	Agvs
	2	(default) +
Backup port available on Agv. Will no be used by the planner. default: 0, range: [0, 100]		

Station Ports

Sorting Sequence : Loading Unloading								
#	Port Name	Port Location	Port Group	Allowed AGVs	Capable AGVs	Port Properties	Transfer Properties	Operation
1	LP1	LP1	0	(All AGVs)	(All AGVs)	☐ Transfer Content ☐ Priority Port	Transfer Timeout: 120 sec Blocking Station: None	↑ ↓ ✕
+ Add port								

#	Group Number	Loading Priority	Unloading Priority	Operation
1	1	5	0	↑ ↓ ×
+ Add group				

#	Carrier ID Prefix	Allowed AGVs	Ports Combination CSV	Operation
+ Add rule				

Planner Settings

Detour Distance
Allowed additional detour distance for each transaction in meter. (set 0 to disable)

Carrier ID Duplicate Action
Action taken when scanner read different carrier id then expected.

Carrier ID Read Failure Action
Action taken when scanner fail to read the expected carrier id.

Carrier Handoff Failure Action
Action taken when carrier handoff failure triggered.

No.	Component	Description
1.	Enable Transaction Manager	Checkbox to enable Transaction Manager. Without enabling this, any transaction added will not be executed by the transaction component.
2.	Goto Task template	Description Task template used to move AMR to a station. The task template used here is expected to inherit the "station" provided in order to instruct the AMR to move the target station.
		Provides station Target station. Type: Station
3.	Loading Task template	Description Task template used for AMR to load a carrier. The task template used here is expected to verify that the given carrier id is at its source port. Then, the task template is expected to perform carrier id scanning and acknowledge the carrier is at its expected source location by executing the "Carrier Scanned at Port" Task Template action. This action will register the carrier on the carrier tables. If any error occurs, the task template is expected to abort the task.
		Provides carrier_id Carrier id to load. Type: String
		source_port Source port to unload from. Type: String
		destination_port Destination port to load into. (AMR Port) Type: String
		destination_port_number AMR port number for the destination port. Type: Integer

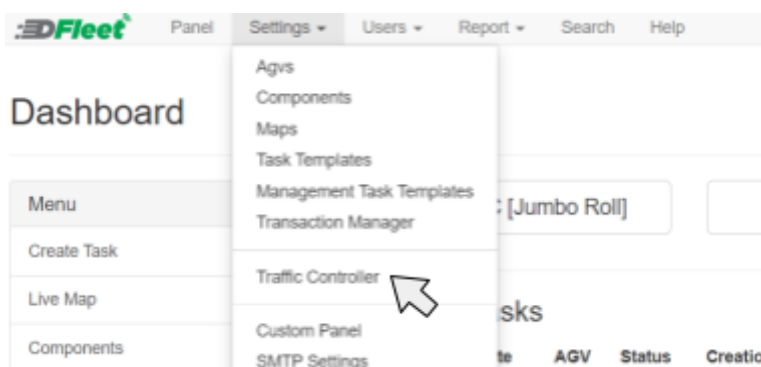
			final_destination_port	Final destination port of this transaction. Type: String
			station	Station of the source port. Type: Station
			transfer_content	Flag indicates for transfer content only. Type: Boolean
			continuous	Flag indicating the next action will be loading/unloading from the same port group. Type: Boolean
			skip_scan	Flag indicating the loading should skip carrier id scanning. Type: Boolean
			continuous_carrier	Flag indicating the next action will be loading/unloading for the same carrier. Type: Boolean
4.	Unloading template	Task	Description	Task template used for AGV to unload a carrier. If any error occurs, the task template is expected to abort the task.
			Provides	
			carrier_id	Carrier id to unload. Type: String
			source_port	Source port to unload from. (AMR Port) Type: String
			source_port_number	AMR port number for the source port. Type: String
			destination_port	Destination port to load into. Type: Integer
			station	Station of the destination port. Type: Station
			transfer_content	Flag indicates for transfer content only. Setting this to true indicates that the content of the carrier is going to be transferred here instead of the carrier itself. Type: Boolean
			continuous	Flag indicating the next action will be loading/unloading from the same port

			group. Type: Boolean	
			skip_scan	Flag indicating the loading should skip carrier id scanning. Type: Boolean
			continuous_carrier	Flag indicating the next action will be loading/unloading for the same carrier. Type: Boolean
5.	Agv Port Count	Description	Number of ports available on AMR. This also means maximum how many carriers that every AMR can carry. Default: 1 Range: [1, 100]	
6.	Agv Backup Port Count	Description	Backup port available on AMR. Will not be used by the planner. Default: 0 Range: [0, 100]	
7.	Station Ports	Description	Configuration table to define Name, Station, Port Group, Allowed AGVs and Transfer Content for the station ports.	
		Parameters	Port Name	Name of the port.
			Port Location	Port station location. (one station may have multiple ports)
			Port Group	The port group to which this port belongs.
			Allowed AGVs	Allowed AMRs for this station port.
			Capable AGVs	Capable AMRs for this station port.
			Port Properties (Transfer Content)	Properties to indicate that the content of the carrier is going to be transferred instead of the carrier itself.
			Port Properties (Priority Port)	Set the port as a priority port.
			Transfer Properties (Transfer Timeout)	The maximum allowed duration for a loading or unloading transfer at a port.
			Transfer Properties (Blocking Station)	A list of other stations whose ports must also be blocked when this port times out.
8.	Loading	Description	Click to view and arrange the priority of port stations for loading tasks. If the station ports are in the same Port Group, then the loading priority is based on their sorting sequence in this table.	
9.	Unloading	Description	Click to view and arrange the priority of port stations for unloading tasks. If the station ports are in the same Port Group, then the unloading priority is based on their sorting sequence in this table.	
10.	Reset Unloading Sorting Config 	Description	Click to reset the unloading sorting configuration to match the loading sorting configuration.	

11.	AGV-Specific Overrides +	Description	Click to add AGV-specific task templates for Goto, Loading, or Unloading tasks, as well as overrides for Port Count, Backup Port Count and Port Location.	
12.	Add alt port name +	Description	Click to add an alternate port name for the station port.	
13.	Port Groups Attribute	Description	Configuration table to define port group numbers and their respective loading and unloading priorities.	
		Parameters	Group Number	Referring to the Port Group defined in the Station Ports configuration.
			Loading Priority	Loading Priority for the respective port group. Port groups with higher loading priority will be loaded first.
			Unloading Priority	Unloading Priority for the respective port group. Port groups with higher unloading priority will be unloaded first.
14.	Multi Port Rule	Description	Configuration Table for Carrier that will reserve multiple ports at the same time.	
		Parameters	Carrier ID Prefix	Prefix for the Carrier ID that requires this type of port handling.
			Allowed AGVs	Allowed AMRs to transfer the carrier.
			Ports Combination CSV	Ports number combination in a comma separated format. The first number will be used as the carrier port location.
15.	Detour Distance	Description	Allowed additional detour distance for each transaction in meter. (set 0 to disable)	
16.	Carrier ID Duplicate Action	Description	Action taken when the scanner reads a different carrier id than expected. Options: Reject: Abort the transaction.	
17.	Carrier ID Read Failure Action	Description	Action taken when the scanner fails to read the expected carrier id. Options: - Remove: Abort the transaction. - Continue: Register transaction with prefix "UNKNOWN_" and skip future scanning of the carrier.	
18.	Carrier Handoff Failure Action	Description	Action taken (Abort / Pause) when carrier handoff failure triggered.	

Section 7. Traffic Controller

Traffic Controller can be accessed by clicking **Settings > Traffic Controller**.



Navigating to Traffic Controller

Traffic Controller

Maximum buffer count

The maximum distance and junction count (n) that can be approved and sent to an AGV (in the navigation plan) ahead of the AGV's current position. Increase this value to reduce intermittent stop due to poor wifi condition, but doing so makes AGV reserve longer path and reduces path sharing efficiency.

⇒ **distance*** m
default: 15, range: [0.01, 1000000.0]

⇒ **junction count***
default: 5, range: [1, 10000]

Trailing mode

The separation to be maintained by one AGV moving behind or intersecting with the path of another AGV.
When two AGVs are one junction apart:

- in *Loose* mode, the rear AGV will start to move to the front AGV only when the front AGV reaches the next junction;
- in *Tight* mode, the rear AGV will start to move to the front AGV once the front AGV has departed its current junction.

⇒ **trailing mode***
default: Loose

Alternative Path Constraints

Configuration for preventing AMR going too far for swapping, via and intermediate junctions.

⇒ **maximum alternative distance***
default: 10000, range: [1, 1000000.0]

☒ **alternative search**
default: True

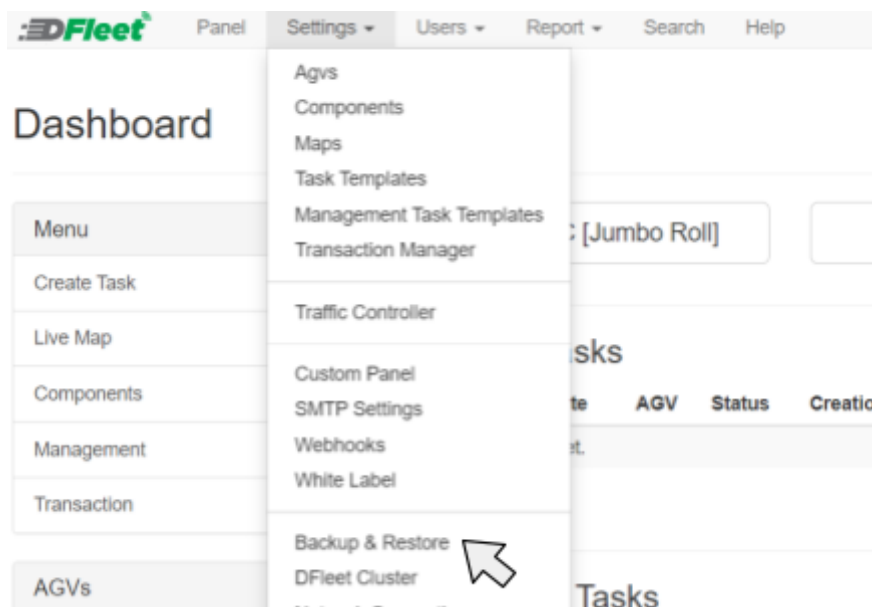
Traffic Controller page

The Traffic Controller offers 3 settings for users to fine-tune path sharing efficiency and traffic behavior.

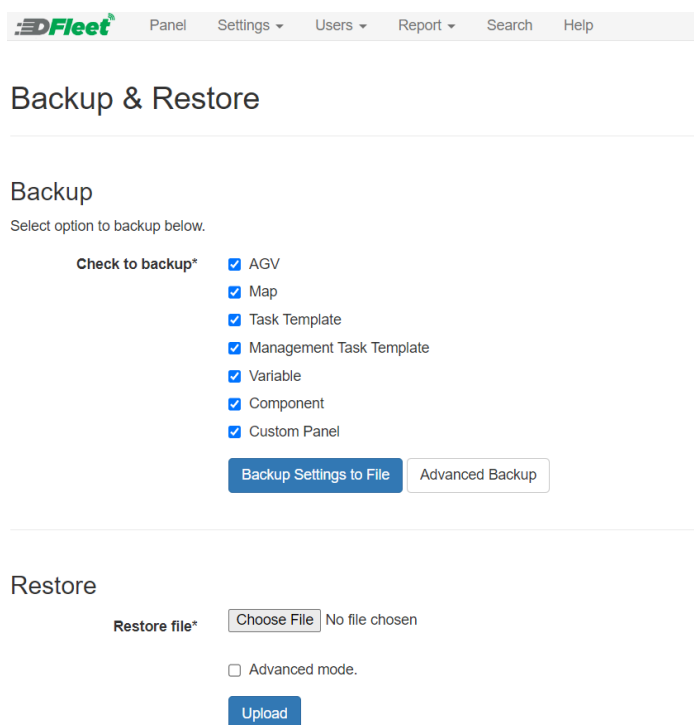
1.	Maximum Buffer Count	Description	The maximum distance and junction count (∩) that can be approved and sent to an AGV (in the navigation plan) ahead of the AMR's current position. Increase this value to reduce intermittent stop due to poor wifi condition, but doing so makes AMR reserve a longer path and reduces path sharing efficiency.	
		Parameters	Distance (int)	Maximum distance (in meter) that can be approved and sent to an AMR ahead
				Default: 15
				Min: 0.01
				Max: 1000000.0
			Junction Count (int)	Maximum junction count that can be approved and sent to an AMR ahead
				Default: 5
				Min: 1
				Max: 10000
2.	Trailing Mode	Description	The separation to be maintained by one AGV moving behind or intersecting with the path of another AGV. When two AGVs are one junction apart: - in Loose mode, the rear AGV will start to move to the front AGV only when the front AGV reaches the next junction; - in Tight mode, the rear AGV will start to move to the front AGV once the front AGV has departed its current junction.	
		Parameters	Trailing Mode	The trailing behaviour when two AGVs are one junction apart.
				Default: Loose
3.	Alternative Path Constraints	Description	Configuration for preventing AMR going too far for swapping, via and intermediate junctions.	
		Parameters	Maximum Alternative Distance (int)	Maximum distance (in meter) that can be allowed for alternative paths.
				Default: 10000
				Min: 1
				Max: 1000000.0
		Parameters	Alternative Search (bool)	Check to allow AMR to search for alternative paths.
				Default: True

Section 8. Backup & Restore

Backup and Restore can only be performed by Supervisor account. It is advisable to perform a backup before editing any setting in DFleet.



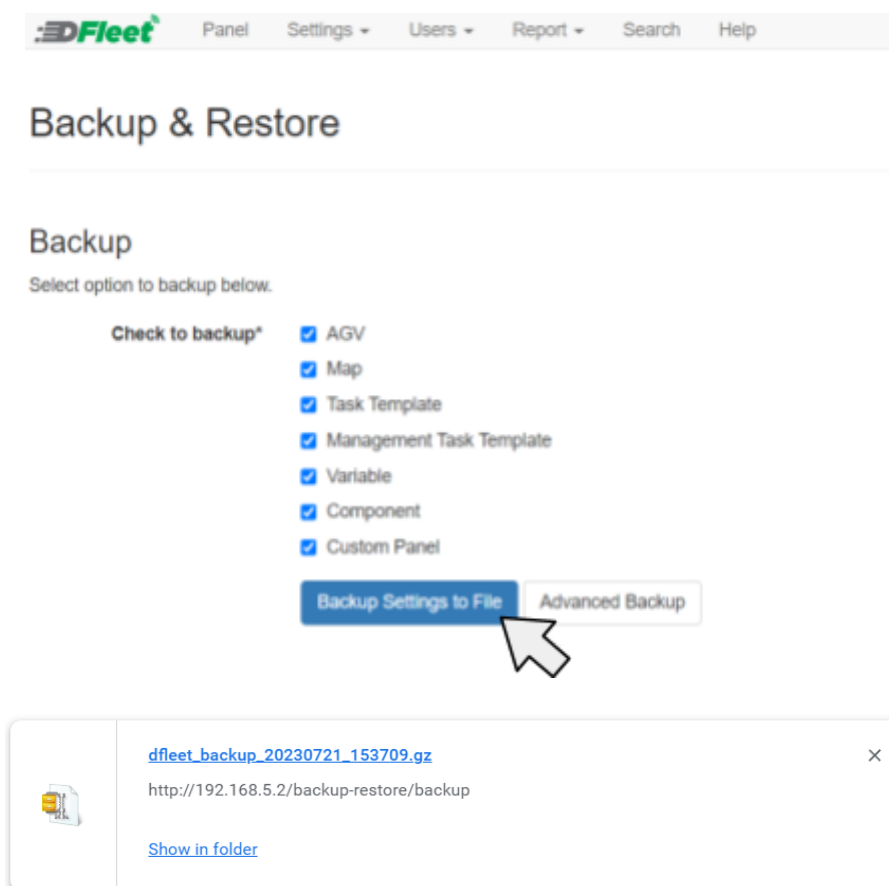
Navigating to Backup and Restore page



Backup and Restore page

Backup

To perform a backup, simply select the components you want to back up, and then click on the "Backup Settings to File" button. The backup file will be automatically downloaded to your PC.



The backup file will have the following naming convention: "dfleet_backup_20230721_153709.gz". In this file name, "20230721" represents the date of the backup, and "153709" represents the time of the backup. This naming convention allows for easy identification of the backup's creation date and time.



Info:

In the normal backup mode, inactive task templates and maps will not be included in the backup. If you want to back up inactive maps and task templates, you should use the Advanced Backup option.

Advanced Backup

Selecting Advanced Backup allows for a more defined selection of the components to be downloaded.

Advanced Backup

Select option to backup below.

Check to backup*

☐ All

☐ ☒ Agv

☐ ☒ Component

☐ ☒ Custom Panel

☐ ☐ Map

☐ ☒ Active Map

☐ ☐ Inactive Map

☐ ☒ Map Param

☐ ☒ Mgmt Task Template

☐ ☒ Mgmt Task Template

☐ ☒ Task Template

☐ ☒ Global Param

☐ ☒ Task Template

☐ ☒ Variable

☒ agv_assignment_policy

☒ skillset

☒ smtp

☒ traffic_controller

☒ transaction_manager

Backup Settings to File

Restore

To restore a backup file, click on the “Choose File” button to select a backup file.

Restore

Restore file* No file chosen

☐ Advanced mode.

Pressing the Upload button will upload the backup file to DFleet and users will be served with the option to select backup data to restore.

Restore (version: 2.2.0-src)

Select available option to restore.

Check to restore*

- ☒ AGV
- ☒ Map
- ☒ Task Template
- ☒ Management Task Template
- ☒ Variable
- ☒ Component
- ☒ Custom Panel

Advanced Restore

Alternatively, by ticking the Advance mode before uploading in the restore panel as shown in Figure below, users will be able to perform more advanced restore options. This mode allows the user to select desired data under the component. It will automatically select data that is dependent on user selected data.

Restore (version: 2.2.0-src)

Select available option to restore.

Mode* Restore (clear local II)

Check to restore*

- ☒ All
- ☒ Agv
- ☒ Component
- ☒ Custom Panel
- ☒ Map
 - ☒ Active Map
 - ☒ Map Param
 - ☒ Teleport
 - ☒ Transition Trigger
- ☒ Mgmt Task Template
- ☒ Task Template
 - ☒ Global Param
 - ☒ Task Template
 - ☒ Variable
- ☒ Variable

Restore

After clicking the Restore button, all the selected data will replace local data. Do handle it with care to avoid replacing important Map or Task Template configuration. Moreover, advanced restore options provide additional modes that allow users to merge local data with the uploaded data. Table below shows available restore mode options with its description.

Mode	Description
Restore	Clean restore mode. Will clear the local component before restoring the selected component's data. This mode behaves similarly to normal mode with the option to select specific data inside the component.
Import (conflict: keep local)	Import mode will keep local conflict resolution. Import mode does not clear local components. It will skip restoring selected data where data with the same id exists locally. Other imported data that depend with this data will update accordingly
Import (conflict: overwrite local)	Import mode with overwrite local conflict resolution. Import mode does not clear local components. Imported data will replace local data if it has similar id.
Import (conflict: keep both)	Import mode will keep both conflict resolution. Import mode does not clear local components. If Imported data id matches with local data, it will be renamed. Other imported data that depend on this data will depend on the new renamed data.

Section 9. DFleet Cluster

DFleet Cluster page provides a platform for users to configure clusters, which are essential for achieving high availability. Clusters are composed of multiple DFleet servers working together to ensure continuous operations. Within a cluster, one server acts as the primary node, taking the lead in managing the cluster's operations. In the event that the primary server experiences an issue or goes offline, another server within the cluster seamlessly takes over to maintain uninterrupted functionality.

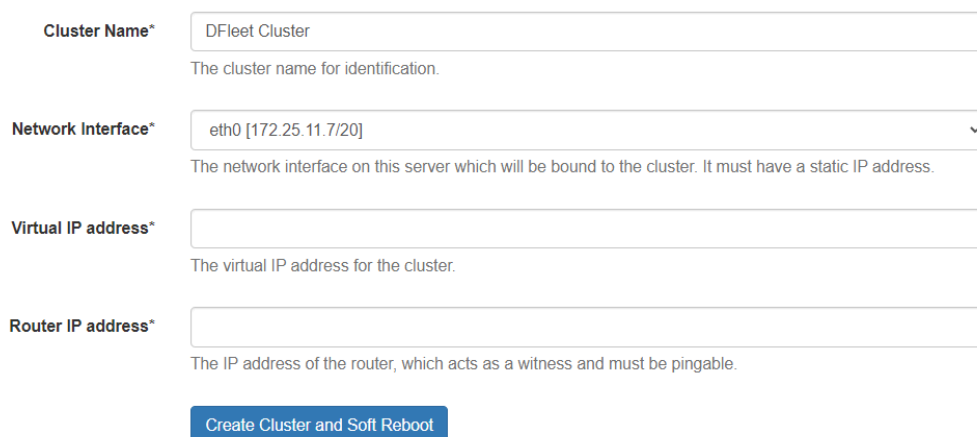
The primary purpose of clustering is to enhance availability and reliability. When data is written within a cluster, it is distributed to all the nodes within the cluster. The write operation requires acknowledgment from at least one replica node to be considered successful. This synchronous data replication approach ensures that the data remains consistent across all nodes, minimizing the risk of data inconsistencies or conflicts.

There is also a time synchronization process between cluster nodes to ensure consistency in timestamps. Additionally, there is an alert feature available for cluster failure, which can be configured through SMTP settings.

On this page, users have the option to create a new cluster or join an existing cluster.



Create a New Cluster





Join an Existing Cluster

Network Interface* 

The network interface on this server which will be bound to the cluster. It must have a static IP address.

Cluster Endpoint*

Specify cluster endpoint URL, which can be obtained from an existing cluster.

Cluster Token*

Specify cluster token, which can be obtained from an existing cluster.

Warning: By joining an existing cluster, all data on this server will be deleted. Please make sure all necessary backup has been done.

[Join Cluster, Clear Data and Soft Reboot](#)

Through this configuration page, users are also allowed to:

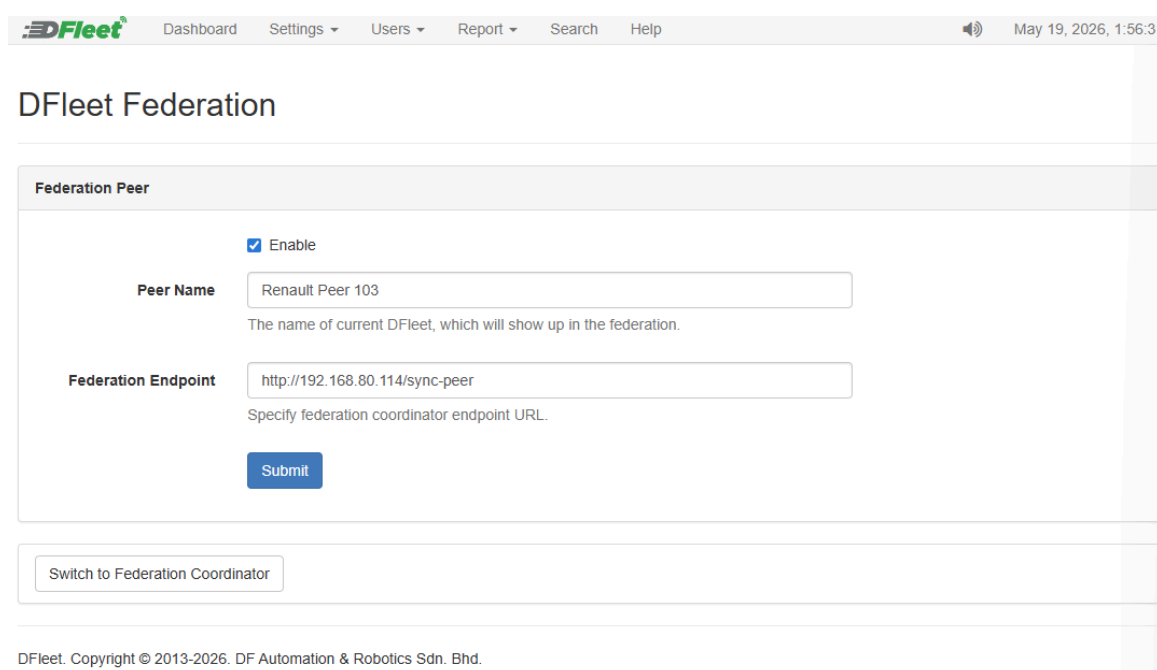
- manually re-assign another node as the cluster's Primary node
- remove a replica node from the cluster
- destroy the cluster.

Section 10. DFleet Federation

The DFleet Federation page provides a platform for users to configure federation settings, where one coordinator oversees multiple independent DFleet instances.

Federation Peer

Federation peer is a standard DFleet instance that manages its own fleet of AGVs on its own map, but it can also publish the live data to the coordinator.



DFleet Federation

Federation Peer

☒ Enable

Peer Name
The name of current DFleet, which will show up in the federation.

Federation Endpoint
Specify federation coordinator endpoint URL.

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Component	Description
Enable	Enable or disable the Federation Peer feature.
Peer Name	The name of the current DFleet, which will show up in the federation.
Federation Endpoint	Specify federation coordinator endpoint URL.. The endpoint URL can be obtained from the DFleet that is running as the Federation Coordinator.
Switch to Federation Coordinator	Change the current DFleet instance to Federation Coordinator mode.

Federation Coordinator

A DFleet that does not manage any AGVs of its own. Instead, it connects to multiple peer DFleet instances and aggregates their state into a single view.


Dashboard
Settings
Users
Report
Help
May 19, 2026, 1:57:24 PM
admin

DFleet Federation

Federation Coordinator

Federation Endpoint: http://192.168.80.114/sync-peer?token=770626611

View peers

Switch to Federation Peer

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Component	Description
Federation Endpoint	Specify the Federation Coordinator endpoint URL. The endpoint URL is used by the Federation Peer to connect to the coordinator.
View peers	Open the Peers page to view all connected peers.
Switch to Federation Peer.	Change the current DFleet instance to Federation Peer mode.

The Peers page in the DFleet coordinator displays all DFleet peers connected to it.

Peers

Federation Endpoint: http://192.168.80.114/sync-peer?token=770626611

Peers (1-3 of 3)

#	Name	Mode	Operation
1	Pier 39 (202390c1-436a-4758-cf33-745f4249ea68)	Trackless	
2	ST Davinci (c0593559-6a21-3611-748f-ea84b84e8e3b)	Trackless	
3	Tracking (4c2d8a5e-78fe-5551-84e3-19e335115f38)	Tracked	

The landing page of the DFleet coordinator shows all Live Maps, AGV status, and Group Reservation status. Only junctions with Group IDs from 9000 to 9999 are used by the DFleet Federation feature.

Section 11. Network Connections

In Network Connections, the user is able to view the connection details of the system. It shows the DFleet hostname, network scheme used and connection status of each available ethernet port.

Network Connections

[Edit Network Configuration](#)
[Edit Hostname](#)
[Edit HTTPS Settings](#)

Hostname:	ubuntu
Network Scheme:	HTTP
eth0 (Main Interface) ⓘ	
MAC Address:	00:15:5D:00:65:1E
Link Speed:	10000 Mbps
IP Address:	172.20.244.212
Subnet Mask:	255.255.240.0
Gateway:	172.20.240.1
DNS Server:	127.0.0.53
Bytes Received:	462,180,940
Bytes Sent:	22,305,556

Network Connection page

Network Configurations Editor

The Network Configuration Editor enables users to view the network settings of each Ethernet port. Users have the flexibility to configure the Ethernet ports as either static or DHCP.

Edit Network Configuration

Main Interface*

enp0s3

enp0s3 ⓘ

Configure IP*

Automatic (DHCP)

☐ Override DNS server addresses

Metric*

0

The metric 0 is the highest priority route, but it is reserved for the main interface. A larger metric value means a lower priority route.

Submit

Network Configuration page

**Notice:**

DFleet listens on port 80 (HTTP), which is accessed by AMR, Station PCs and maintenance personnel.

Hostname Editor

Hostname editor allows users to change the hostname reported by DFleet to the network it is connected to.

Edit Hostname

Hostname*

HTTPS Settings Editor

This page allows users to switch between HTTPS or HTTP mode. Users are able to upload manually generated private keys and certificates or generate automatically by selecting the Generate Certificate button. These configurations are required to enable HTTPS mode.

Edit HTTPS Settings

Network Scheme*

HTTPS scheme require public certificate bundle and private key.

Private Key: (KEY)

Private key should not have a password.

**Certificate Bundle:
(CRT)**

Section 12. Date & Time

Date and Time Configuration

Date and time*	30/06/2023 16:42
Time zone*	Asia/Kuala_Lumpur
	☐ Auto synchronize with network time server
Local NTP Server	
	The IP address of a preferred network time server for synchronization (optional).
Synchronization status	Disabled.

Warning: A soft reboot will be performed after updating the date and time. Please make sure DFleet is not executing any task.

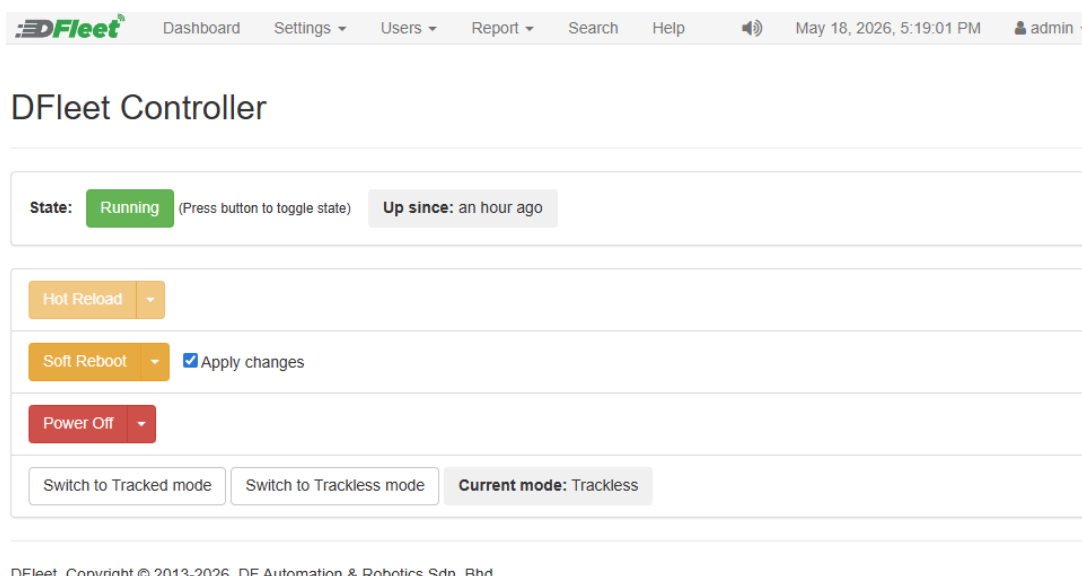
[Update and Soft Reboot](#)

The Date and Time Configuration page provides users the capability to configure date, time and time zone of DFleet. There is an option to enable the "Auto synchronize with network time server" checkbox, which allows the DFleet server to automatically synchronize its date and time with a network time server.

If users have a preferred network time server for time synchronization purposes, they can set it in the "Local NTP Server" field. The "Synchronization Status" field will display the IP address of the server to which the DFleet server is synchronized.

If the "Update and Soft Reboot" button is clicked, the system will immediately perform a soft reboot. Hence, users have to make sure DFleet is not executing any task at that moment.

Section 13. DFleet Controller



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DFleet Controller is the place to reboot the DFleet for any changes made or to turn it off. We can also view the state of the DFleet and toggle it to either “Pause” or “Running” state.

The hot reload button is used to reload the changes without the need to perform a soft reboot of the DFleet. This feature is useful for updating the changes of the AMRs without requiring the stopping of all AMRs. Hot reload covers map, task templates, params, variables, teleports, transition triggers, skillsets, ocs, and assignment policies. Beside the Hot Reload button, there's a dropdown option for Force Hot Reload to reload the changes immediately.

Beside the Soft Reboot button, there's a dropdown option for Force Soft Reboot:

- Force Soft Reboot: Immediately restarts the DFleet, regardless of the AGVs' current tasks.
- Soft Reboot: Waits for all AGVs to complete their current tasks before rebooting. During this time, DFleet stops assigning new tasks.

Next to the Soft Reboot button, there is also a checkbox to indicate whether the soft reboot should apply changes. By default, when restarting the executor (such as by restarting the PC server), changes will not be applied to ensure that they are only applied when intended. If you restart through the 'Soft Reboot' button, the changes will only be applied if the 'Apply Changes' option is selected.

DFleet current mode can be switched from Tracked to Trackless or vice versa.



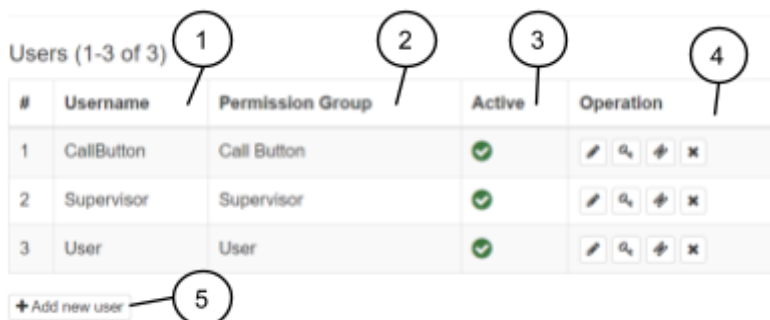
Caution:

To force soft reboot the DFleet, pause the DFleet controller and ensure all AMR return to its home station.

Section 14. Users, Groups & Permissions

Users

Users



The screenshot shows a 'Users' management page with a table of users and an 'Add new user' button. Numbered callouts identify the following elements:

- 1: Username column header
- 2: Permission Group column header
- 3: Active column header
- 4: Operation column header
- 5: '+ Add new user' button

#	Username	Permission Group	Active	Operation
1	CallButton	Call Button	✓	
2	Supervisor	Supervisor	✓	
3	User	User	✓	

+ Add new user

Under the Users page, the Supervisor can create new users, delete users, change user password and protection pins. Authority tokens can also be gained here.

No.	Component	Description
1.	Username	Username assigned to the account.
2.	Permission Group	Permission groups determine the level of access each account has.
3.	Active	Deactivate instead of deleting the account if you only need to disable the account temporarily.
4.	Operation	Open edit user page. Which allows the supervisor to edit Username , active state, and permission group of user.
		Open change user password page. Can directly assign a new password without the need of the old password.
		Open page that provides Authorization Token of user.
		Delete user currently selected.
5.	Add new user	Add a new user.

Groups

Groups

Groups (1-4 of 4)

#	Group	Operation
1	Guest default	
2	Call Button default	
3	User default	
4	Supervisor default	

[+ Add new group](#)

Different access levels are given to different Users. By default, there are a total 4 groups with different access levels to DFleet. To add a new group, click on the “Add new group” button as shown above.

Permissions

Permissions

DFleet Boot
Panel
Task Operation
System
Component
Users

Can pause DFleet controller

☐ Guest
☐ Call Button
☐ User
☒ Supervisor

Can resume DFleet controller

☐ Guest
☐ Call Button
☐ User
☒ Supervisor

Can perform soft reboot

☐ Guest
☐ Call Button
☐ User
☒ Supervisor

Can perform hard reboot

☐ Guest
☐ Call Button
☐ User
☒ Supervisor

Can perform poweroff

☐ Guest
☐ Call Button
☐ User
☒ Supervisor

Update

Reset to Factory Default...

The permissions page allows defining the access level of DFleet for different groups.



Notice:

Permissions page only accessible by “admin” account.

Create New User

To create a new user, click on the “Add new user” button.

Add new user

Username*

Required. 150 characters or fewer. Letters, digits and @/./+/-/_ only.

Permission group*

Call Button

▼

Password*

- Your password can't be too similar to your other personal information.
- Your password must contain at least 8 characters.
- Your password can't be a commonly used password.
- Your password can't be entirely numeric.

Password confirmation*

Enter the same password as before, for verification.

Create

Do take note that after creating a new user, access of task templates is needed to be given to this new user accordingly.

Delete User

After clicking on the “Delete user button”, a pop up confirmation will be asking for confirmation about this deletion.

Delete user "User"

Are you sure you want to delete user "User"?

Delete

Cancel

Change User Password

Click on the “Change Password” button to change the user’s password. Another page will pop out to ask for a new password.

Change password of user "CallButton"

New password*

- Your password can't be too similar to your other personal information.
- Your password must contain at least 8 characters.
- Your password can't be a commonly used password.
- Your password can't be entirely numeric.

New password confirmation*

Default Permission Group

Access level for DFleet is divided into 4 different account type:

1. Call Button
2. User
3. Supervisor

Call Button

Call Button account is basically used to login into DFleet Call Button device. Call Button account has the permission to view the list of current running, queued tasks and create tasks that are being assigned to this account type. The configuration of the task assignment for all call buttons are available under the Task Template Page (only configurable by Supervisor account).

User

User account is basically a normal login account for users of the AMR. User accounts have the permission to view the list of current running, queued tasks and create tasks that are being assigned to this account. This account is advised to use Terminals for showing the status of AMR. The configuration of the task assignment for all user are available under the Task Template Page (only configurable by Supervisor account)

Supervisor

Supervisor accounts in DFleet have extensive control over the system, with the exception of the ability to view and modify the permission list.

Identity Provider

The Identity Provider Settings are used to connect DFleet to an external OIDC identity provider. The OIDC URLs must be registered with the identity provider for the OIDC handshake to work.

Identity Provider Settings

OIDC URLs

Redirect URL

`http://192.168.1.8/oidc/callback`

Use this URL as the redirect/callback URI in your OIDC provider configuration.

Logout Redirect URL

`http://192.168.1.8/login`

Register this URL as a post-logout redirect URI in your OIDC provider so users return here after "Logout All".

Back-Channel Logout URL

`http://192.168.1.8/oidc/backchannel-logout`

Register this URL with your OIDC provider as the back-channel logout URI. The provider will POST a signed logout token to it when a session ends.

Identity Provider

Provider*

Disabled

Enable or disable OIDC for authentication.

OIDC Configuration

Login Button Text

Login with Organization Account

Custom text for the organization login button.

Well-Known URL

OIDC provider well-known configuration URL (example: `https://.well-known/openid-configuration`).

Client ID

OIDC client identifier.

Client Secret

OIDC client secret.

Scope

openid profile email

OIDC scopes to request (space-separated).

☒ Verify SSL

Verify SSL certificate when contacting the OIDC provider. Disable only for self-signed test environments.

Claim Mappings

Username Claim
 OIDC claim name for username. Ensure the value is unique for each user.

Display Name Claim
 OIDC claim name for user display name.

Groups Claim
 OIDC claim name for user groups.

Permission Claim
 OIDC claim name whose values carry permission codes. Leave empty to skip permission sync.

Permission Prefix
 Stripped from each claim value to produce the Django permission codename (e.g. "dfleet1:agvccs.view_settings" -> "agvccs.view_settings"). If empty, claim values are used as-is.

Group Configuration

Default User Group*
 Default DFleet group for users without mapped groups.

Group Mappings

#	OIDC Group	DFleet Group	Actions
1	Enter OIDC group name	-----	

[+ Add Group Mapping](#)

[Submit](#)

No.	Section	Component	Description
1.	OIDC URLs	Redirect URL	Use this URL as the redirect/callback URI in your OIDC provider configuration.
		Logout Redirect URL	Register this URL as a post-logout redirect URI in your OIDC provider so users return here after "Logout All".
		Back-Channel Logout URL	Register this URL with your OIDC provider as the back-channel logout URI. The provider will POST a signed logout token to it when a session ends.
2	Identity Provider	Provider	Enable or disable OIDC authentication. If disabled, users can log in only with local DFleet credentials, and all other fields will have no effect.
3.	OIDC Configuration	Login Button Text	Custom text for the organization login button.
		Well-Known URL	OIDC provider well-known configuration URL (example: https://.well-known/openid-configuration).
		Client ID	OIDC client identifier.
		Client Secret	OIDC client secret.
		Scope	OIDC scopes to request (space-separated).
		Verify SSL	Verify SSL certificate when contacting the OIDC provider. Disable only for self-signed test environments.

4.	Claim Mappings	Username Claim	OIDC claim name for username. Ensure the value is unique for each user.
		Display Name Claim	OIDC claim name for user display name.
		Groups Claim	OIDC claim name for user groups.
		Permission Claim	OIDC claim name whose values carry permission codes. Leave empty to skip permission sync.
		Permission Prefix	Stripped from each claim value to produce the Django permission codename (e.g. "dfleet1:agvccs.view_settings" -> "agvccs.view_settings"). If empty, claim values are used as-is.
5.	Group Configuration	Default User Group	Default DFleet group for users without mapped groups.
		Group Mappings	Map the OIDC group name to the DFleet group.

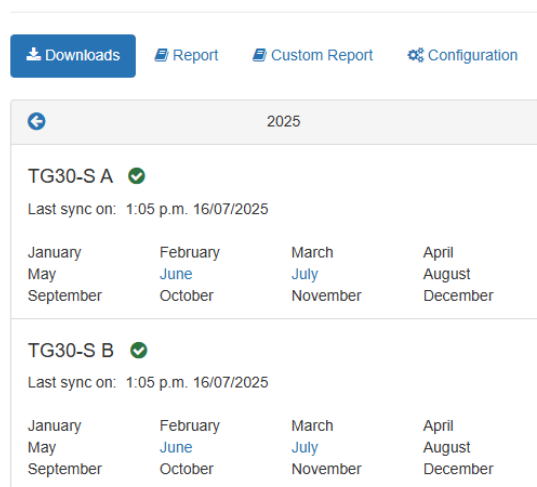
Section 15. Reports

AGV Activities

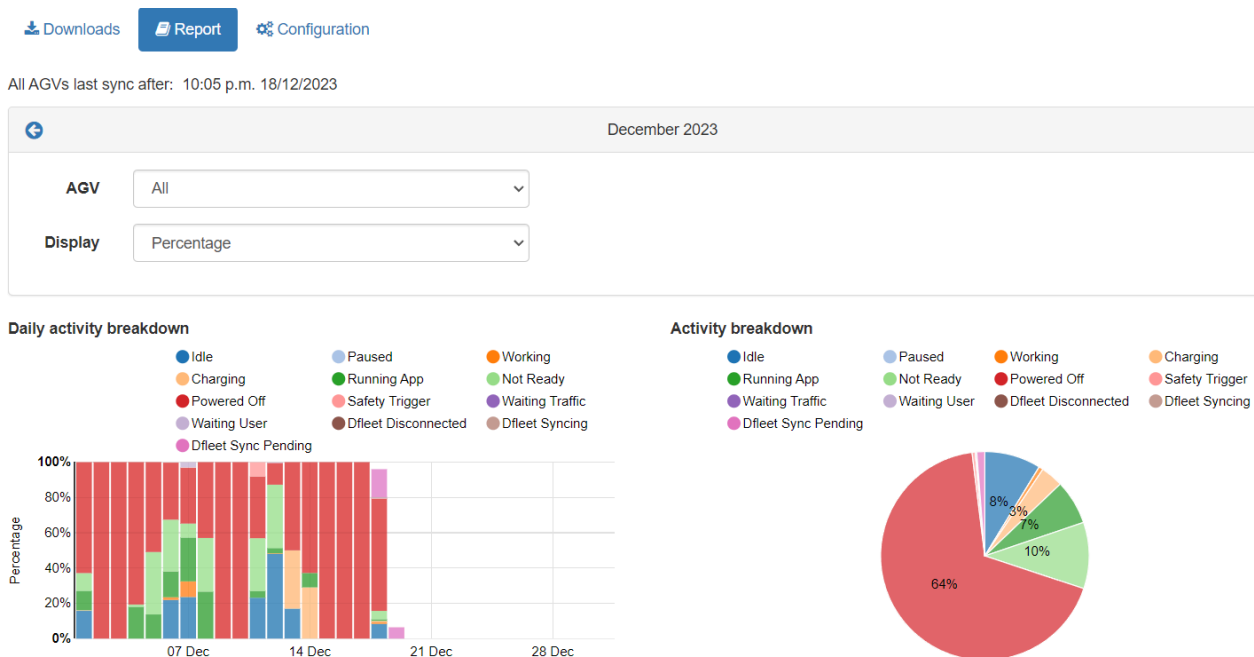
The AGV Activities feature offers users a comprehensive overview of AMR operational activities. It provides insights into performance metrics such as time spent on running jobs and idle periods.

Users could download AMR activities for a month in CSV format, facilitating easy access for analysis or integration with other tools.

AGV Activities



The Report page visualizes monthly AMR activities through Daily and Overall Activity Breakdown charts, offering a clear insight into trends and patterns.



On the Custom Report page, users can specify a custom date range for generating reports.

AGV Activities

[Downloads](#)
[Report](#)
[Custom Report](#)
[Configuration](#)

Agv*

Start time*

End time*

[Generate Report](#)

On the Configuration page, users have the flexibility to exclude specific AMRs from the report and modify the labels associated with each activity code.

AGV Activities

[Downloads](#)
[Report](#)
[Configuration](#)

AGVs			
Name	Active	Report View	Last Sync
Linde Stacker	✓	Include	3:05 PM 11/24/2023
Titan-san	✓	Include	3:05 AM 12/19/2023

Activity Labels		
Activity	Activity Code	Label (for reporting)
Idle	0	Idle
Paused	1	Paused
Working	2	Working
Charging	3	Charging
Running App	10	Running App

The following table presents the list of AGV Activity Codes, arranged based on their precedence levels, from high to low.

	Activity	Activity Code	Running Task?	Remarks
Task Runner Inactive	Powered Off / Not Ready / Running App	81 / 80 / 10	No	
	Unplanned Maintenance, Planned Maintenance, System Failure, Hardware Failure	82 / 83 / 84 / 85	No	Applicable only when downtime tracker is enabled

Task Runner Active	Idle	0	No	
	Suspended	1	Ambiguous	
	Working	2	Yes	
	Charging and Idle	3	Ambiguous	
	Charging and Suspended	31	Ambiguous	
	Charging and Working	32	Yes	
	Charging and Paused	33	Ambiguous	
	Charging and Running App	34	Ambiguous	
	Charging and Not Running	35	Ambiguous	
	Paused	4	Ambiguous	
	Safety error codes (Bumper blocked, External Safety Trigger, ...)	100 ... 199	Yes	
	Waiting Traffic	200	Yes	
	Waiting User	201	Yes	
	FMS error codes (FMS disconnected, FMS incompatible, FMS syncing, FMS sync pending)	300 / 301 / 302 / 303	Ambiguous	
	FMS error codes while charging (FMS disconnected, FMS incompatible, FMS syncing, FMS sync pending)	310 / 311 / 312 / 313	Ambiguous	
	FMS error codes while working (FMS disconnected, FMS incompatible, FMS syncing, FMS sync pending)	320 / 321 / 322 / 323	Ambiguous	

	FMS error codes while charging and working (FMS disconnected, FMS incompatible, FMS syncing, FMS sync pending)	330 / 331 / 332 / 333	Ambiguous	
--	--	-----------------------	-----------	--

Completed Tasks

List down all completed tasks.

Completed Tasks

[List](#)
[Downloads](#)
[Report](#)
[Custom Report](#)

Completed Tasks (1-13 of 13)

#	Name	Template	Agv	Status	Progress History	Creation Time	Start Time	End Time	Duration	Owner	Battery Usage (%)
1	AGV Idle Task	[0] ARV Homing	SIM 02	Completed	5:54 p.m.: TM Project is Paused 5:51 p.m.: TM Project is Not Running	5:50 p.m. 12/05/2026	5:50 p.m. 12/05/2026	5:54 p.m. 12/05/2026	3 minutes	-	0
2	[0] ARV Navigate to	[0] ARV Navigate to Station: ZF PSAWC Out	SIM 02	Completed	Progress: TM Project is Not Running	5:49 p.m. 12/05/2026	5:49 p.m. 12/05/2026	5:50 p.m. 12/05/2026	0 minutes	admin	0
3	AGV Idle Task	[0] ARV Homing	SIM 02	Aborted		5:49 p.m. 12/05/2026	5:49 p.m. 12/05/2026	5:49 p.m. 12/05/2026	0 minutes	-	0
4	[0] ARV Navigate to	[0] ARV Navigate to Station: ZF PSAWA ENTRY	SIM 02	Aborted	5:48 p.m.: TM Project is Paused Progress: TM Project is Paused	5:48 p.m. 12/05/2026	5:48 p.m. 12/05/2026	5:49 p.m. 12/05/2026	1 minute	admin	0

Users could also download a list of completed tasks in a given month in csv format.

Completed Tasks

[List](#)
[Downloads](#)
[Report](#)
[Custom Report](#)

2025

January

February

March

April

May

June

July

August

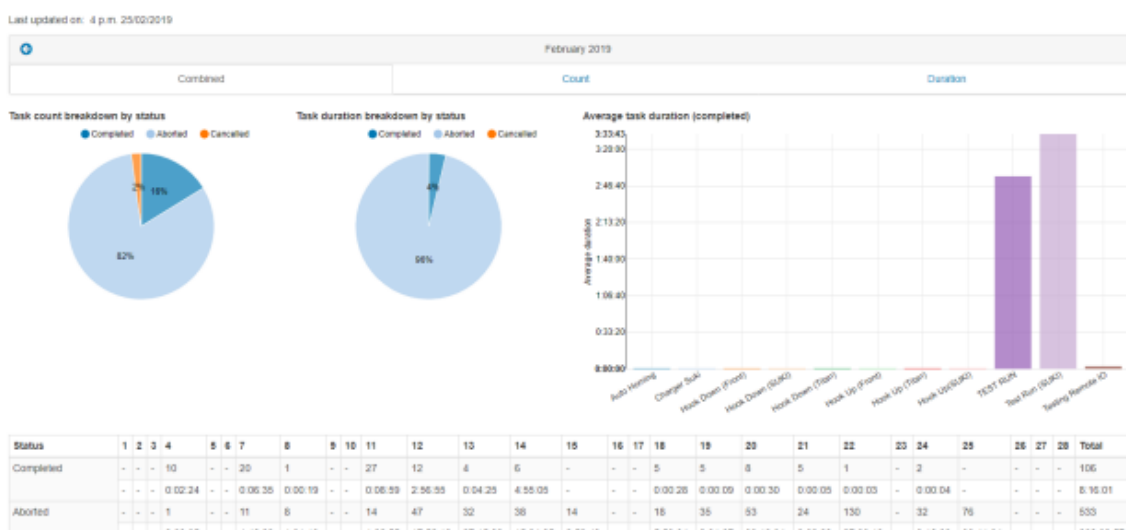
September

October

November

December

On the Report page, users could view the latest tabulated and visualised forms of the count, duration, or combined analysis of both for the completed task report of the current month.



Lastly, users could also generate customized reports using the Custom Report feature.

Completed Tasks

[List](#)
[Downloads](#)
[Report](#)
[Custom Report](#)

Start time*

End time*

☒ Filter by agv

Agv

☒ Filter by owner

Owner

[Generate Report](#)

Completed Transactions

All completed transactions will be displayed in the Completed Transactions page.

Completed Transactions

Completed Transactions (1-100 of 124)

#	Name	Transfer	Agv	Status	Error	Creation Time	Start Time	Transfer Time	End Time	Owner
1	46ed8abf-073f-4064-a3fb-113ff2301350	item2: a2 → b2	SIM 01	Completed	-	11:06 a.m. 12/12/2022	11:06 a.m. 12/12/2022	11:07 a.m. 12/12/2022	11:08 a.m. 12/12/2022	-
2	00241a41-8ed2-4c8a-8f60-3607afbc9d1	item1: a1 → b1	SIM 01	Completed	-	11:06 a.m. 12/12/2022	11:06 a.m. 12/12/2022	11:06 a.m. 12/12/2022	11:08 a.m. 12/12/2022	-
3	8945f1db-0124-49d7-83d2-4b370f9f37b2	item2: a2 → b2	SIM 01	Completed	-	11:01 a.m. 12/12/2022	11:01 a.m. 12/12/2022	11:01 a.m. 12/12/2022	11:03 a.m. 12/12/2022	-
4	ef9c02f2-730a-480c-8e51-4df1e05756dd	item1: a1 → b1	SIM 01	Completed	-	11:01 a.m. 12/12/2022	11:01 a.m. 12/12/2022	11:01 a.m. 12/12/2022	11:02 a.m. 12/12/2022	-
5	6fc9fedf-b739-414a-8556-8b07c3e00604	item2: a2 → b2	SIM 01	Cancelled	-	11 a.m. 12/12/2022	11 a.m. 12/12/2022		11:01 a.m. 12/12/2022	-
6	de148139-2aee-420f-9107-28b05dbfa5ad	item1: a1 → b1	SIM 01	Aborted	-	11 a.m. 12/12/2022	11 a.m. 12/12/2022	11:01 a.m. 12/12/2022	11:01 a.m. 12/12/2022	-
7	c2b4b136-2b37-4dd5-81a8-eb28f718d8b8	item2: a2 → b2	SIM 01	Completed	-	10:57 a.m. 12/12/2022	10:57 a.m. 12/12/2022	10:58 a.m. 12/12/2022	10:59 a.m. 12/12/2022	-
8	6eb48140-1ca6-4163-bc68-6b8bc436cacb	item1: a1 → b1	SIM 01	Completed	-	10:57 a.m. 12/12/2022	10:57 a.m. 12/12/2022	10:57 a.m. 12/12/2022	10:59 a.m. 12/12/2022	-

Similar to AGV Activities and Completed Tasks, the Completed Transactions page also includes features to download data by month, generate standard reports, and generate custom reports.

Completed Transactions

[List](#)
[Downloads](#)
[Report](#)
[Custom Report](#)

Start time*

End time*

☒ Filter by agv

Agv

☒ Filter by owner

Owner

[Generate Report](#)

System Monitor

Provide current status of the DFleet system specifically the CPU usage, CPU Frequency, Memory, and Hard Disk usage.

System Monitor

CPU Usage: 11.005%

CPU Freq: 2893.026 MHz

Memory Usage:

	total	used	free	shared	buff/cache	available
Mem:	3.8Gi	1.8Gi	1.6Gi	1.0Mi	438Mi	1.9Gi
Swap:	3.8Gi	0B	3.8Gi			

Hard Disk Usage:

Filesystem	Size	Used	Avail	Use%	Mounted on
udev	1.9G	0	1.9G	0%	/dev
tmpfs	394M	1.1M	393M	1%	/run
/dev/sda2	44G	9.9G	32G	24%	/
tmpfs	2.0G	0	2.0G	0%	/dev/shm
tmpfs	5.0M	8.0K	5.0M	1%	/run/lock
tmpfs	2.0G	0	2.0G	0%	/sys/fs/cgroup
/dev/loop0	64M	64M	0	100%	/snap/core20/2769
/dev/loop1	64M	64M	0	100%	/snap/core20/2866
/dev/loop2	92M	92M	0	100%	/snap/lxd/24061
/dev/loop3	92M	92M	0	100%	/snap/lxd/38688
/dev/loop4	49M	49M	0	100%	/snap/snapd/26382
/dev/loop5	50M	50M	0	100%	/snap/snapd/26865

Message Inspector

The Message Inspector is a live message broker viewer and debugging tool. It allows users to subscribe to broker topics in real time and monitor the raw message stream directly from the UI. The Topic filter has two dropdowns — one to select the instance (DFleet and/or specific AMRs) and one to select the topic types — which work together to display only messages from the selected instances on the selected topics.


Dashboard Settings Users Report Search Help May 18, 2026, 5:11:52 PM admin

Message Inspector

```

"1bb68a3819339b371aaa6e0dcd8336b9", "run_id": "1779089388.240127", "skillset": "bda47228cff3da97feaa273b6f2e4f9e",
"station_names": "aad806ecb739798700ccbe6c341f35c8", "task_templates": "66cfa9e8ea50e434bb93488488a4220a", "teleports":
"d751713988987e9331980363e24189ce", "transaction_enabled": "68934a3e9455fa72420237eb05902327", "transition_triggers":
"d751713988987e9331980363e24189ce", "variables": "d751713988987e9331980363e24189ce", "error_code": 0, "fms_status":
"Ready", "location": null, "location_hint": "", "mileage": 1716.7, "motion": null, "name": "SIM 01", "pose": null,
"prev_location": null, "run_id": "1779089388.240127", "safety_message": null, "skillset_md5":
"bda47228cff3da97feaa273b6f2e4f9e", "status": "Not Ready", "task_counter": 0, "task_name": null, "token":
"5ae50680ae6f0c14bc778a4c4eccf0bf7f17cd8a", "user_message": null, "uuid": "00000000-0000-0000-0001-000000000000",
"velocity": [0, 0, 0 ]}
2026-05-18 17:11:52.104 {"action": "", "agv_ip": "/", "battery": 88, "charging": false, "dimension_profile": 0,
"downloadables_md5": {"agv_home": "Home 11", "executor_cfg": "4c773a481281e3b82581e70297ae79a3", "fms_skillset":
"11c6dd730df0746b7a5f35970b471be3", "global_params": "5f4184edf87823444d8b942c4cbbf825", "map_params":
"c7482453fc0ee0747628011bcf0d8f24", "map_structure": "e869e716e01b25b1b30298831ea1f04c", "ocg":
"1bb68a3819339b371aaa6e0dcd8336b9", "run_id": "1779089388.240127", "skillset": "bda47228cff3da97feaa273b6f2e4f9e",
"station_names": "aad806ecb739798700ccbe6c341f35c8", "task_templates": "66cfa9e8ea50e434bb93488488a4220a", "teleports":
"d751713988987e9331980363e24189ce", "transaction_enabled": "68934a3e9455fa72420237eb05902327", "transition_triggers":
"d751713988987e9331980363e24189ce", "variables": "d751713988987e9331980363e24189ce", "error_code": 0, "fms_status":
"Ready", "location": null, "location_hint": "", "mileage": 1716.7, "motion": null, "name": "SIM 01", "pose": null,
"prev_location": null, "run_id": "1779089388.240127", "safety_message": null, "skillset_md5":
"bda47228cff3da97feaa273b6f2e4f9e", "status": "Not Ready", "task_counter": 0, "task_name": null, "token":
"5ae50680ae6f0c14bc778a4c4eccf0bf7f17cd8a", "user_message": null, "uuid": "00000000-0000-0000-0001-000000000000",
"velocity": [0, 0, 0 ]}

```

Clear messages
☒ Auto-scroll

Topic SIM 01 agv.agv_update

DFleet. Copyright © 2013-2026. DF Automation & Robotics Sdn. Bhd.



Audit Log

The audit Log feature allows users to access live updates of log messages related to user login and action events. The complete log can also be downloaded in CSV format.

Audit Log

FILTERS

Download CSV

Audit Log (1-300 of 385)

#	Timestamp	Username	Event Type	Event Summary
1	5:02 p.m.	admin	System	User login. (ip=192.168.1.13)
2	5:02 p.m.	User	System	User logout. (ip=192.168.1.13)
3	5:02 p.m.	User	System	User login. (ip=192.168.1.13)
4	5:01 p.m.	Supervisor	System	User logout. (ip=192.168.1.13)
5	5:01 p.m.	Supervisor	System	User login. (ip=192.168.1.13)
6	5:01 p.m.	admin	System	User logout. (ip=192.168.1.13)
7	3:29 p.m.	admin	System	Soft rebooting (apply_changes=True).
8	3:29 p.m.	admin	Config	Map "Example Map 01" layout updated.
9	3:19 p.m.	admin	System	Soft rebooting (apply_changes=True).

Custom Log

List out the custom logs of the current DFleet system. Capable of applying filters to the custom log message.

Custom Log

2023-05-25 17:26:10.012	INFO	[DFleet]	Fire Alarm End
2023-05-25 17:26:36.507	INFO	[DFleet]	Fire Alarm End
2023-05-25 17:29:08.334	INFO	[DFleet]	Fire Alarm End
2023-05-25 17:32:11.151	INFO	[DFleet]	Fire Alarm End
2023-05-25 17:33:24.263	INFO	[DFleet]	Fire Alarm End
2023-05-25 17:36:43.971	INFO	[DFleet]	Fire Alarm End
2023-05-25 17:44:01.703	INFO	[DFleet]	Fire Alarm End
2023-05-25 17:45:48.734	INFO	[DFleet]	Fire Alarm End
2023-05-25 17:46:38.650	INFO	[DFleet]	Fire Alarm End
2023-05-25 17:47:03.524	INFO	[DFleet]	Fire Alarm End
2023-05-25 18:05:35.466	INFO	[DFleet]	Fire Alarm End
2023-05-25 18:37:34.928	INFO	[DFleet]	Fire Alarm End
2023-05-25 18:51:34.644	INFO	[DFleet]	Fire Alarm End
2023-07-21 14:33:08.969	ERROR	[AMR 001]	Unable to find a valid path to destination.
2023-07-21 14:33:20.013	ERROR	[AMR 001]	Unable to find a valid path to destination.
2023-07-21 14:33:58.203	ERROR	[AMR 001]	Unable to find a valid path to destination.

☒ Auto-scroll

Message Filter Nothing selected DFleet, AMR 001

Download custom log

System Log

List down all previous logs of the system recorded.

Index of /logs/

../	17-Jul-2023 17:17	-
cron_scripts/	19-Jul-2023 07:56	-
scripts/	21-Jul-2023 16:00	12570
celery.log	21-Jul-2023 07:30	5090
celery.log.1.gz	20-Jul-2023 07:30	5639
celery.log.2.gz	19-Jul-2023 07:30	6256
celery.log.3.gz	18-Jul-2023 07:30	3041
celery.log.4.gz	17-Jul-2023 17:30	1676
celerybeat.log	21-Jul-2023 16:00	12016
celerybeat.log.1.gz	21-Jul-2023 07:55	2198
celerybeat.log.2.gz	20-Jul-2023 07:55	2590
celerybeat.log.3.gz	19-Jul-2023 07:55	3371
celerybeat.log.4.gz	18-Jul-2023 07:55	1656
celerybeat.log.5.gz	17-Jul-2023 17:45	1175
executor.log	21-Jul-2023 07:56	0
executor.log.1.gz	20-Jul-2023 14:19	109
executor.log.2.gz	19-Jul-2023 10:15	26539
executor.log.3.gz	18-Jul-2023 14:53	444
executor.log.4.gz	17-Jul-2023 17:42	770

Download Log


Panel Settings ▾ Users ▾ Report ▾ Search Help June 30, 2023, 4:59:57 PM admin ▾

Download Log

Start time*

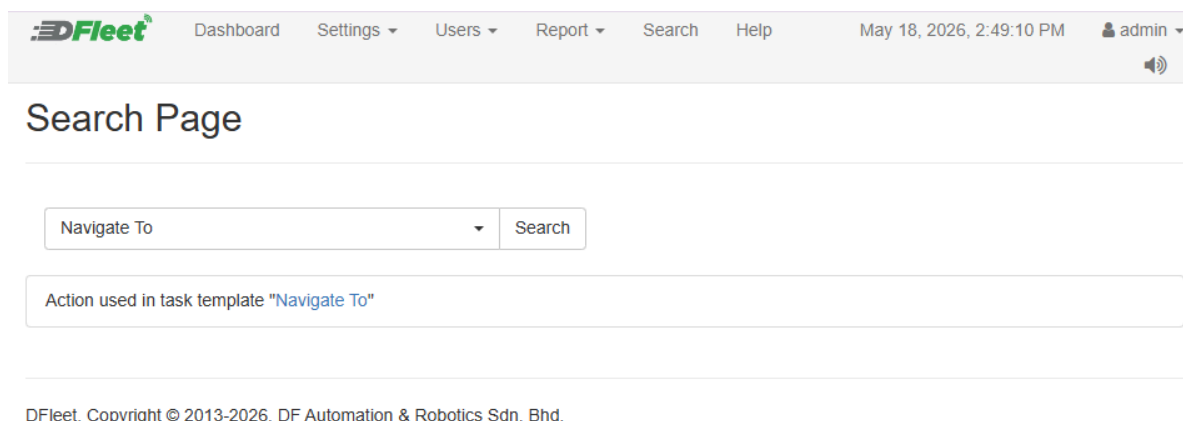
End time*

DFleet. Copyright © 2013-2023. DF Automation & Robotics Sdn. Bhd.

This feature allows users to select a date range and download a single bundled file that contains all logs within the date range.

Section 16. Search

The search bar allows users to search for built-in actions, task templates, stations, global parameters, registers and global variables.



The screenshot shows the DF Fleet web interface. At the top is a navigation bar with the DF Fleet logo, links for Dashboard, Settings, Users, Report, Search, and Help, the date and time (May 18, 2026, 2:49:10 PM), and the user name (admin). Below the navigation bar is the title "Search Page". Underneath the title is a search bar with a dropdown menu showing "Navigate To" and a "Search" button. Below the search bar is a result box containing the text "Action used in task template 'Navigate To'". At the bottom of the page is a copyright notice: "DFleet. Copyright © 2013-2026. DF Automation & Robotics Sdn. Bhd."

The result will be displayed as shown above if the searched object is being found within task templates, task triggers or maps.

Section 17. Help


Dashboard
Settings
Users
Report
Search
Help
May 18, 2026, 2:49:58 PM
admin

Help

About DFleet



4.0.0~rc1+nightly.20260513.193313~9477cff-0focal

Executor: 4.0.0~rc1+nightly.20260509.194357~dc917cb

Kiosk: 4.0.0~rc1+nightly.20260423.193314~d356e40-0focal

[Check for updates...](#)

DFleet is the AGV fleet management system developed by DF Automation.

It automates the control of multiple AGVs, which include the following:

- dispatching tasks;
- controlling traffic;
- providing a dashboard to display the AGVs' status; and
- providing a backend to configure all the AGVs.

This product has been licensed to Zeng Yu Lam (zylam@dfautomation.com.my) for trial purpose only. You have 3 day(s) remaining to evaluate this software.

[Change license key](#)

Remote Assistance

Status: Disconnected.

[Start](#) [Stop](#)

Contact Us

DF Automation & Robotics Sdn. Bhd.

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Taman Impian Emas
81300 Skudai, Johor, Malaysia.
Tel: +607-5623547

Sales and Support: sales@dfautomation.com

Help page contains some summaries on DFleet version and DF contact information. It also contains these features:

- Check for updates
- Change license key
- Remote Assistance

Advance Features

Section 1. Custom Panel

Custom panel features provide a single page option for customization. Users are required to write their own html page with the assistance of ready-made tags of the DFleet system.

Custom Panel

☒ Enable custom panel view.

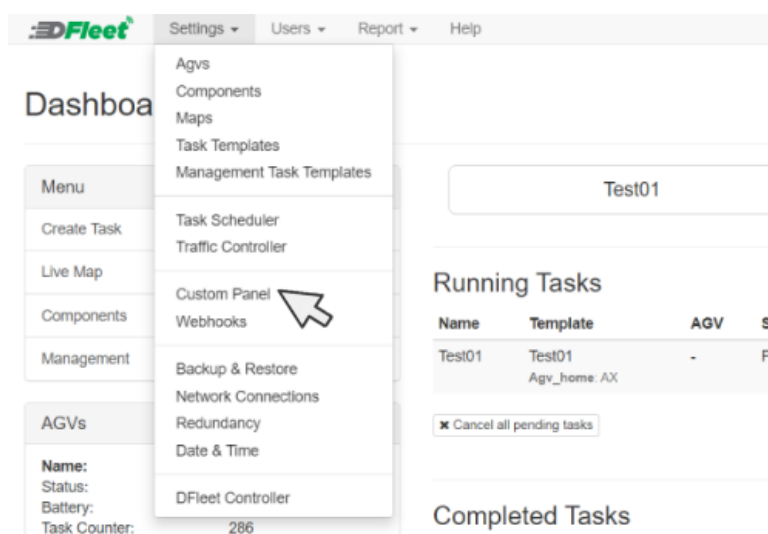
Custom panel file

Currently: [customPanelExample.html](#) ☐ Clear

Change:

No file chosen

Upload custom panel html and enable custom panel view



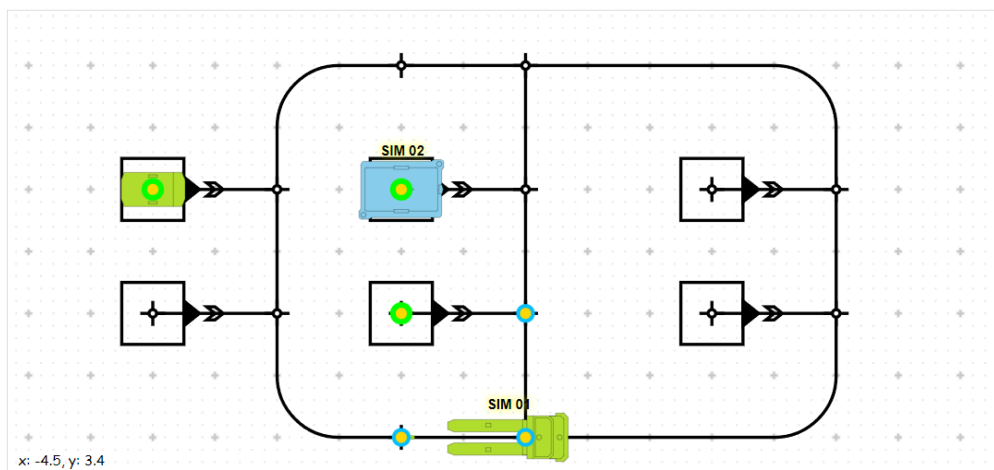
Navigate to Custom Panel

Running Tasks

Name	Template	AGV	Status	Creation Time	Start Time	ETA Station	ETA	Owner	
Nav To	Nav To St. D	SIM 01	In Progress	4:41 PM	4:41 PM	-	-	admin	
Nav To	Nav To St. F	SIM 02	In Progress	4:41 PM	4:41 PM	-	-	admin	

Live Map

Example Map 01



Example of Custom Panel

```
<div style="margin: 10px;">
  <h1>Running Tasks</h1>
  {% running_tasks %}
  <h1>Live Map</h1>
  {% live_map %}
</div>
```

Simple example of custom panel HTML

Available Tag

Tag is a marker for DFleet to take action based on the tag usage.

Disconnected Indicator

Display bootstrap well when DFleet disconnected.

```
{% disconnected_indicator %}
```

Live Map

Fill the parent container with a live map viewer.

```
{% live_map mode_3d=False %}
```

Parameter	Description
mode_3d	Display live map in 3D or 2D mode. Default value is False

AMR Status

Insert AMR status table that lists all active AMR and its status.

```
{% agv_status show_link=True show_status=True show_battery=True show_task_counter=True show_last_seen=True show_task_runner_ctrl=True %}
```

Parameter	Description
show_link	Display URL link to AMR task panel. Default value is True
show_status	Display AMR status. Default value is True
show_battery	Display AMR battery information. Default value is True
show_task_counter	Display AMR task counter information. Default value is True
show_last_seen	Display AMR last seen information. Default value is True
show_task_runner_ctrl	Display AMR task runner play/pause button. Default value is True

AMR Card

Insert multiple AMR cards that show status for each active AMRs.

```
{% agv_card %}
```

AMR Summary

Insert AMR summary card that shows stats of active AMRs status.

```
{% agv_summary %}
```

AMR Battery

Insert AMR battery card that shows battery stats of active AMRs.

```
{% agv_battery %}
```

Running Tasks

Insert list of running tasks in table form.

```
{% running_tasks simple=False %}
```

Parameter	Description
simple	Display running tasks table in simplified or complete with control button. Default value is False

Completed Tasks

Insert list of recently completed tasks in table form.

```
{% completed_tasks %}
```

Task Card

Insert Task card that shows task completed stats.

```
{% task_stats %}
```

Task Breakdown Card

Insert Task breakdown card that shows completed task stats.

```
{% task_breakdown %}
```

Recent Custom Logs

Insert custom logs list view.

```
{% log_custom_recent %}
```

System Info Bar

Insert system information bar display.

```
{% system_info %}
```

Controller Info Row

Insert controller up time row.

```
{% controller_row %}
```

User Group Conditional Logic

Provide a way for users to customize content display in a custom panel based on user group.

```
{% if_ug "<group_csv_list>" %}  
...  
{% elif_ug "<group_csv_list>" %}  
...  
{% else_ug %}  
...  
{% endif_ug %}
```

<group_csv_list> is a list of user groups where the condition is true if the current user group were included in the list. Each group must be separated by symbol.

Live Register

Change parent node text based on register value.

```
{% live_register register="FA0" %}
```

Parameter	Description
register	(required) DFleet register id. (FA0-FA9, FB0-FB9, FC0-FC9, FP0-FP9)

Live Remote IO

Change parent node text based on remote I/O register value.

```
{% live_remote_io device=1 register_mode=0 %}
```

Parameter	Description
device	(required) Remote io device id. (1-20)
register_mode	Set 0 for input register and 1 for output register. Default value is 0.

Live Remote IO Error

Change parent node text based on remote io error status.

```
{% live_remote_io_error device=1 %}
```

Parameter	Description
device	(required) Remote io device id. (1-20)

Running Transactions

Insert list of running transactions in table form.

```
{% running_transactions simple=False %}
```

Parameter	Description
simple	Display running transaction table in simplified or complete with control button. Default value is False

Completed Transactions

Insert list of recently completed transactions in table form.

```
{% completed_transactions %}
```

Transaction Card

Insert Transaction card that shows transaction completed stats.

```
{% transaction_stats %}
```

Transaction Breakdown Card

Insert Transaction breakdown card that shows completed transaction stats.

```
{% transaction_breakdown %}
```

Active Carriers

Insert list of active carriers in table form, and provide buttons to add, modify or delete carriers if the users have the respective permission.

```
{% active_carriers %}
```

Blocked Ports

Insert blocked ports table.

```
{% blocked_ports %}
```

SecsGem Terminal

Insert a SecsGem terminal window.

```
{% secsgem_terminal %}
```

SecsGem Row

Insert rows of secsgem server info.

```
{% secsgem_row %}
```

Live Device Monitoring

Change parent node text based on device monitoring matrix value.

```
{% live_device_monitoring metric="enable" %}
```

Parameter	Description
metric	One of the following metric: enable - 1 if the component is enabled, otherwise 0. agvs.good - number of connected agv or '-' if disabled. agvs.total - total number of agv available. rios.good - number of connected remote io devices or '-' if disabled. rios.total - total number of remote io devices available. devs.good - number of connected network devices or '-' if disabled. devs.total - total number of connected network devices configured.

Device Monitoring Row

Insert rows of device monitoring metrics.

```
{% device_monitoring_row %}
```

Javascripts API

Inside the custom panel, there is a ***window.DFleet*** object that is accessible globally that exposes certain functionality to interact with DFleet backend. This section will detail out all the available interfaces to be consumed by custom panel scripts.



Notice:

The data channel might receive different data than expected. Users are expected to validate or filter the required data only by checking if the data "id" property is matching or if the property name exists on the received channel data.

Following is an example use case where the agv channel is used to monitor and update custom panel information based on agv status.

```
1 function onAgvUpdate(d) {  
2     if (d.id === 'agv_update') {  
3         let agv = DFleet.agv.Agv(d.agv);  
4         // do some operation with current agv info  
5     }  
6 }  
7  
8 DFleet.agv.channel.subscribe(onAgvUpdate);
```

DFleet.apiUrl (property)

String property that holds the root REST API URL for the DFleet.

DFleet.agv (object)

Expose interfaces to obtain AMRs information.

DFleet.agv.getAgvs() (function)

Function to get list of all available AMRs. Return a promise that resolves into a javascript object with the following properties.

- id: 'agv_list'
- agvs: *array of AMRs information.

DFleet.agv.channel.subscribe(cb) (function)

DFleet.agv.channel.unsubscribe(cb) (function)

Subscribe and unsubscribe function to register callback for agv update channel.

Parameters:

- cb (function(d)): function type that accepts a single parameter. This function will be registered/unregistered for callback when data arrives on the channel.

Mainly used to listen to 'agv_update' data id which will be called when there is new data received from AMR. Following are the properties of the data

- id: 'agv_update'
- agv: the new AMR information.

DFleet.agv.Agv (class)

Agv class to properly parse AMR information and expose all the information. To be used with AMR information obtained from agv channel or the getAgvs function.

Properties available are similar to REST API agv endpoint except for the following:

- last_seen: Momentjs instance of the time when AMR last contacted DFleet.

DFleet.task (object)

Expose interface to obtain task information.

DFleet.task.channel.subscribe(cb) (function)

DFleet.task.channel.unsubscribe(cb) (function)

Subscribe and unsubscribe function to register callback when data arrives at the task channel.

Parameters:

- cb (function(d)): function type that accepts a single parameter. This function will be registered/unregistered for callback when data arrives on the channel.

Mainly used to listen to 'task_update' data id. This data provides a recent task completed list and currently running task in the queue. Following are the data properties.

- id: 'task_update'
- runningTasks: List of Task information that is in running state.
- completedTasks: List of Task information that is recently turned into completed state.

DFleet.task.Task (class)

Task class to properly parse Task information received from the channel. Properties available are similar to REST API task endpoint except for the following properties:

- params: Parameter object which is passed into the task during the creation of this task.
- status: Status code of the task. Use the StatusName enum to convert it into a status name.
- created: Momentjs instance of the time when the task was created.
- start_after: Momentjs instance of the time restriction before the task can be started. Null if not applicable.
- run_start: Momentjs instance of the starting time of the task.
- run_end: Momentjs instance of the completion time of the task.
- Status: Object mapping from status name to status code.
- StatusName: Object mapping from status code to status name.

DFleet.CPNT.register (object)

Expose interface to get DFleet register information.

DFleet.CPNT.register.getRegisters() (function)

Function to get list of all available DFleet registers. Return a promise object that resolve into an object that have following properties:

- id: 'list_register'
- registers: list of register information in following format
 - id: register name.
 - value: current integer value of the register.

DFleet.CPNT.register.channel.subscribe(cb) (function)

DFleet.CPNT.register.channel.unsubscribe(cb) (function)

Subscribe and unsubscribe function to register callback to the register channel.

Parameters:

- cb (function(d)): function type that accepts a single parameter. This function will be registered/unregistered for callback when data arrives on the channel.

Mainly used to listen into changes in register value. When the register value changes, the changed event will be sent to the channel with following data properties:

- name: register name.
- value: latest register value.

DFleet.CPNT.remotelo (object)

Expose interface to get DFleet remote IO information.

DFleet.CPNT.remotelo.getDevices() (function)

Function to get list of configured remote io devices on DFleet. Return a promise object that resolve into an object with following properties:

- id: 'list_devices'
- devices: List of remote IO object information. Each remote IO object provide following information:
 - ip_address: ip address of the remote io.
 - model: model name of the remote io.
 - name: the name given for the remote io.
 - read_cache: the read cache time configured.
 - remarks: remarks provided for the remote io.

DFleet.CPNT.remotelo.channel.subscribe(cb) (function)

DFleet.CPNT.remotelo.channel.unsubscribe(cb) (function)

Subscribe and unsubscribe function to register callback to the remote io channel.

Parameters:

- cb (function(d)): function type that accepts a single parameter. This function will be registered/unregistered for callback when data arrives on the channel.

Mainly used to listen into changes in remote io device state change. When the remote io state changes, the changed event will be sent to the channel with following data properties:

- id: 'device_state'
- device_id: device number on the remote io list.
- error: Error message if any or null if no error.
- inputs: List of input pin status in boolean value.
- outputs: List of output pin status in boolean value.

Section 2. SMTP Settings

The SMTP settings are specifically for the alert feature of DFleet Clusters. When there is only one (Primary) server alive in the cluster, indicating a sync error state, a popup notification will be displayed and an alarm will sound on the user interface. Additionally, if the SMTP settings are configured, an email notification will be sent to the user.

SMTP Settings

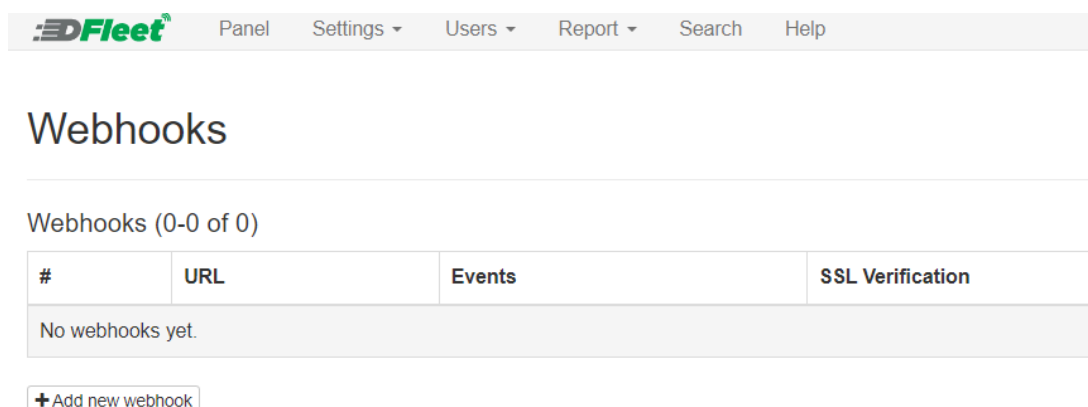
From Email Address*	
	This email address will be used in the "From" field.
From Name*	
	This text will be used in the "From" field.
SMTP Host*	
	Your mail server.
Encryption Type*	SSL/TLS ▼
	For most servers SSL/TLS is the recommended option.
SMTP Port*	465
	The port to your mail server.
SMTP Username*	
	The username to login to your mail server.
SMTP Password*	
	The password to login to your mail server.

Once the SMTP settings are configured, users can utilize the Test Email feature to verify the SMTP settings.

Test Email

To	
	Enter the recipient's email address
Subject	Test
	Enter a subject for your message.
Message	Test message from DFleet.
	Write your email message.
	<button>Submit</button><button>Submit and Send Test Email</button><button>Delete Settings...</button>

Section 3. Webhooks



The screenshot shows the DF Fleet web interface. At the top is a navigation bar with the logo and links for Panel, Settings, Users, Report, Search, and Help. Below this is a heading 'Webhooks'. Under the heading, it says 'Webhooks (0-0 of 0)'. There is a table with four columns: '#', 'URL', 'Events', and 'SSL Verification'. The table is currently empty, with a message 'No webhooks yet.' below the header. At the bottom of the table area is a button labeled '+ Add new webhook'.

Webhooks are custom HTTP callbacks that you define. They are usually triggered by an event, such as creating a new task or an AGV completing a task. When the event occurs, the software makes an HTTP request to the URL configured for the webhook. The action to take may be anything. For example, you can use webhooks to:

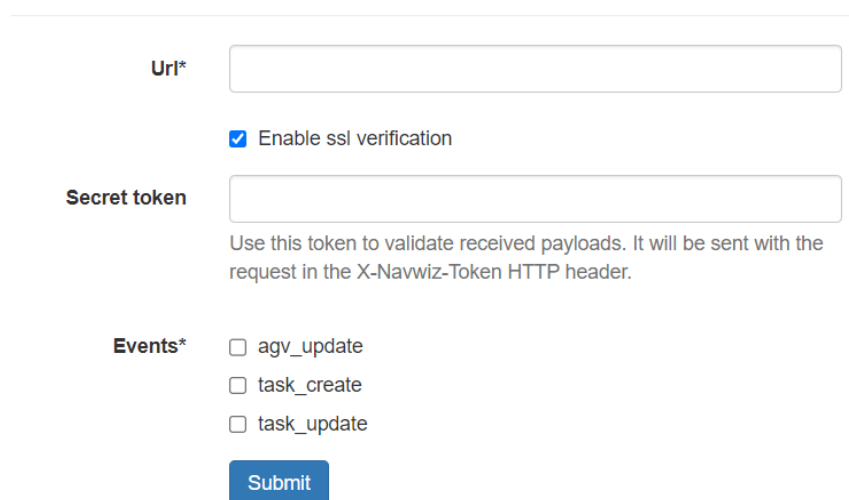
1. Receive task progress update
2. Update database on goods delivery status
3. Send notification when AGV is safety triggered or needs assistance

Adding a new webhook

To configure a webhook:

1. Select “Add new webhook”.

Add new webhook



The form for adding a new webhook contains the following fields and options:

- Uri***: A text input field.
- ☒ **Enable ssl verification**: A checkbox that is currently checked.
- Secret token**: A text input field. Below it is a note: "Use this token to validate received payloads. It will be sent with the request in the X-Navwiz-Token HTTP header."
- Events***: A group of three checkboxes:
 - ☐ agv_update
 - ☐ task_create
 - ☐ task_update
- Submit**: A blue button at the bottom.

2. In the URL field, enter the URL of the webhook endpoint. The URL must be percentage-encoded, if necessary.
3. In the Secret token field, enter the secret token to validate payloads.
4. In the Trigger section, select the events to trigger the webhook.
5. Optional. Clear the Enable SSL verification checkbox to disable SSL verification.
6. Click Submit.

Content of the webhook

The content of a webhook differs based on the event type.

agv_update event

For agv_update event, the request body is as follow:

Parameter	Type	Description
uuid	string	The UUID of the AMR.
name	string	The name of the AMR.
status	string	The status of the AMR. Can be one of the following: • Not Ready • Idle • Working • Paused • Running App • Sync Error
location	array	The expected location of the AMR after completing the current motion: represented as: [junction_id, direction] For tracked AMR, the direction can be 0 (North), 1 (South), 2 (East), and 3 (West). For trackless AMR, the direction is the heading in degree with respect to the map coordinates.
prev_location	string	The expected location of the AMR after completing the current motion: represented as: [junction_id, direction] For tracked AMR, the direction can be 0 (North), 1 (South), 2 (East), and 3 (West). For trackless AMR, the direction is the heading in degree with respect to the map coordinates.
pose	string	(For trackless AMR only) The current position of AMR with respect to the map coordinate, represented as: [x, y, heading] Measurements are specified in meters and radians.
motion	string	The current motion of the AMR. Can be one of the following: • null • Forward • Reverse • Rotate Left • Rotate Right • Branch Forward Left (For tracked AMR only) • Branch Forward Right (For tracked AMR only)

		• Branch Reverse Left (For tracked AMR only) • Branch Reverse Right (For tracked AMR only)
velocity	array	The speed of the AMR, represented as: [linear_velocity, 0.0, angular_velocity] Measurements are specified in m/s and rad/s.
safety_message	object	A message object containing the message and type of the safety check which is currently being triggered. null if no safety check is triggered.
user_message	object	A message object containing the message and type of the user popup which is currently being displayed. null if no user popup is active.
battery	float	The battery percentage of the AMR.
task_counter	integer	The number of tasks which have been executed by the AMR.
agv_ip	string	The IP address of the AMR.

Example:

```
{
  "uuid": "cb625ca1-1e9b-490c-8b76-dae58fd6722b",
  "name": "AMR 1",
  "status": "Working",
  "location": [100023, 3],
  "prev_location": [100022, 3],
  "pose": [103.45, -32.12, 1.234],
  "motion": "Forward",
  "velocity": [0.42, 0.0, -0.13],
  "safety_message":
    [
      {
        "message": "The following safety checks are triggered:\n\nObstacle blocked",
        "type": "safety-triggered"
      }
    ],
  "user_message": null,
  "battery": 87.86,
  "task_counter": 5040,
  "agv_ip": "172.16.31.101"
}
```

task_create & task_update

For task_create and task_update events, the request body is as follow:

Parameter	Type	Description
id	integer	The ID of the task.
name	string	The name of the task.
task_template_id	integer	The ID of the task template associated with this task.
task_template	string	The name of the task template associated with this task.

params	string	A json-encoded parameter object which is passed into the task template during the creation of this task.
agv_id	string	The UUID of the AMR assigned to this task. null if no AMR is assigned yet.
agv	string	The name of the AMR assigned to this task. "" if no AMR is assigned yet.
status	string	The status of the task. Can be one of the following: • suspended • pending • cancelling • in_progress • aborting • completed • aborted • cancelled
progress	string	The progress message (if any) set in the task template.
sequence	integer	The sequence number to determine which task will be run first. The lower (more negative) the sequence number, the higher the priority. Defaults to 0. (Applicable for pending tasks only)
owner	string	The name of the user who creates, and therefore owns the task.
start_after	string	An ISO 8601 datetime string indicating a time restriction prior to which the task cannot be started. null if not applicable.
created	string	An ISO 8601 datetime string indicating the creation time of the task.
run_start	string	An ISO 8601 datetime string indicating the creation time of the task.
run_end	string	An ISO 8601 datetime string indicating the completion time of the task.

Example:

```
{
  "id": 1091,
  "name": "Transfer 5306",
  "task_template_id": 87,
  "task_template": "S1 -> S2",
  "params": "0",
  "agv_id": null,
  "agv": "",
  "status": "pending",
  "progress": "",
  "sequence": 0,
  "owner": "user",
  "start_after": null,
  "created": "2018-11-01T13:15:34+08:00",
  "run_start": null,
  "run_end": null
}
```

Additional Content

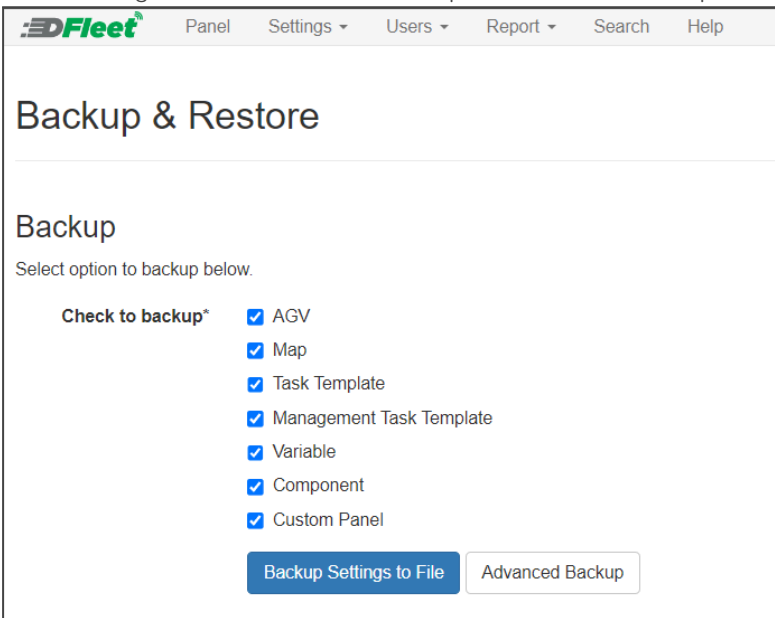
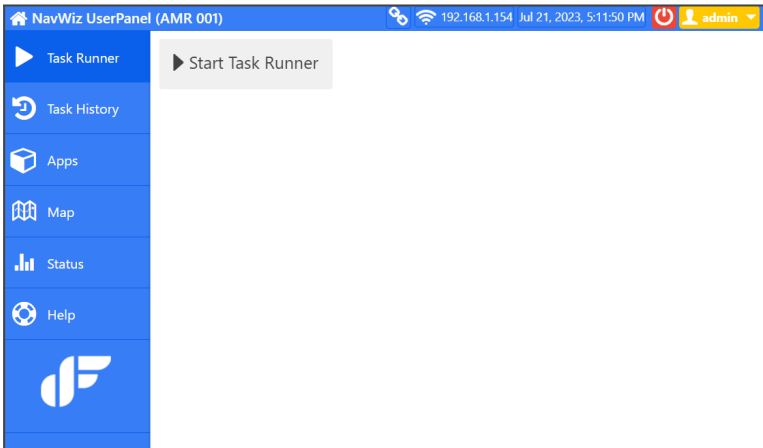
Section 1: AMR Transfer Between DFleet

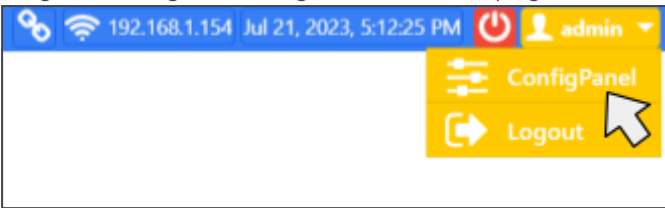
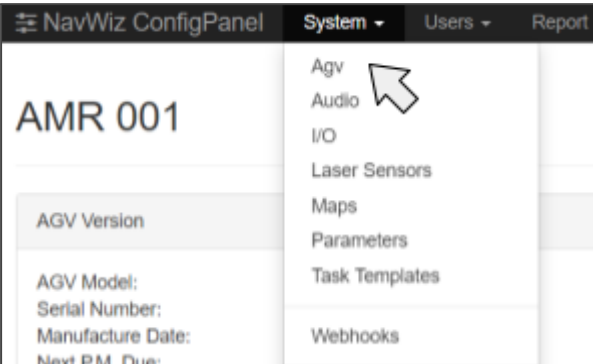
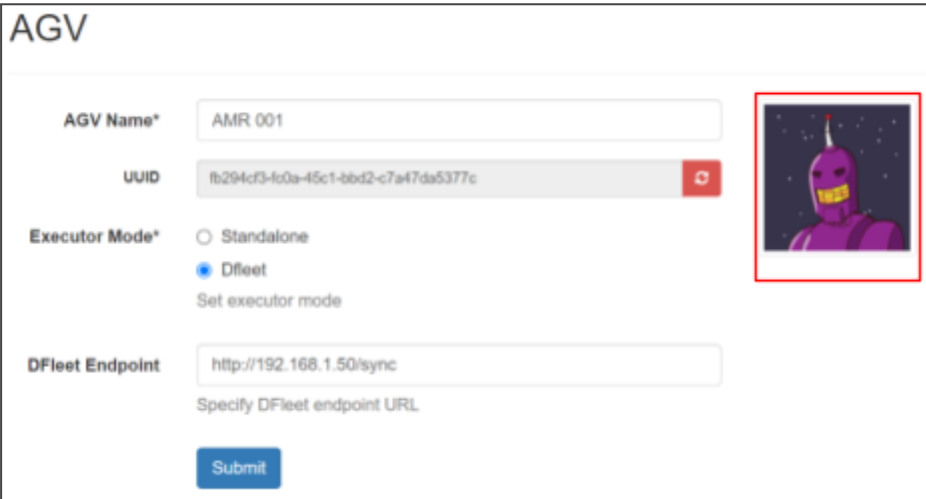
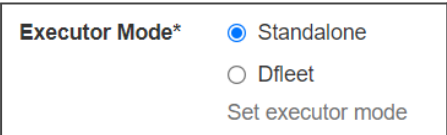
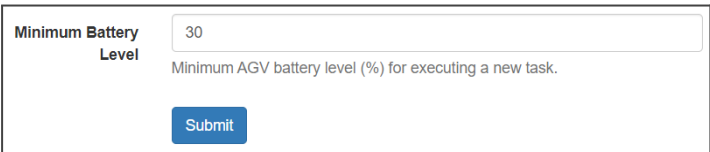


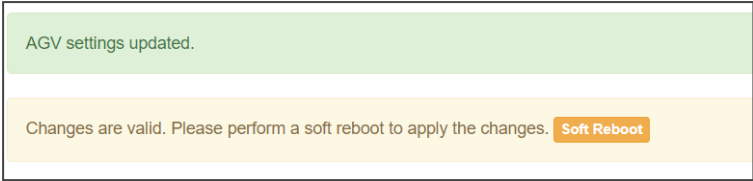
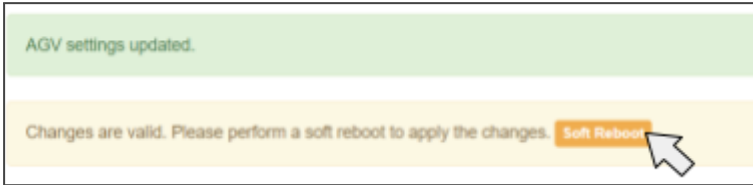
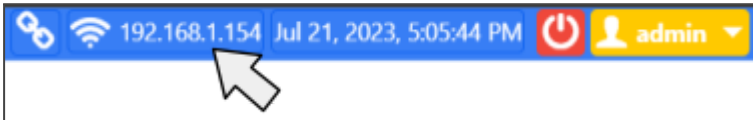
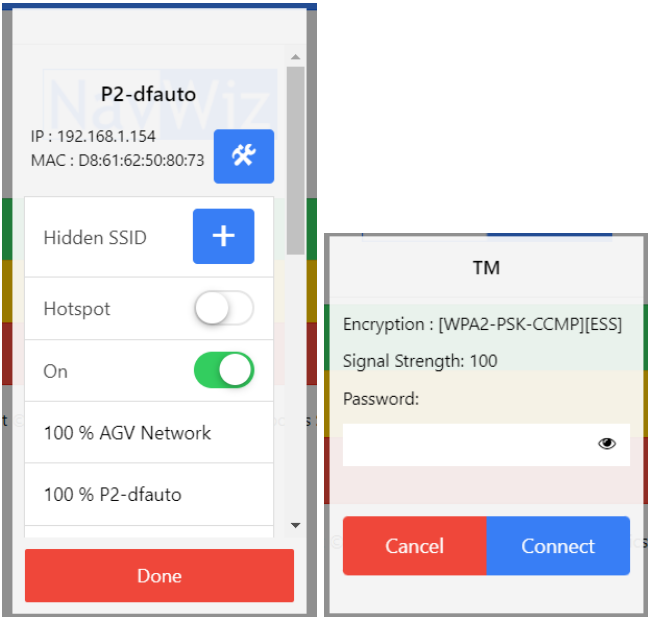
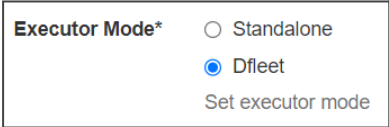
Caution:

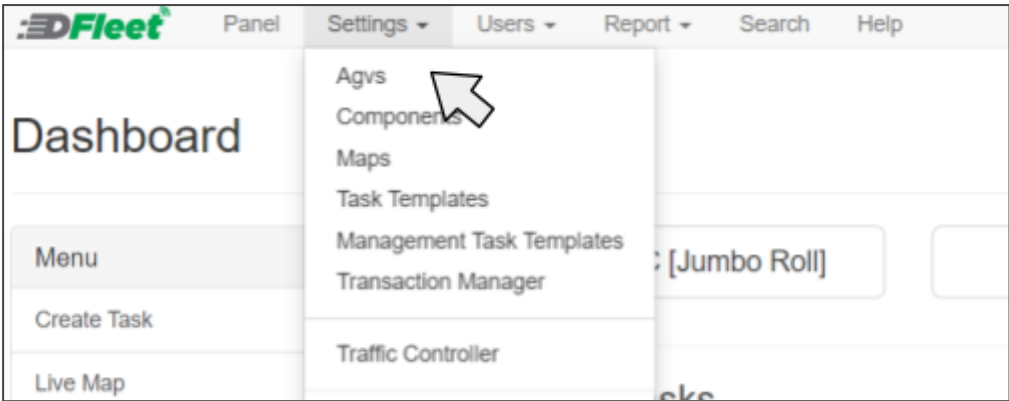
It is not advisable to add or replace existing AMR line-up inside an DFleet system without first consulting appropriate DF personnel as it may bring forth several unexpected discrepancies in the system.

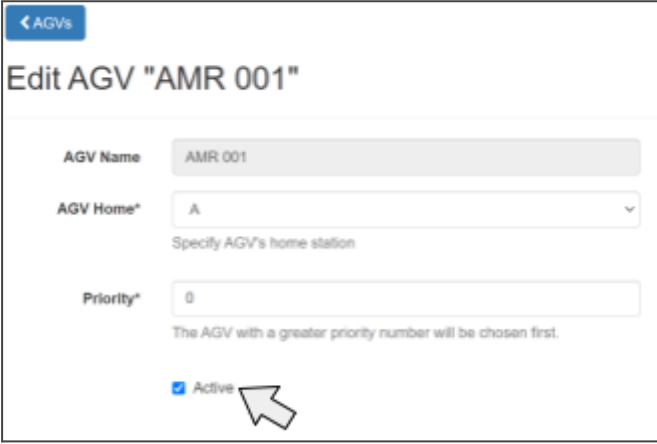
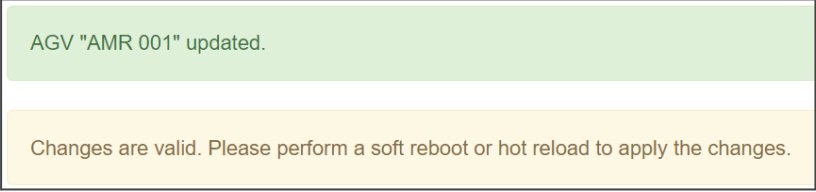
For this example, assume there exist two different lines of AMR from two different DFleet. In the situation where it is required to move an AMR already registered inside an DFleet to a different DFleet will involve a few steps to be adhered:

Step 1	Before starting the transfer. It is of utmost importance to start a backup of the DFleet system beforehand.  The backup file would be a lifesaver in the event of any unprecedented error or malfunction of the system after following through this guide.
Step 2	Now that the system has been backed up. Connect to respective AMR NavWiz through a web browser wirelessly using IP. 

Step 3	Navigate to ConfigPanel through the link on the top right corner of NavWiz interface. 
Step 4	Inside the ConfigPanel, navigate to from the tab on System > Agv top right corner of interface. 
Step 5	 Inside the AGV page, ensure that the AMR skill set matches with the skill sets of the AMRs in the second DFleet. If the skill sets do not match, the AMR won't be able to integrate into the second DFleet successfully. If the skill sets do not match, please consult with DF staff for further measures.
Step 6	Inside the Agv page, change the Executor Mode from Dfleet to Standalone. 
Step 7	Submit change by pressing the Submit button on the bottom left of the interface. 

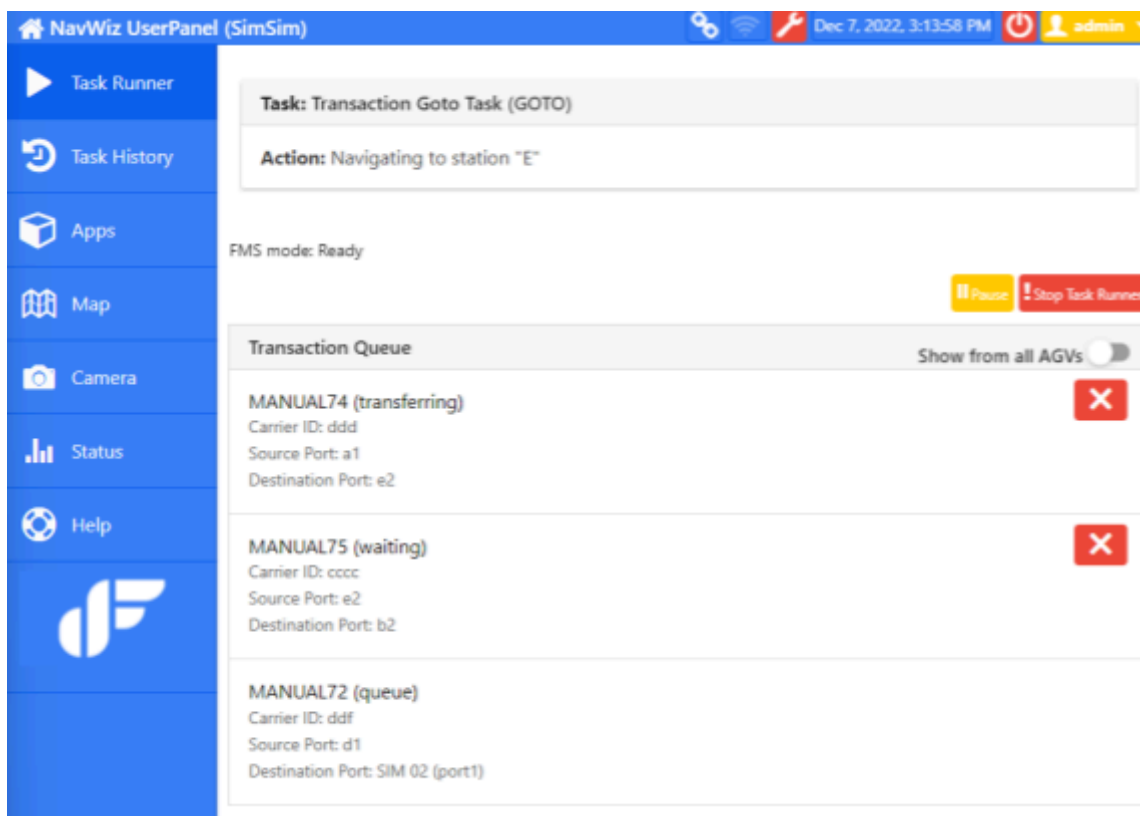
Step 8	A notification will pop up to confirm the change and request the user to reboot the AMR. 
Step 9	Click on the "Soft Reboot" button to reboot the AMR. 
Step 10	Once AMR is rebooted, change the Wi-Fi setting of the AMR by navigating to the top bar of the NavWiz Interface. 
Step 11	A popup will open where you can change the Wi-Fi connection to the respective DFleet network. 
Step 13	Next navigate back to the Agv page on ConfigPanel, change executor mode back to DFleet from Standalone. 

Step 14	Before submitting the change, open the new DFleet Interface and navigate to Settings > Agvs 
Step 15	Inside the Agvs page copy the DFleet Endpoint and paste into the AGV DFleet Endpoint. AGVs DFleet Version: 2.2.0 (tracked) DFleet Endpoint: <code>http://192.168.0.56/sync</code> AGV AGV Name* AMR 001 UUID <code>fb294cf3-fc0a-45c1-bbd2-c7a47da5377c</code> Executor Mode* ☐ Standalone ☒ DFleet Set executor mode DFleet Endpoint http://192.168.0.56/sync Specify DFleet endpoint URL Submit
Step 16	Submit and wait for the pop up notification to confirm the change. AGV settings updated and AGV is successfully paired with DFleet server. Current AGV status: Activated Changes are valid. Perform soft reboot or hot reload to apply the changes. Soft Reboot

Step 18	As you may have noticed, it is stated that the AMR status is currently Deactivated. To change this, open the DFleet Interface and navigate to again and check the AMR status. 
Step 19	Navigate to the AGV editor through the pencil icon on the right. 
Step 20	Inside the AGV editor, tick the Active option to activate the AMR and submit the change. 
Step 21	Notification will pop up to confirm the change accompanied with either “Changes are valid” or “Invalid changes” messages.  If there is an error message, users need to follow the notification respectively to solve the subsequent error before being able to fully integrate the AMR into the DFleet system.
Step 22	Some of the expected errors may be due to the AGV id mismatched. Thus supervisors need to manually reset the id of the AGV especially in the event of AGV replacement in the system as the AGV UUID are unique for each AGV. In the event of AGV addition, then the supervisor may need to assign the AGV into the specific task template which the AGV would be interacting with. Lastly, in the event that the supervisors are unable to resolve the given issue that rises after the change, immediately contact any DF staff currently in charge of the site or directly contact any relevant staff of DF for assistance.

Section 2: AMR in Transaction Mode

During the execution of transaction tasks, NavWiz switches to the transaction display mode, allowing users to monitor the current action of the AMR. Additionally, users have the option to cancel and abort transactions directly from this page.



The screenshot displays the NavWiz UserPanel (SimSim) interface. The top status bar shows the date and time as Dec 7, 2022, 3:13:58 PM, and the user as admin. The left sidebar contains navigation options: Task Runner, Task History, Apps, Map, Camera, Status, and Help. The main content area shows the current task: "Task: Transaction Goto Task (GOTO)" with the action "Action: Navigating to station 'E'". Below this, the FMS mode is set to "Ready". A "Pause" button and a "Stop Task Runner" button are visible. The "Transaction Queue" section lists three tasks:

Task Name	Carrier ID	Source Port	Destination Port	Action
MANUAL74 (transferring)	ddd	a1	e2	Cancel (X)
MANUAL75 (waiting)	cccc	e2	b2	Cancel (X)
MANUAL72 (queue)	ddf	d1	SIM 02 (port1)	

The "Transaction Queue" section also includes a toggle for "Show from all AGVs".

Disclaimer

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Change Log

Version	Date	Changes
1.0	2025-07-16	Initial Draft
1.1	2025-10-09	Update additional panel tags.



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